Linear Algebra

Lecture notes, Chapters 1–6 and Appendices

Based on Friedberg, Insel and Spence, *Linear Algebra* (4th ed.)

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Chapter 1. Vector Spaces

1.1 Introduction

Motivation. Eight rules obeyed by adding and scaling arrows are all that linear algebra uses.

Parallelogram law: with P at the origin, $x = (a_1, a_2)$, $y = (b_1, b_2)$, the diagonal is $x + y = (a_1 + b_1, a_2 + b_2)$; also $tx = (ta_1, ta_2)$.

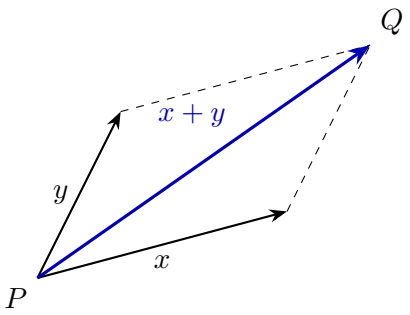


Figure. *Parallelogram law.*

For all vectors x, y, z and reals a, b:

1. $x + y = y + x$; 2. $(x + y) + z = x + (y + z)$; 3. there is 0 with $x + 0 = x$; 4. each x has y with $x + y = 0$;

5. $1x = x$; 6. $(ab)x = a(bx)$; 7. $a(x + y) = ax + ay$; 8. $(a + b)x = ax + bx$.

Lines and planes. Line through A, B (endpoints of u, v): $x = u + t(v - u)$. Plane through noncollinear A, B, C, with u, v from A to B, C: $x = A + su + tv$.

Example 1

$A = (-2, 0, 1)$, $B = (4, 5, 3)$: $x = (-2, 0, 1) + t(6, 5, 2)$.

Example 2

$A = (1, 0, 2)$, $B = (-3, -2, 4)$, $C = (1, 8, -5)$: $x = (1, 0, 2) + s(-4, -2, 2) + t(0, 8, -7)$.

1.2 Vector spaces

Motivation. A vector space is any set obeying the eight rules.

Definition (Vector space)

A **vector space** V over a field $\mathbb{F}$ (see Appendix C) is a set with an addition $V \times V \to V$ and a scalar multiplication $\mathbb{F} \times V \to V$ such that for all $x, y, z \in V$, $a, b \in \mathbb{F}$:
(VS 1) $x + y = y + x$; (VS 2) $(x + y) + z = x + (y + z)$;
(VS 3) there is $0 \in V$ with $x + 0 = x$ for all x; (VS 4) for each x there is y with $x + y = 0$;
(VS 5) $1x = x$; (VS 6) $(ab)x = a(bx)$;
(VS 7) $a(x + y) = ax + ay$; (VS 8) $(a + b)x = ax + bx$.
Elements of $\mathbb{F}$ are **scalars**, elements of V **vectors**.

An $m \times n$ **matrix** A has entries A_{ij} (row i, column j); A_{ii} are the **diagonal entries**, O is the **zero matrix**; A is **square** if $m = n$.

Example 1

$\mathbb{F}^n$, n-tuples with coordinatewise operations; in $\mathbb{C}^2$, $(1+i, 2) + (2-3i, 4i) = (3-2i, 2+4i)$, $i(1+i, 2) = (-1+i, 2i)$.

Example 2

$\mathrm{M}_{m\times n}(\mathbb{F})$: $(A+B)_{ij} = A_{ij} + B_{ij}$, $(cA)_{ij} = cA_{ij}$.

Example 3

$\mathcal{F}(S, \mathbb{F})$ $(S \neq \emptyset)$: $(f+g)(s) = f(s) + g(s)$, $(cf)(s) = cf(s)$.

The **degree** of $f(x) = a_n x^n + \cdots + a_0$ is the largest k with $a_k \neq 0$; the **zero polynomial** has degree -1.

Example 4

$\mathsf{P}(\mathbb{F})$, polynomials with coefficientwise operations.

Example 5

Sequences in $\mathbb{F}$ with finitely many nonzero terms, termwise operations.

Example 6

On $\mathbb{R}^2$ with $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 - b_2)$, $c(a_1, a_2) = (ca_1, ca_2)$: (VS 1), (VS 2), (VS 8) fail.

Example 7

On $\mathbb{R}^2$ with $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, 0)$, $c(a_1, a_2) = (ca_1, 0)$: (VS 3), (VS 4), (VS 5) fail.

Theorem 1.1 (Cancellation law for vector addition)

If $x + z = y + z$ in V, then $x = y$.

Proof. Take v with $z + v = 0$ (VS 4). Then $x = x + 0 = x + (z + v) = (x + z) + v = (y + z) + v = y + (z + v) = y + 0 = y$ by (VS 2), (VS 3). ■

Corollary 1

The vector 0 of (VS 3) is unique.

Proof [Supplement]. If $0'$ also satisfies (VS 3), then $0 + 0' = 0 = 0 + 0$ (by (VS 1), (VS 3)), so $0' = 0$ by Theorem 1.1. ■

Corollary 2

The vector y of (VS 4) is unique.

Proof [Supplement]. If $x+y=0=x+y'$, then $y+x=y'+x$ by (VS 1), so $y=y'$ by Theorem 1.1. ■

0 is the **zero vector**, and the y of (VS 4) is the **additive inverse** $-x$.

Theorem 1.2

In any vector space V:
(a) $0x=0$ for each $x\in V$.
(b) $(-a)x=-(ax)=a(-x)$ for each $a\in\mathbb{F}$, $x\in V$.
(c) $a0=0$ for each $a\in\mathbb{F}$.

Proof.

(a) $0x+0x=(0+0)x=0x=0x+0=0+0x$, so $0x=0$ by Theorem 1.1.

(b) $ax+(-a)x=(a+(-a))x=0x=0$, so $(-a)x=-(ax)$ by Corollary 2. In particular $(-1)x=-x$, and $a(-x)=a((-1)x)=(a(-1))x=(-a)x$ by (VS 6).

(c) [Supplement] $a0+a0=a(0+0)=a0=a0+0$, so $a0=0$ by Theorem 1.1. ■

1.3 Subspaces

Motivation. Three closure checks decide whether a subset is a vector space in its own right.

Definition (Subspace)

A subset W of V is a **subspace** if it is a vector space over $\mathbb{F}$ with the operations of V. V and the **zero subspace** $\{0\}$ are subspaces.

(VS 1, 2, 5–8) are inherited by any subset.

Theorem 1.3

$W\subseteq V$ is a subspace iff (a) $0\in W$,
(b) $x+y\in W$ for $x,y\in W$, and (c) $cx\in W$ for $c\in\mathbb{F}$, $x\in W$.

Proof.

Step 1 (only if). (b), (c) are closure; the zero $0'$ of W has $x+0'=x=x+0$, so $0'=0\in W$ (Theorem 1.1).

Step 2 (if). Inverses: $-x=(-1)x\in W$ by (c) and Theorem 1.2; the other axioms are inherited. ■

The **transpose** has $(A^t)_{ij}=A_{ji}$; $(aA+bB)^t=aA^t+bB^t$. A is **symmetric** if $A^t=A$; symmetric matrices form a subspace of $\mathrm{M}_{n\times n}(\mathbb{F})$.

Example 1

$\mathsf{P}_n(\mathbb{F})$ (degree $\le n$) $\subseteq\mathsf{P}(\mathbb{F})$.

Example 2

$C(\mathbb{R})$ (continuous functions) $\subseteq \mathcal{F}(\mathbb{R}, \mathbb{R})$.

Example 3

Diagonal matrices ($M_{ij} = 0$ for $i \neq j$) form a subspace.

Example 4

The **trace** $\operatorname{tr}(M) = M_{11} + \cdots + M_{nn}$ is linear; trace-zero matrices form a subspace.

Example 5

Matrices with nonnegative entries: not a subspace (not closed under $-1\cdot$).

Theorem 1.4

Any intersection of subspaces of V is a subspace of V.

Proof. Each member contains 0, $x + y$ and ax whenever it contains x, y; so does the intersection W. Apply Theorem 1.3. ■

A union of two subspaces is a subspace iff one contains the other. The **sum** $W_1 + W_2 = \{x + y : x \in W_1, y \in W_2\}$ is the smallest subspace containing both. V is the **direct sum** $W_1 \oplus W_2$ if $W_1 + W_2 = V$ and $W_1 \cap W_2 = \{0\}$.

1.4 Linear combinations and systems of linear equations

Motivation. Testing for a linear combination means solving a linear system.

Definition (Linear combination)

For nonempty $S \subseteq V$, v is a **linear combination** of vectors of S if $v = a_1u_1 + \cdots + a_nu_n$ for some finitely many $u_i \in S$ and scalars a_i (the **coefficients**). $0 = 0u$ is a linear combination of any nonempty S.

Example 1

Vitamin vectors of foods lie in $\mathbb{R}^5$; mixing foods combines them linearly.

Elimination. Swapping equations, scaling one by $c \neq 0$, and adding a multiple of one to another preserve solutions. Reduce until each leading coefficient is 1, its variable appears nowhere else, and leading variables move right; $0 = c \neq 0$ means no solution.

Example 2

$2x^3 - 2x^2 + 12x - 6 = a(x^3 - 2x^2 - 5x - 3) + b(3x^3 - 5x^2 - 4x - 9)$ means

$$a + 3b = 2, \quad -2a - 5b = -2, \quad -5a - 4b = 12, \quad -3a - 9b = -6,$$

solved by $a = -4$, $b = 2$. For $3x^3 - 2x^2 + 7x + 8$ the system reduces to $0 = 1$: not a

combination.

Definition (Span)

The **span** $\operatorname{span}(S)$ of nonempty $S \subseteq V$ is the set of all linear combinations of vectors in S; $\operatorname{span}(\emptyset) = \{0\}$.

Theorem 1.5

$\operatorname{span}(S)$ is a subspace of V, and every subspace containing S contains $\operatorname{span}(S)$.

Proof. Clear for $S = \emptyset$. Otherwise $0 = 0z$, $\sum a_i u_i + \sum b_j v_j$ and $c\sum a_i u_i = \sum (ca_i)u_i$ are combinations (Theorem 1.3). A subspace $W \supseteq S$ is closed under the operations, so $W \ni \sum a_i u_i$. ■

Definition (Generating set)

S **generates** (**spans**) V if $\operatorname{span}(S) = V$.

Example 3

$(1,1,0), (1,0,1), (0,1,1)$ generate $\mathbb{R}^3$:

$$(a_1, a_2, a_3) = \tfrac{1}{2}(a_1 + a_2 - a_3)(1,1,0) + \tfrac{1}{2}(a_1 - a_2 + a_3)(1,0,1) + \tfrac{1}{2}(-a_1 + a_2 + a_3)(0,1,1).$$

Example 4

$x^2 + 3x - 2$, $2x^2 + 5x - 3$, $-x^2 - 4x + 4$ generate $\mathsf{P}_2(\mathbb{R})$ (solve a 3×3 system).

Example 5

The matrices $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ generate $\mathrm{M}_{2\times 2}(\mathbb{R})$. Matrices with equal diagonal entries cannot: their combinations keep that property.

1.5 Linear dependence and linear independence

Motivation. Independence means no vector is redundant: 0 has only the trivial representation.

Definition (Linear dependence)

$S \subseteq V$ is **linearly dependent** if there are finitely many distinct $u_1, \ldots, u_n \in S$ and scalars $a_1, \ldots, a_n$, not all zero, with $a_1 u_1 + \cdots + a_n u_n = 0$ (a **nontrivial representation** of 0). Any set containing 0 is dependent $(1 \cdot 0 = 0)$.

Example 1

Dependent in $\mathbb{R}^4$: $4(1,3,-4,2) - 3(2,2,-4,0) + 2(1,-3,2,-4) + 0(-1,0,1,0) = 0$.

Example 2

$M_1 = \begin{pmatrix} 1 & -3 & 2 \\ -4 & 0 & 5 \end{pmatrix}$, $M_2 = \begin{pmatrix} -3 & 7 & 4 \\ 6 & -2 & -7 \end{pmatrix}$, $M_3 = \begin{pmatrix} -2 & 3 & 11 \\ -1 & -3 & 2 \end{pmatrix}$: $5M_1 + 3M_2 - 2M_3 = O$.

Definition (Linear independence)

S is **linearly independent** if it is not linearly dependent: $\sum a_i u_i = 0$ with distinct $u_i \in S$ forces all $a_i = 0$.

$\emptyset$ is independent; $\{u\}$ is independent iff $u \neq 0$ (if $au = 0$, $a \neq 0$, then $u = a^{-1}(au) = 0$).

Example 3

$\{(1,0,0,-1), (0,1,0,-1), (0,0,1,-1), (0,0,0,1)\}$: $\sum a_i u_i = (a_1, a_2, a_3, a_4 - a_1 - a_2 - a_3) = 0$ forces all $a_i = 0$.

Example 4

$p_k = x^k + \cdots + x^n$ $(0 \leq k \leq n)$: in $\sum a_k p_k = 0$ the x^m-coefficient is $a_0 + \cdots + a_m$, so $a_0 = a_1 = \cdots = 0$.

Theorem 1.6

If $S_1 \subseteq S_2 \subseteq V$ and S_1 is linearly dependent, so is S_2.

Proof. A nontrivial representation of 0 by vectors of S_1 is one by vectors of S_2. ■

Corollary

If $S_1 \subseteq S_2 \subseteq V$ and S_2 is linearly independent, so is S_1.

Proof. Contrapositive of Theorem 1.6. ■

Theorem 1.7

Let S be linearly independent and $v \notin S$. Then $S \cup \{v\}$ is linearly dependent iff $v \in \text{span}(S)$.

Proof.

Step 1 (only if). In a nontrivial $\sum a_i u_i = 0$ ($u_i \in S \cup \{v\}$ distinct), independence of S forces $u_1 = v$, $a_1 \neq 0$; so $v = -a_1^{-1} \sum_{i \geq 2} a_i u_i \in \text{span}(S)$.

Step 2 (if). $v = \sum b_i v_i$ ($v_i \in S$) gives the nontrivial $\sum b_i v_i + (-1)v = 0$. ■

1.6 Bases and dimension

Motivation. Independent sets never outnumber generating sets, so all bases have one size: the dimension.

Definition (Basis)

A **basis** β for V is a linearly independent subset that generates V.

Example 1

$\emptyset$ is a basis for the zero space.

Example 2

$e_1 = (1, 0, \dots, 0), \dots, e_n = (0, \dots, 0, 1)$ form the **standard basis** of $\mathbb{F}^n$.

Example 3

The matrices E^{ij} (single 1 in position (i, j)) form a basis of $\mathrm{M}_{m\times n}(\mathbb{F})$.

Example 4

$\{1, x, \dots, x^n\}$ is the **standard basis** of $\mathsf{P}_n(\mathbb{F})$.

Example 5

$\{1, x, x^2, \dots\}$ is a basis of $\mathsf{P}(\mathbb{F})$.

Theorem 1.8

$\beta = \{u_1, \dots, u_n\}$ is a basis for V iff each $v \in V$ can be written uniquely as $v = a_1u_1 + \cdots + a_nu_n$.

Proof.

Step 1 (only if). Existence: β generates. $\sum a_iu_i = \sum b_iu_i \Rightarrow \sum(a_i - b_i)u_i = 0 \Rightarrow a_i = b_i$.

Step 2 (if) [Supplement]. β generates; uniqueness at $v = 0 = \sum 0u_i$ is independence. ■

Theorem 1.9

If V is generated by a finite set S, then some subset of S is a basis for V. Hence V has a finite basis.

Proof. If $S \subseteq \{0\}$, $\emptyset$ is a basis. Else grow $\{u_1\}$ $(u_1 \neq 0)$ by vectors of S keeping independence, until (finiteness) β has $\beta \cup \{v\}$ dependent for all $v \in S - \beta$; so $S \subseteq \mathrm{span}(\beta)$ (Theorem 1.7) and $V = \mathrm{span}(S) \subseteq \mathrm{span}(\beta)$ (Theorem 1.5). ■

Example 6

$S = \{(2, -3, 5), (8, -12, 20), (1, 0, -2), (0, 2, -1), (7, 2, 0)\}$: drop $(8, -12, 20) = 4(2, -3, 5)$ and $(7, 2, 0) = 2(2, -3, 5) + 3(1, 0, -2) + 4(0, 2, -1)$; basis $\{(2, -3, 5), (1, 0, -2), (0, 2, -1)\}$ of $\mathbb{R}^3$.

Theorem 1.10 (Replacement theorem)

Let V be generated by a set G of exactly n vectors and let L be linearly independent with exactly m vectors. Then $m \leq n$, and there is $H \subseteq G$ with exactly $n - m$ vectors such that $L \cup H$ generates V.

Proof. Induction on m; $m = 0$: $H = G$. For independent $L = \{v_1, \dots, v_{m+1}\}$, $\{v_1, \dots, v_m\}$ is independent (Corollary), so by induction $m \leq n$ and $\{v_1, \dots, v_m, u_1, \dots, u_{n-m}\}$ generates V for

some $u_i \in G$:

$$a_1 v_1 + \cdots + a_m v_m + b_1 u_1 + \cdots + b_{n-m} u_{n-m} = v_{m+1}. \tag{1.1}$$

If $n = m$ or all $b_i = 0$, $v_{m+1} \in \operatorname{span}\{v_1, \ldots, v_m\}$, contradicting Theorem 1.7. So $n \geq m + 1$ and, say, $b_1 \neq 0$. Solving (1.1) for u_1: $u_1 \in \operatorname{span}(L \cup H)$, $H = \{u_2, \ldots, u_{n-m}\}$. Then $\operatorname{span}(L \cup H) \ni v_1, \ldots, v_m, u_1, \ldots, u_{n-m}$, so it is V (Theorem 1.5). ■

Corollary 1

If V has a finite basis, every basis of V has the same number of vectors.

Proof. With $|\beta| = n$, every basis γ is finite with $m \leq n$ (Theorem 1.10 on finite subsets of γ); reversing roles, $n \leq m$. ■

Definition (Dimension)

V is **finite-dimensional** if it has a finite basis; the common number of vectors in a basis is its **dimension** $\dim(V)$. Otherwise V is **infinite-dimensional**.

Example 7

$\dim\{0\} = 0$.

Example 8

$\dim \mathbb{F}^n = n$.

Example 9

$\dim \mathrm{M}_{m\times n}(\mathbb{F}) = mn$.

Example 10

$\dim \mathrm{P}_n(\mathbb{F}) = n + 1$.

Example 11

$\mathbb{C}$ over $\mathbb{C}$ has dimension 1 (basis $\{1\}$).

Example 12

$\mathbb{C}$ over $\mathbb{R}$ has dimension 2 (basis $\{1, i\}$).

Corollary 2

Let $\dim(V) = n$.
(a) Every finite generating set has at least n vectors, and a generating set of exactly n vectors is a basis.
(b) Every linearly independent set of exactly n vectors is a basis.
(c) Every linearly independent subset of V extends to a basis.

Proof. Let β be a basis, $|\beta| = n$.

(a) Generating $G \supseteq$ basis H (Theorem 1.9), $|H| = n$ (Corollary 1); so $|G| \geq n$, and $|G| = n \Rightarrow G = H$.

(b) Theorem 1.10 with $G = \beta$, $|L| = n$: $H = \emptyset$, so L generates.

(c) For independent L, $|L| = m \leq n$, Theorem 1.10: $L \cup H$ generates with $|H| = n - m$, so $|L \cup H| \leq n$; by (a) it is a basis. ■

Example 13

By (a), the three polynomials of Example 4, Section 1.4, form a basis of $\mathsf{P}_2(\mathbb{R})$.

Example 14

By (a), the four matrices of Example 5, Section 1.4, form a basis of $\mathrm{M}_{2\times 2}(\mathbb{R})$.

Example 15

By (b), the four vectors of Example 3, Section 1.5, form a basis of $\mathbb{R}^4$.

Example 16

By (b), $\{p_0, \ldots, p_n\}$ of Example 4, Section 1.5, is a basis of $\mathsf{P}_n(\mathbb{F})$.

Theorem 1.11

If W is a subspace of a finite-dimensional V, then W is finite-dimensional and $\dim W \leq \dim V$; if $\dim W = \dim V$, then $W = V$.

Proof. Let $\dim V = n$, $W \neq \{0\}$. Grow an independent $\{x_1, \ldots, x_k\} \subseteq W$ from $x_1 \neq 0$; by Theorem 1.10 this stops at $k \leq n$, when no vector of W can be added. By Theorem 1.7 it generates W: $\dim W = k \leq n$. If $k = n$, it is a basis of V (Corollary 2(b)), so $W = V$. ■

Example 17

$W = \{(a_1, \ldots, a_5) \in \mathbb{F}^5 : a_1 + a_3 + a_5 = 0,\ a_2 = a_4\}$ has basis $\{(-1, 0, 1, 0, 0), (-1, 0, 0, 0, 1), (0, 1, 0, 1, 0)\}$, so $\dim W = 3$.

Example 18

The diagonal $n \times n$ matrices have basis $\{E^{11}, \ldots, E^{nn}\}$; dimension n.

Example 19

Symmetric $n \times n$ matrices: basis $\{A^{ij} : i \leq j\}$ (1 at (i, j), (j, i), else 0); dimension $\frac{1}{2}n(n+1)$.

Corollary

If W is a subspace of a finite-dimensional V, any basis for W extends to a basis for V.

Proof. A basis of W is an independent subset of V; apply Corollary 2(c) of Theorem 1.10. ■

Example 20

Even polynomials in $\mathsf{P}_{18}(\mathbb{F})$: basis $\{1, x^2, \ldots, x^{18}\}$, dimension 10.

Subspaces of $\mathbb{R}^3$: $\{0\}$, lines and planes through 0, $\mathbb{R}^3$ (dimensions 0–3).

1.6.1 The Lagrange interpolation formula

Let $c_0, \ldots, c_n$ be distinct scalars of an infinite field $\mathbb{F}$, and

$$f_i(x) = \prod_{k \neq i} \frac{x - c_k}{c_i - c_k} \qquad (0 \leq i \leq n),$$

the **Lagrange polynomials**: $f_i(c_j) = \delta_{ij}$. Evaluating $\sum a_i f_i = 0$ at c_j gives $a_j = 0$, so $\{f_0, \ldots, f_n\}$ is a basis of $\mathsf{P}_n(\mathbb{F})$ (Corollary 2(b)); evaluating $g = \sum b_i f_i$ gives $b_j = g(c_j)$:

$$g = \sum_{i=0}^{n} g(c_i) f_i \qquad (g \in \mathsf{P}_n(\mathbb{F})),$$

the **Lagrange interpolation formula**. So $\sum b_i f_i$ is the unique $g \in \mathsf{P}_n(\mathbb{F})$ with $g(c_j) = b_j$, and a nonzero $g \in \mathsf{P}_n(\mathbb{F})$ has at most n zeros.

1.7 Maximal linearly independent subsets *

Motivation. The Maximal Principle gives every space, even infinite-dimensional, a basis.

Definition (Maximal element; chain)

A member M of a family $\mathcal{F}$ of sets is **maximal** if no member of $\mathcal{F}$ other than M contains M. A collection $\mathcal{C}$ of sets is a **chain** if for all $A, B \in \mathcal{C}$, $A \subseteq B$ or $B \subseteq A$.

Example 1

In the power set of S, S is the (unique) maximal element.

Example 2

For disjoint nonempty S, T, $\mathcal{P}(S) \cup \mathcal{P}(T)$ has maximal elements S, T.

Example 3

Finite subsets of an infinite set have no maximal element.

Example 4

$A_n = \{1, \ldots, n\}$, $n \geq 1$, form a chain.

Maximal Principle (assumed; ≡ Axiom of Choice). If every chain $\mathcal{C} \subseteq \mathcal{F}$ has all its members contained in one member of $\mathcal{F}$, then $\mathcal{F}$ has a maximal member.

Definition (Maximal linearly independent subset)

$B \subseteq S$ is a **maximal linearly independent subset of** S if (a) B is independent and (b) the only independent subset of S containing B is B.

Example 5

By Example 2 of Section 1.4, any two elements of $S = \{2x^3 - 2x^2 + 12x - 6, x^3 - 2x^2 - 5x - 3, 3x^3 - 5x^2 - 4x - 9\}$ form a maximal linearly independent subset.

A basis of V is a maximal independent subset of V (Theorem 1.7).

Theorem 1.12

If S generates V and β is a maximal linearly independent subset of S, then β is a basis for V.

Proof. Need $S \subseteq \text{span}(\beta)$ (Theorem 1.5): $v \in S - \text{span}(\beta)$ would make $\beta \cup \{v\}$ independent (Theorem 1.7). ■

Theorem 1.13

If S is a linearly independent subset of V, some maximal linearly independent subset of V contains S.

Proof. Let $\mathcal{F}$ be the independent sets $\supseteq S$. For a chain $\mathcal{C} \subseteq \mathcal{F}$ put $U = \bigcup \mathcal{C}$ ($U = S$ if $\mathcal{C} = \emptyset$). If $\sum a_i u_i = 0$, $u_i \in A_i \in \mathcal{C}$ distinct, some $A_k \supseteq$ all A_i, so all $a_i = 0$: $U \in \mathcal{F}$. The Maximal Principle gives a maximal member of $\mathcal{F}$; it is maximal independent in V, as independent supersets contain S. ■

Corollary

Every vector space has a basis.

Proof. Apply Theorem 1.13 to $S = \emptyset$, and Theorem 1.12 with $S = V$. ■

All bases of a space have the same cardinality (cited).

Chapter 2. Linear Transformations and Matrices

2.1 Linear transformations, null spaces, and ranges

Motivation. Linear maps preserve the vector operations and are fixed by their values on a basis.

Definition (Linear transformation)

$\mathsf{T}: V \to W$ is **linear** if $\mathsf{T}(x+y) = \mathsf{T}(x) + \mathsf{T}(y)$ and $\mathsf{T}(cx) = c\mathsf{T}(x)$ for all $x, y \in V$, $c \in \mathbb{F}$.

Then $\mathsf{T}(0) = 0$, $\mathsf{T}(x-y) = \mathsf{T}(x) - \mathsf{T}(y)$, $\mathsf{T}(\sum a_i x_i) = \sum a_i \mathsf{T}(x_i)$; linearity $\iff$ $\mathsf{T}(cx+y) = c\mathsf{T}(x) + \mathsf{T}(y)$.

Example 1

$\mathsf{T}(a_1, a_2) = (2a_1 + a_2, a_1)$: for $x = (b_1, b_2)$, $y = (d_1, d_2)$, $\mathsf{T}(cx+y) = (2(cb_1 + d_1) + cb_2 + d_2,\ cb_1 + d_1) = c\mathsf{T}(x) + \mathsf{T}(y)$.

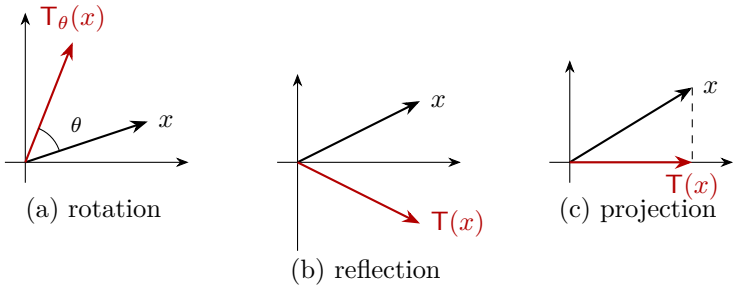


Figure. *Examples 2–4.*

Example 2

Rotation by θ: $r(\cos\alpha, \sin\alpha) \mapsto r(\cos(\alpha+\theta), \sin(\alpha+\theta))$, i.e. $\mathsf{T}_\theta(a_1, a_2) = (a_1\cos\theta - a_2\sin\theta,\ a_1\sin\theta + a_2\cos\theta)$.

Example 3

Reflection about the x-axis: $\mathsf{T}(a_1, a_2) = (a_1, -a_2)$.

Example 4

Projection on the x-axis: $\mathsf{T}(a_1, a_2) = (a_1, 0)$.

Example 5

$\mathsf{T}(A) = A^t$, $\mathrm{M}_{m\times n}(\mathbb{F}) \to \mathrm{M}_{n\times m}(\mathbb{F})$.

Example 6

$\mathsf{T}(f) = f'$, $\mathsf{P}_n(\mathbb{R}) \to \mathsf{P}_{n-1}(\mathbb{R})$: $(ag+h)' = ag' + h'$.

Example 7

$\mathsf{T}(f) = \int_a^b f(t)\,dt$, $C(\mathbb{R}) \to \mathbb{R}$.

The **identity** $\mathsf{I}_V(x) = x$ and **zero transformation** $\mathsf{T}_0(x) = 0$ are linear.

Definition (Null space, range)

For linear $\mathsf{T} : V \to W$, the **null space** (kernel) is $\mathrm{N}(\mathsf{T}) = \{x \in V : \mathsf{T}(x) = 0\}$ and the **range** (image) is $\mathrm{R}(\mathsf{T}) = \{\mathsf{T}(x) : x \in V\}$.

Example 8

$\mathrm{N}(\mathsf{I}) = \{0\}$, $\mathrm{R}(\mathsf{I}) = V$; $\mathrm{N}(\mathsf{T}_0) = V$, $\mathrm{R}(\mathsf{T}_0) = \{0\}$.

Example 9

For $\mathsf{T}(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$ on $\mathbb{R}^3$: $\mathrm{N}(\mathsf{T}) = \{(a, a, 0) : a \in \mathbb{R}\}$ and $\mathrm{R}(\mathsf{T}) = \mathbb{R}^2$.

Theorem 2.1

For linear $\mathsf{T} : V \to W$, $\mathrm{N}(\mathsf{T})$ and $\mathrm{R}(\mathsf{T})$ are subspaces of V and W.

Proof. $\mathsf{T}(0) = 0$ puts 0 in both. For $x, y \in \mathrm{N}(\mathsf{T})$: $\mathsf{T}(x + y) = \mathsf{T}(cx) = 0$. In $\mathrm{R}(\mathsf{T})$: $\mathsf{T}(x) + \mathsf{T}(y) = \mathsf{T}(x + y)$, $c\mathsf{T}(x) = \mathsf{T}(cx)$. Apply Theorem 1.3. ■

Theorem 2.2

If $\beta = \{v_1, \ldots, v_n\}$ is a basis for V and $\mathsf{T} : V \to W$ is linear, then $\mathrm{R}(\mathsf{T}) = \mathrm{span}(\mathsf{T}(\beta)) = \mathrm{span}\{\mathsf{T}(v_1), \ldots, \mathsf{T}(v_n)\}$.

Proof. $\mathrm{span}(\mathsf{T}(\beta)) \subseteq \mathrm{R}(\mathsf{T})$ by Theorem 1.5; conversely $\mathsf{T}(\sum a_i v_i) = \sum a_i \mathsf{T}(v_i)$. ■

(Also for infinite bases.)

Example 10

$\mathsf{T} : \mathsf{P}_2(\mathbb{R}) \to \mathrm{M}_{2\times 2}(\mathbb{R})$, $\mathsf{T}(f) = \begin{pmatrix} f(1)-f(2) & 0 \\ 0 & f(0) \end{pmatrix}$: $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{\mathsf{T}(1), \mathsf{T}(x), \mathsf{T}(x^2)\} = \mathrm{span}\left\{\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} -1 & 0 \\ 0 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} -3 & 0 \\ 0 & 0 \end{smallmatrix}\right)\right\}$, of dimension 2.

Definition (Nullity, rank)

If $\mathrm{N}(\mathsf{T})$, $\mathrm{R}(\mathsf{T})$ are finite-dimensional, $\mathrm{nullity}(\mathsf{T}) = \dim \mathrm{N}(\mathsf{T})$ and $\mathrm{rank}(\mathsf{T}) = \dim \mathrm{R}(\mathsf{T})$.

Theorem 2.3 (Dimension theorem)

If $\mathsf{T} : V \to W$ is linear and V is finite-dimensional, then $\mathrm{nullity}(\mathsf{T}) + \mathrm{rank}(\mathsf{T}) = \dim(V)$.

Proof. Extend a basis $\{v_1, \ldots, v_k\}$ of $\mathrm{N}(\mathsf{T})$ to a basis $\{v_1, \ldots, v_n\}$ of V (Corollary); we show $S = \{\mathsf{T}(v_{k+1}), \ldots, \mathsf{T}(v_n)\}$ is a basis of $\mathrm{R}(\mathsf{T})$ with $n - k$ distinct elements.

Step 1 (spans). $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{\mathsf{T}(v_1), \ldots, \mathsf{T}(v_n)\} = \mathrm{span}(S)$ (Theorem 2.2; $\mathsf{T}(v_i) = 0$ for $i \le k$).

Step 2 (independent). $\sum_{i>k} b_i \mathsf{T}(v_i) = 0 \Rightarrow \sum_{i>k} b_i v_i \in \mathrm{N}(\mathsf{T}) \Rightarrow \sum_{i>k} b_i v_i = \sum_{i \le k} c_i v_i \Rightarrow$ all $b_i = 0$ (also giving distinctness). ■

Theorem 2.4

Linear T is one-to-one iff $\mathrm{N}(\mathsf{T}) = \{0\}$.

Proof. One-to-one: $\mathsf{T}(x) = 0 = \mathsf{T}(0) \Rightarrow x = 0$. Conversely $\mathsf{T}(x) = \mathsf{T}(y) \Rightarrow \mathsf{T}(x - y) = 0 \Rightarrow x = y$. ■

Theorem 2.5

If $\dim V = \dim W < \infty$ and $\mathsf{T} : V \to W$ is linear, the following are equivalent: (a) T is one-to-one;
(b) T is onto;
(c) $\mathrm{rank}(\mathsf{T}) = \dim(V)$.

Proof. By Theorems 2.3 and 2.4: T one-to-one $\iff$ $\mathrm{N}(\mathsf{T}) = \{0\}$ $\iff$ $\mathrm{nullity}(\mathsf{T}) = 0$ $\iff$ $\mathrm{rank}(\mathsf{T}) = \dim V = \dim W$ $\iff$ $\mathrm{R}(\mathsf{T}) = W$, the last step by Theorem 1.11. ■

Finite dimension and linearity are both essential.

Example 11

$\mathsf{T} : \mathsf{P}_2(\mathbb{R}) \to \mathsf{P}_3(\mathbb{R})$, $\mathsf{T}(f) = 2f' + \int_0^x 3f(t)\,dt$: $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{3x,\ 2 + \frac{3}{2}x^2,\ 4x + x^3\}$ has rank $3 < 4$: not onto; nullity $3 - 3 = 0$: one-to-one.

Example 12

$\mathsf{T}(a_1, a_2) = (a_1 + a_2, a_1)$ on $\mathbb{F}^2$: $\mathrm{N}(\mathsf{T}) = \{0\}$, so onto (Theorem 2.5).

A one-to-one linear T maps S to an independent set iff S is independent.

Example 13

With one-to-one $\mathsf{T}(a_0 + a_1 x + a_2 x^2) = (a_0, a_1, a_2)$, $\{2 - x + 3x^2,\ x + x^2,\ 1 - 2x^2\}$ is independent since $\{(2, -1, 3), (0, 1, 1), (1, 0, -2)\}$ is.

Theorem 2.6

Let $\{v_1, \ldots, v_n\}$ be a basis for V. For any $w_1, \ldots, w_n \in W$ there is exactly one linear $\mathsf{T} : V \to W$ with $\mathsf{T}(v_i) = w_i$.

Proof. For $x = \sum a_i v_i$ (unique coefficients, Theorem 1.8) set $\mathsf{T}(x) = \sum a_i w_i$.

Step 1 (linear). If $y = \sum b_i v_i$, then $cx + y = \sum (ca_i + b_i) v_i$, so $\mathsf{T}(cx + y) = \sum (ca_i + b_i) w_i = c\mathsf{T}(x) + \mathsf{T}(y)$.

Step 2 (values). $\mathsf{T}(v_i) = w_i$.

Step 3 (unique). A linear U with $\mathsf{U}(v_i) = w_i$ satisfies $\mathsf{U}(x) = \sum a_i \mathsf{U}(v_i) = \sum a_i w_i = \mathsf{T}(x)$. ■

Corollary

If $\{v_1, \ldots, v_n\}$ is a basis for V and linear $\mathsf{U}, \mathsf{T} : V \to W$ agree on each v_i, then $\mathsf{U} = \mathsf{T}$.

Proof. Uniqueness in Theorem 2.6. ■

Example 14

Linear U with $\mathsf{U}(1,2) = (3,3)$, $\mathsf{U}(1,1) = (1,3)$ equals $\mathsf{T}(a_1, a_2) = (2a_2 - a_1, 3a_1)$, which agrees on the basis $\{(1,2),(1,1)\}$.

2.2 The matrix representation of a linear transformation

Motivation. Ordered bases turn vectors into columns and linear maps into matrices.

Definition (Ordered basis; coordinate vector)

An **ordered basis** is a basis with a fixed order. For $\beta = \{u_1, \ldots, u_n\}$ and $x = \sum a_i u_i$, the **coordinate vector** is $[x]_\beta = (a_1, \ldots, a_n)^t \in \mathbb{F}^n$. The **standard ordered bases** are $\{e_1, \ldots, e_n\}$ of $\mathbb{F}^n$ and $\{1, x, \ldots, x^n\}$ of $\mathsf{P}_n(\mathbb{F})$.

Example 1

$\{e_1, e_2, e_3\} \neq \{e_2, e_1, e_3\}$ as ordered bases of $\mathbb{F}^3$.

Example 2

$f = 4 + 6x - 7x^2$, standard β of $\mathsf{P}_2(\mathbb{R})$: $[f]_\beta = (4, 6, -7)^t$.

Definition (Matrix representation)

Let $\beta = \{v_1, \ldots, v_n\}$, $\gamma = \{w_1, \ldots, w_m\}$ be ordered bases of V, W and $\mathsf{T} : V \to W$ linear, with $\mathsf{T}(v_j) = \sum_{i=1}^m a_{ij} w_i$. The $m \times n$ matrix $A = (a_{ij}) = [\mathsf{T}]_\beta^\gamma$ is the **matrix representation** of T; its j-th column is $[\mathsf{T}(v_j)]_\gamma$. If $V = W$, $\beta = \gamma$, write $[\mathsf{T}]_\beta$.

By the Corollary to Theorem 2.6, $[\mathsf{U}]_\beta^\gamma = [\mathsf{T}]_\beta^\gamma$ implies $\mathsf{U} = \mathsf{T}$.

Example 3

$\mathsf{T}(a_1, a_2) = (a_1 + 3a_2, 0, 2a_1 - 4a_2)$, standard bases: $\mathsf{T}(e_1) = (1,0,2)$, $\mathsf{T}(e_2) = (3,0,-4)$, $[\mathsf{T}]_\beta^\gamma = \begin{pmatrix} 1 & 3 \\ 0 & 0 \\ 2 & -4 \end{pmatrix}$. With $\gamma' = \{e_3, e_2, e_1\}$ the rows are reversed.

Example 4

$\mathsf{T}(f) = f'$, $\mathsf{P}_3(\mathbb{R}) \to \mathsf{P}_2(\mathbb{R})$, standard bases: $\mathsf{T}(1) = 0$, $\mathsf{T}(x) = 1$, $\mathsf{T}(x^2) = 2x$, $\mathsf{T}(x^3) = 3x^2$, $[\mathsf{T}]_\beta^\gamma = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$.

Definition (Sum and scalar multiple of maps)

$(\mathsf{T}+\mathsf{U})(x) = \mathsf{T}(x)+\mathsf{U}(x)$ and $(a\mathsf{T})(x) = a\mathsf{T}(x)$.

Theorem 2.7

Let $\mathsf{T}, \mathsf{U} : V \to W$ be linear.
(a) $a\mathsf{T}+\mathsf{U}$ is linear for all $a \in \mathbb{F}$.
(b) With these operations, the linear maps $V \to W$ form a vector space over $\mathbb{F}$.

Proof.

(a) $(a\mathsf{T}+\mathsf{U})(cx+y) = a\mathsf{T}(cx+y)+\mathsf{U}(cx+y) = c(a\mathsf{T}+\mathsf{U})(x)+(a\mathsf{T}+\mathsf{U})(y)$.

(b) By (a) the linear maps form a subspace of $\mathcal{F}(V,W)$ (it contains T_0). ■

$\mathcal{L}(V,W)$ denotes this space, and $\mathcal{L}(V) = \mathcal{L}(V,V)$.

Theorem 2.8

For finite-dimensional V, W with ordered bases β, γ and linear $\mathsf{T}, \mathsf{U} : V \to W$:
(a) $[\mathsf{T}+\mathsf{U}]_\beta^\gamma = [\mathsf{T}]_\beta^\gamma + [\mathsf{U}]_\beta^\gamma$;
(b) $[a\mathsf{T}]_\beta^\gamma = a[\mathsf{T}]_\beta^\gamma$.

Proof. If $\mathsf{T}(v_j) = \sum_i a_{ij}w_i$ and $\mathsf{U}(v_j) = \sum_i b_{ij}w_i$, then $(\mathsf{T}+\mathsf{U})(v_j) = \sum_i (a_{ij}+b_{ij})w_i$ and $(a\mathsf{T})(v_j) = \sum_i aa_{ij}w_i$. ■

Example 5

$\mathsf{T}(a_1,a_2) = (a_1+3a_2, 0, 2a_1-4a_2)$, $\mathsf{U}(a_1,a_2) = (a_1-a_2, 2a_1, 3a_1+2a_2)$: $(\mathsf{T}+\mathsf{U})(a_1,a_2) = (2a_1+2a_2, 2a_1, 5a_1-2a_2)$ and $\begin{pmatrix} 2 & 2 \\ 2 & 0 \\ 5 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & 0 \\ 2 & -4 \end{pmatrix} + \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 3 & 2 \end{pmatrix}$.

2.3 Composition of linear transformations and matrix multiplication

Motivation. Matrix multiplication is defined so that it matches composition of maps.

Theorem 2.9

If $\mathsf{T} : V \to W$ and $\mathsf{U} : W \to Z$ are linear, so is $\mathsf{UT} : V \to Z$.

Proof. $\mathsf{UT}(ax+y) = \mathsf{U}(a\mathsf{T}(x)+\mathsf{T}(y)) = a\mathsf{UT}(x)+\mathsf{UT}(y)$. ■

Theorem 2.10

For $\mathsf{T}, \mathsf{U}_1, \mathsf{U}_2 \in \mathcal{L}(V)$:
(a) $\mathsf{T}(\mathsf{U}_1+\mathsf{U}_2) = \mathsf{TU}_1+\mathsf{TU}_2$ and $(\mathsf{U}_1+\mathsf{U}_2)\mathsf{T} = \mathsf{U}_1\mathsf{T}+\mathsf{U}_2\mathsf{T}$;
(b) $\mathsf{T}(\mathsf{U}_1\mathsf{U}_2) = (\mathsf{TU}_1)\mathsf{U}_2$;
(c) $\mathsf{TI} = \mathsf{IT} = \mathsf{T}$;
(d) $a(\mathsf{U}_1\mathsf{U}_2) = (a\mathsf{U}_1)\mathsf{U}_2 = \mathsf{U}_1(a\mathsf{U}_2)$.

Proof [Supplement]. Evaluate at x. (a) $\mathsf{T}(\mathsf{U}_1x+\mathsf{U}_2x) = \mathsf{TU}_1x+\mathsf{TU}_2x$ (linearity); $(\mathsf{U}_1+\mathsf{U}_2)(\mathsf{T}x) = \mathsf{U}_1\mathsf{T}x+\mathsf{U}_2\mathsf{T}x$.

(b) Composition is associative.

(c) Clear.

(d) $a\mathsf{U}_1(\mathsf{U}_2x) = (a\mathsf{U}_1)(\mathsf{U}_2x) = \mathsf{U}_1(a\mathsf{U}_2x)$ by linearity of U_1. ■

Let $\mathsf{T}: V \to W$, $\mathsf{U}: W \to Z$ with ordered bases $\alpha = \{v_j\}$, $\beta = \{w_k\}$, $\gamma = \{z_i\}$, $A = [\mathsf{U}]_\beta^\gamma$, $B = [\mathsf{T}]_\alpha^\beta$. Then

$$\mathsf{UT}(v_j) = \mathsf{U}\Big(\sum_k B_{kj} w_k\Big) = \sum_k B_{kj} \sum_i A_{ik} z_i = \sum_i \Big(\sum_k A_{ik} B_{kj}\Big) z_i.$$

Definition (Matrix product)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ and $B \in \mathrm{M}_{n\times p}(\mathbb{F})$, $AB \in \mathrm{M}_{m\times p}(\mathbb{F})$ has $(AB)_{ij} = \sum_{k=1}^n A_{ik}B_{kj}$.

Example 1

$\begin{pmatrix}1&2&1\\0&4&-1\end{pmatrix}\begin{pmatrix}4\\2\\5\end{pmatrix} = \begin{pmatrix}13\\3\end{pmatrix}$. Not commutative: $\begin{pmatrix}1&1\\0&0\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix} = \begin{pmatrix}1&1\\0&0\end{pmatrix}$ but $\begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}1&1\\0&0\end{pmatrix} = \begin{pmatrix}0&0\\1&1\end{pmatrix}$. Also $(AB)^t = B^tA^t$, since both have (i,j) entry $\sum_k A_{jk}B_{ki}$.

Theorem 2.11

With α, β, γ ordered bases of finite-dimensional V, W, Z and linear $\mathsf{T}: V \to W$, $\mathsf{U}: W \to Z$: $[\mathsf{UT}]_\alpha^\gamma = [\mathsf{U}]_\beta^\gamma [\mathsf{T}]_\alpha^\beta$.

Proof. By the computation above, $([\mathsf{UT}]_\alpha^\gamma)_{ij} = \sum_k A_{ik}B_{kj}$. ■

Corollary

For $\mathsf{T}, \mathsf{U} \in \mathcal{L}(V)$ and an ordered basis β, $[\mathsf{UT}]_\beta = [\mathsf{U}]_\beta[\mathsf{T}]_\beta$.

Proof. Theorem 2.11 with $\alpha = \beta = \gamma$. ■

Example 2

$\mathsf{U}(f) = f'$ ($\mathsf{P}_3 \to \mathsf{P}_2$) and $\mathsf{T}(f) = \int_0^x f(t)\,dt$ ($\mathsf{P}_2 \to \mathsf{P}_3$) give $\mathsf{UT} = \mathsf{I}$:

$$[\mathsf{U}][\mathsf{T}] = \begin{pmatrix}0&1&0&0\\0&0&2&0\\0&0&0&3\end{pmatrix}\begin{pmatrix}0&0&0\\1&0&0\\0&\frac12&0\\0&0&\frac13\end{pmatrix} = I_3.$$

Definition (Kronecker delta; identity matrix)

$\delta_{ij} = 1$ if $i = j$ and 0 otherwise; the $n \times n$ **identity matrix** is $(I_n)_{ij} = \delta_{ij}$.

Theorem 2.12

Let A be $m \times n$, B, C be $n \times p$, D, E be $q \times m$. Then
(a) $A(B + C) = AB + AC$ and $(D + E)A = DA + EA$;
(b) $a(AB) = (aA)B = A(aB)$;
(c) $I_mA = A = AI_n$;
(d) $[\mathsf{I}_V]_\beta = I_n$ for an n-dimensional V with ordered basis β.

Proof.

(a) $(A(B+C))_{ij} = \sum_k A_{ik}(B_{kj}+C_{kj}) = (AB)_{ij} + (AC)_{ij}$; similarly for $(D+E)A$.

(b) [Supplement] All three have entries $a\sum_k A_{ik}B_{kj}$.

(c) $(I_m A)_{ij} = \sum_k \delta_{ik}A_{kj} = A_{ij}$, and similarly for AI_n.

(d) [Supplement] $\mathsf{I}_V(v_j) = v_j$, whose coordinate vector is e_j. ■

Corollary

$A(\sum_{i=1}^k a_iB_i) = \sum_i a_iAB_i$ and $(\sum_i a_iC_i)A = \sum_i a_iC_iA$.

Proof. Induction on k using Theorem 2.12(a), (b). ■

Powers: $A^0 = I$, $A^k = A^{k-1}A$. No cancellation: $A = \left(\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right) \neq O$ has $A^2 = O$.

Theorem 2.13

Let A be $m\times n$, B be $n\times p$, and u_j, v_j the j-th columns of AB and B. Then (a) $u_j = Av_j$; (b) $v_j = Be_j$.

Proof.

(a) $(u_j)_i = (AB)_{ij} = \sum_k A_{ik}B_{kj} = (Av_j)_i$.

(b) (a) for $BI_p = B$: column j is Be_j. ■

Likewise row i of AB is (row i of A)$\cdot B$.

Theorem 2.14

For linear $\mathsf{T}: V\to W$ with ordered bases β,γ and $u\in V$: $[\mathsf{T}(u)]_\gamma = [\mathsf{T}]_\beta^\gamma[u]_\beta$.

Proof. Let $f(a) = au$ $(\mathbb{F}\to V)$, $g(a) = a\mathsf{T}(u)$ $(\mathbb{F}\to W)$, $\alpha = \{1\}$. Then $g = \mathsf{T}f$, $[f]_\alpha^\beta = [u]_\beta$, $[g]_\alpha^\gamma = [\mathsf{T}(u)]_\gamma$, and by Theorem 2.11 $[\mathsf{T}(u)]_\gamma = [\mathsf{T}f]_\alpha^\gamma = [\mathsf{T}]_\beta^\gamma[f]_\alpha^\beta = [\mathsf{T}]_\beta^\gamma[u]_\beta$. ■

Example 3

$\mathsf{T}(f) = f'$, $p = 2-4x+x^2+3x^3$: $[\mathsf{T}]_\beta^\gamma(2,-4,1,3)^t = (-4,2,9)^t = [p']_\gamma$.

Definition (Left-multiplication transformation)

For $A\in \mathrm{M}_{m\times n}(\mathbb{F})$, $\mathsf{L}_A:\mathbb{F}^n\to\mathbb{F}^m$ is $\mathsf{L}_A(x) = Ax$.

Example 4

For $A = \left(\begin{smallmatrix}1&2&1\\0&1&2\end{smallmatrix}\right)$, $\mathsf{L}_A(1,3,-1)^t = (6,1)^t$.

Theorem 2.15

L_A is linear. For $B\in\mathrm{M}_{m\times n}(\mathbb{F})$ and standard ordered bases β,γ of $\mathbb{F}^n,\mathbb{F}^m$:
(a) $[\mathsf{L}_A]_\beta^\gamma = A$;
(b) $\mathsf{L}_A = \mathsf{L}_B$ iff $A = B$;

(c) $\mathsf{L}_{A+B} = \mathsf{L}_A + \mathsf{L}_B$ and $\mathsf{L}_{aA} = a\mathsf{L}_A$;
(d) if $\mathsf{T} : \mathbb{F}^n \to \mathbb{F}^m$ is linear, there is a unique C with $\mathsf{T} = \mathsf{L}_C$, namely $C = [\mathsf{T}]_\beta^\gamma$;
(e) if E is $n \times p$, then $\mathsf{L}_{AE} = \mathsf{L}_A\mathsf{L}_E$;
(f) if $m = n$, then $\mathsf{L}_{I_n} = \mathsf{I}_{\mathbb{F}^n}$.

Proof. Linearity follows from Theorem 2.12.

(a) Column j of $[\mathsf{L}_A]_\beta^\gamma$ is Ae_j, column j of A (Theorem 2.13).

(b) If $\mathsf{L}_A = \mathsf{L}_B$, then $A = [\mathsf{L}_A]_\beta^\gamma = [\mathsf{L}_B]_\beta^\gamma = B$ by (a).

(c) [Supplement] $(A+B)x = Ax + Bx$ and $(aA)x = a(Ax)$.

(d) Put $C = [\mathsf{T}]_\beta^\gamma$. By Theorem 2.14, $\mathsf{T}(x) = [\mathsf{T}(x)]_\gamma = C[x]_\beta = Cx = \mathsf{L}_C(x)$. Uniqueness is (b).

(e) $(AE)e_j = A(Ee_j)$ (Theorem 2.13): L_{AE}, $\mathsf{L}_A\mathsf{L}_E$ agree on a basis (Corollary to Theorem 2.6).

(f) [Supplement] $I_n x = x$. ■

Theorem 2.16

If $A(BC)$ is defined, so is $(AB)C$, and $A(BC) = (AB)C$.

Proof. By Theorem 2.15(e) and associativity of composition, $\mathsf{L}_{A(BC)} = \mathsf{L}_A\mathsf{L}_{BC} = \mathsf{L}_A(\mathsf{L}_B\mathsf{L}_C) = (\mathsf{L}_A\mathsf{L}_B)\mathsf{L}_C = \mathsf{L}_{(AB)C}$; apply (b). ■

Incidence matrices. For a relation on $\{1, \ldots, n\}$, $A_{ij} = 1$ if $i \to j$, else 0; $(A^k)_{ij}$ counts k-step paths $i \to \cdots \to j$. For a **dominance relation** (exactly one of $i \to j$, $j \to i$ for $i \neq j$; $A_{ii} = 0$), row sums of $A + A^2$ rank the members.

2.4 Invertibility and isomorphisms

Motivation. Finite-dimensional spaces over $\mathbb{F}$ are classified up to isomorphism by dimension.

Definition (Inverse)

$\mathsf{U} : W \to V$ is an **inverse** of linear $\mathsf{T} : V \to W$ if $\mathsf{TU} = \mathsf{I}_W$ and $\mathsf{UT} = \mathsf{I}_V$; T is then **invertible**, the inverse is unique and denoted T^{-1}.

$(\mathsf{TU})^{-1} = \mathsf{U}^{-1}\mathsf{T}^{-1}$, $(\mathsf{T}^{-1})^{-1} = \mathsf{T}$; invertible $\iff$ one-to-one and onto, so for $\dim V = \dim W < \infty$: invertible $\iff$ $\operatorname{rank}(\mathsf{T}) = \dim V$ (Theorem 2.5).

Example 1

$\mathsf{T}(a + bx) = (a, a + b)$ from $\mathsf{P}_1(\mathbb{R})$ to $\mathbb{R}^2$ has $\mathsf{T}^{-1}(c, d) = c + (d - c)x$.

Theorem 2.17

If $\mathsf{T} : V \to W$ is linear and invertible, then T^{-1} is linear.

Proof. With $x_i = \mathsf{T}^{-1}(y_i)$, $\mathsf{T}^{-1}(cy_1 + y_2) = \mathsf{T}^{-1}(c\mathsf{T}(x_1) + \mathsf{T}(x_2)) = \mathsf{T}^{-1}(\mathsf{T}(cx_1 + x_2)) = cx_1 + x_2 = c\mathsf{T}^{-1}(y_1) + \mathsf{T}^{-1}(y_2)$. ■

Definition (Invertible matrix)

$A \in \mathrm{M}_{n\times n}(\mathbb{F})$ is **invertible** if $AB = BA = I$ for some B. Then B is unique ($C = CI = C(AB) = (CA)B = B$), the **inverse** A^{-1}.

Example 2

$\left(\begin{smallmatrix} 5 & 7 \\ 2 & 3 \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} 3 & -7 \\ -2 & 5 \end{smallmatrix}\right)$.

Lemma

Let $\mathsf{T} : V \to W$ be an invertible linear map. Then V is finite-dimensional iff W is, and then $\dim V = \dim W$.

Proof. A finite basis β of V gives $W = \mathrm{R}(\mathsf{T}) = \operatorname{span}\mathsf{T}(\beta)$ (Theorem 2.2); conversely use T^{-1}. Then $\operatorname{nullity}\mathsf{T} = 0$, $\operatorname{rank}\mathsf{T} = \dim W$, so $\dim V = \dim W$ (Theorem 2.3). ■

Theorem 2.18

Let V, W be finite-dimensional with ordered bases β, γ and $\mathsf{T} : V \to W$ linear. Then T is invertible iff $[\mathsf{T}]_\beta^\gamma$ is invertible, and then $[\mathsf{T}^{-1}]_\gamma^\beta = ([\mathsf{T}]_\beta^\gamma)^{-1}$.

Proof.

Step 1 (only if). $\dim V = \dim W = n$ (Lemma). By Theorems 2.11, 2.12(d), $I_n = [\mathsf{I}_V]_\beta = [\mathsf{T}^{-1}\mathsf{T}]_\beta = [\mathsf{T}^{-1}]_\gamma^\beta[\mathsf{T}]_\beta^\gamma$, and likewise $[\mathsf{T}]_\beta^\gamma[\mathsf{T}^{-1}]_\gamma^\beta = I_n$.

Step 2 (if). $A = [\mathsf{T}]_\beta^\gamma$ has inverse B, so A is square: $\dim V = \dim W = n$. Theorem 2.6 gives linear U with $\mathsf{U}(w_j) = \sum_i B_{ij}v_i$, i.e. $[\mathsf{U}]_\gamma^\beta = B$. Then $[\mathsf{UT}]_\beta = BA = I_n = [\mathsf{I}_V]_\beta$, so $\mathsf{UT} = \mathsf{I}_V$; likewise $\mathsf{TU} = \mathsf{I}_W$. ■

Example 3

Example 1, standard bases: $[\mathsf{T}] = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, $[\mathsf{T}^{-1}] = \left(\begin{smallmatrix} 1 & 0 \\ -1 & 1 \end{smallmatrix}\right)$, mutually inverse.

Corollary 1

For $\mathsf{T} \in \mathcal{L}(V)$, V finite-dimensional with ordered basis β: T is invertible iff $[\mathsf{T}]_\beta$ is, and $[\mathsf{T}^{-1}]_\beta = ([\mathsf{T}]_\beta)^{-1}$.

Proof. Theorem 2.18 with $W = V$, $\gamma = \beta$. ■

Corollary 2

$A \in \mathrm{M}_{n\times n}(\mathbb{F})$ is invertible iff L_A is, and $(\mathsf{L}_A)^{-1} = \mathsf{L}_{A^{-1}}$.

Proof. Theorem 2.18 with standard bases and Theorem 2.15(a), (d): $[(\mathsf{L}_A)^{-1}] = A^{-1}$, so $(\mathsf{L}_A)^{-1} = \mathsf{L}_{A^{-1}}$. ■

Definition (Isomorphism)

V is **isomorphic** to W ($V \cong W$) if there is an invertible linear $\mathsf{T} : V \to W$, an **isomorphism**. "Is isomorphic to" is an equivalence relation.

Example 4

$\mathsf{T}(a_1, a_2) = a_1 + a_2 x$ is an isomorphism $\mathbb{F}^2 \to \mathsf{P}_1(\mathbb{F})$.

Example 5

$\mathsf{T}(f) = \begin{pmatrix} f(1) & f(2) \\ f(3) & f(4) \end{pmatrix}$ from $\mathsf{P}_3(\mathbb{R})$ to $\mathrm{M}_{2\times 2}(\mathbb{R})$ is linear, $\mathsf{T}(f) = O \Rightarrow f = 0$ (four zeros; Lagrange), and both dimensions are 4: an isomorphism (Theorem 2.5).

Theorem 2.19

Finite-dimensional V, W over $\mathbb{F}$ are isomorphic iff $\dim V = \dim W$.

Proof. "Only if": the Lemma. "If": with bases $\{v_i\}$, $\{w_i\}$, take linear $\mathsf{T}(v_i) = w_i$ (Theorem 2.6); $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{w_i\} = W$, so T is onto, hence one-to-one (Theorem 2.5). ■

Corollary

$V \cong \mathbb{F}^n$ iff $\dim V = n$.

Proof. $\dim \mathbb{F}^n = n$; apply Theorem 2.19 (if $V \cong \mathbb{F}^n$, V is finite-dimensional by the Lemma). ■

Theorem 2.20

Let $\dim V = n$, $\dim W = m$ with ordered bases β, γ. Then $\Phi : \mathcal{L}(V, W) \to \mathrm{M}_{m\times n}(\mathbb{F})$, $\Phi(\mathsf{T}) = [\mathsf{T}]_\beta^\gamma$, is an isomorphism.

Proof. Φ is linear (Theorem 2.8); for each A there is exactly one linear T with $\mathsf{T}(v_j) = \sum_i A_{ij} w_i$, i.e. $\Phi(\mathsf{T}) = A$ (Theorem 2.6). ■

Corollary

$\dim \mathcal{L}(V, W) = mn$.

Proof. Theorems 2.20 and 2.19. ■

Definition (Standard representation)

For an ordered basis β of n-dimensional V, $\phi_\beta : V \to \mathbb{F}^n$, $\phi_\beta(x) = [x]_\beta$, is the **standard representation** of V with respect to β.

Example 6

$\beta = \{(1,0),(0,1)\}$, $\gamma = \{(1,2),(3,4)\}$, $x = (1,-2) = -5(1,2) + 2(3,4)$: $\phi_\beta(x) = (1,-2)^t$, $\phi_\gamma(x) = (-5,2)^t$.

Theorem 2.21

ϕ_β is an isomorphism.

Proof [Supplement]. $[cx+y]_\beta = c[x]_\beta + [y]_\beta$ (uniqueness, Theorem 1.8), $\phi_\beta(x) = 0 \Rightarrow x = 0$, and $\dim V = \dim \mathbb{F}^n$; apply Theorem 2.5. ■

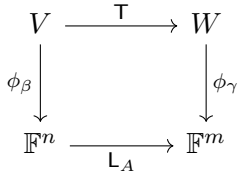


Figure. *With $A = [\mathsf{T}]^\gamma_\beta$, $\mathsf{L}_A\phi_\beta = \phi_\gamma\mathsf{T}$ (Theorem 2.14).*

Example 7

$\mathsf{T}(p) = p'$, $p = 2 + x - 3x^2 + 5x^3$: $\mathsf{L}_A\phi_\beta(p) = A(2,1,-3,5)^t = (1,-6,15)^t = \phi_\gamma(p')$.

2.5 The change of coordinate matrix

Motivation. A change of basis multiplies coordinates by Q and conjugates operator matrices by Q.

Theorem 2.22

Let β, β' be ordered bases of finite-dimensional V and $Q = [\mathsf{I}_V]^\beta_{\beta'}$. Then (a) Q is invertible; (b) $[v]_\beta = Q[v]_{\beta'}$ for all $v \in V$.

Proof.

(a) I_V is invertible, so Q is by Theorem 2.18.

(b) $[v]_\beta = [\mathsf{I}_V(v)]_\beta = [\mathsf{I}_V]^\beta_{\beta'}[v]_{\beta'}$ by Theorem 2.14. ■

Q is the **change of coordinate matrix** ($\beta' \to \beta$); column j is $[x'_j]_\beta$; Q^{-1} goes $\beta \to \beta'$.

Example 1

$\beta = \{(1,1),(1,-1)\}$, $\beta' = \{(2,4),(3,1)\}$: $(2,4) = 3(1,1) - (1,-1)$, $(3,1) = 2(1,1) + (1,-1)$, $Q = \left(\begin{smallmatrix} 3 & 2 \\ -1 & 1 \end{smallmatrix}\right)$.

A linear map $\mathsf{T} : V \to V$ is a **linear operator**.

Theorem 2.23

Let T be a linear operator on finite-dimensional V with ordered bases β, β', and $Q = [\mathsf{I}_V]^\beta_{\beta'}$. Then $[\mathsf{T}]_{\beta'} = Q^{-1}[\mathsf{T}]_\beta Q$.

Proof. $\mathsf{T} = \mathsf{IT} = \mathsf{TI}$, so by Theorem 2.11, $Q[\mathsf{T}]_{\beta'} = [\mathsf{I}]^\beta_{\beta'}[\mathsf{T}]^{\beta'}_{\beta'} = [\mathsf{T}]^\beta_{\beta'} = [\mathsf{T}]^\beta_\beta[\mathsf{I}]^\beta_{\beta'} = [\mathsf{T}]_\beta Q$. Multiply by Q^{-1}. ■

Example 2

$\mathsf{T}(a,b) = (3a-b, a+3b)$, β, β' of Example 1: $[\mathsf{T}]_\beta = \left(\begin{smallmatrix} 3 & 1 \\ -1 & 3 \end{smallmatrix}\right)$, $Q^{-1} = \frac{1}{5}\left(\begin{smallmatrix} 1 & -2 \\ 1 & 3 \end{smallmatrix}\right)$, and

$$[\mathsf{T}]_{\beta'} = Q^{-1}[\mathsf{T}]_\beta Q = \begin{pmatrix} 4 & 1 \\ -2 & 2 \end{pmatrix}.$$

Check: $\mathsf{T}(2,4) = (2,14) = 4(2,4) - 2(3,1)$.

Example 3

Reflection about $y = 2x$: $\beta' = \{(1,2), (-2,1)\}$, $\mathsf{T}(1,2) = (1,2)$, $\mathsf{T}(-2,1) = -(-2,1)$, $[\mathsf{T}]_{\beta'} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. With β standard, $Q = \left(\begin{smallmatrix} 1 & -2 \\ 2 & 1 \end{smallmatrix}\right)$ and

$$[\mathsf{T}]_\beta = Q[\mathsf{T}]_{\beta'}Q^{-1} = \frac{1}{5}\begin{pmatrix} -3 & 4 \\ 4 & 3 \end{pmatrix}, \qquad \mathsf{T}(a,b) = \tfrac{1}{5}(-3a+4b,\ 4a+3b).$$

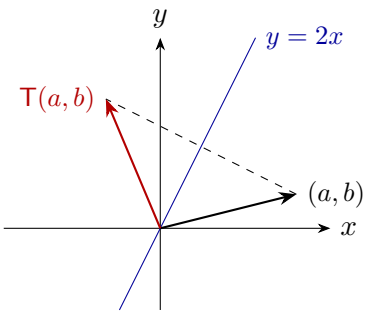


Figure. *Example 3: reflection about $y = 2x$.*

Corollary

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$ and an ordered basis γ of $\mathbb{F}^n$, $[\mathsf{L}_A]_\gamma = Q^{-1}AQ$, where the j-th column of Q is the j-th vector of γ.

Proof. Theorem 2.23 with β standard: $[\mathsf{L}_A]_\beta = A$ and $Q = [\mathsf{I}]^\beta_\gamma$ has columns the vectors of γ. ■

Example 4

For $\gamma = \{(-1,0,0), (2,1,0), (1,1,1)\}$,

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 3 \\ 0 & -1 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad [\mathsf{L}_A]_\gamma = Q^{-1}AQ = \begin{pmatrix} 0 & 2 & 8 \\ -1 & 4 & 6 \\ 0 & -1 & -1 \end{pmatrix}.$$

Definition (Similar matrices)

B is **similar** to A if $B = Q^{-1}AQ$ for some invertible Q (an equivalence relation). By Theorem 2.23, matrices of one operator are similar.

2.6 Dual spaces *

Motivation. Linear functionals form V^*, whose dual maps have transposed matrices.

A **linear functional** on V is a linear map $V \to \mathbb{F}$.

Example 1

$h(x) = \frac{1}{2\pi}\int_0^{2\pi} x(t)g(t)\,dt$ (g continuous); $g = \sin nt, \cos nt$ give Fourier coefficients.

Example 2

$\mathrm{tr} : \mathrm{M}_{n\times n}(\mathbb{F}) \to \mathbb{F}$ is a linear functional.

Example 3

The **i-th coordinate function** $f_i(x) = a_i$, $[x]_\beta = (a_1, \ldots, a_n)^t$, has $f_i(x_j) = \delta_{ij}$.

Definition (Dual space)

$V^* = \mathcal{L}(V, \mathbb{F})$. If $\dim V < \infty$, $\dim V^* = \dim V \cdot \dim \mathbb{F} = \dim V$ (Corollary to Theorem 2.20), so $V \cong V^*$. The **double dual** is $V^{**} = (V^*)^*$.

Theorem 2.24

For an ordered basis $\beta = \{x_1, \ldots, x_n\}$ with coordinate functions f_i, $\beta^* = \{f_1, \ldots, f_n\}$ is an ordered basis of V^*, and $f = \sum_{i=1}^n f(x_i)f_i$ for every $f \in V^*$.

Proof. As $\dim V^* = n$, generating suffices (Corollary 2 of Theorem 1.10). $g = \sum f(x_i)f_i$ has $g(x_j) = f(x_j)$, so $g = f$ (Corollary to Theorem 2.6). ■

β^* ($f_i(x_j) = \delta_{ij}$) is the **dual basis** of β.

Example 4

$\beta = \{(2,1),(3,1)\}$: $f_i(x_j) = \delta_{ij}$ gives $f_1(x,y) = -x + 3y$ and $f_2(x,y) = x - 2y$.

Theorem 2.25

Let V, W be finite-dimensional with ordered bases β, γ. For linear $\mathsf{T} : V \to W$, the map $\mathsf{T}^t : W^* \to V^*$, $\mathsf{T}^t(g) = g\mathsf{T}$, is linear and $[\mathsf{T}^t]_{\gamma^*}^{\beta^*} = ([\mathsf{T}]_\beta^\gamma)^t$.

Proof. $g\mathsf{T}$ is linear (Theorem 2.9), and $\mathsf{T}^t(ag + h) = (ag + h)\mathsf{T} = a\,g\mathsf{T} + h\mathsf{T}$. Let $\beta = \{x_s\}$, $\gamma = \{y_k\}$, $\beta^* = \{f_s\}$, $\gamma^* = \{g_j\}$, $A = [\mathsf{T}]_\beta^\gamma$. By Theorem 2.24, $\mathsf{T}^t(g_j) = g_j\mathsf{T} = \sum_s (g_j\mathsf{T})(x_s)f_s$, so the (i,j) entry of $[\mathsf{T}^t]_{\gamma^*}^{\beta^*}$ is

$$(g_j\mathsf{T})(x_i) = g_j\Big(\sum_k A_{ki}y_k\Big) = A_{ji}.$$

■

Example 5

$\mathsf{T}(p) = (p(0), p(2))$, $\mathsf{P}_1(\mathbb{R}) \to \mathbb{R}^2$: $[\mathsf{T}]_\beta^\gamma = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 2 \end{smallmatrix}\right)$; $\mathsf{T}^t(g_1)\colon p \mapsto p(0)$ is f_1, $\mathsf{T}^t(g_2)\colon a+bx \mapsto a+2b$ is $f_1 + 2f_2$, so $[\mathsf{T}^t] = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 2 \end{smallmatrix}\right) = ([\mathsf{T}]_\beta^\gamma)^t$.

For $x \in V$ define $\hat{x} \in V^{**}$ by $\hat{x}(f) = f(x)$; $\hat{x}$ is linear in f.

Lemma

If $\dim V < \infty$, $x \in V$ and $\hat{x}(f) = 0$ for all $f \in V^*$, then $x = 0$.

Proof. If $x \neq 0$, extend to a basis $\{x_1 = x, \ldots, x_n\}$; its dual f_1 has $f_1(x) = 1$. ■

Theorem 2.26

For finite-dimensional V, $\psi : V \to V^{**}$, $\psi(x) = \hat{x}$, is an isomorphism.

Proof. Linear: $\widehat{cx+y}(f) = c\hat{x}(f) + \hat{y}(f)$; one-to-one (Lemma); $\dim V^{**} = \dim V$; apply Theorem 2.5. ■

(ψ is basis-free: a *natural* isomorphism.)

Corollary

If V is finite-dimensional, every ordered basis of V^* is the dual basis of some basis of V.

Proof. Write the dual of a basis $\{f_j\}$ of V^* as $\{\hat{x}_i\}$ (Theorems 2.24, 2.26); $\{x_i\}$ is a basis (ψ isomorphism) and $\delta_{ij} = \hat{x}_i(f_j) = f_j(x_i)$. ■

2.7 Homogeneous linear differential equations with constant coefficients *

Motivation. Solutions of an nth-order equation form $\mathrm{N}(p(D))$, of dimension n, with a basis of exponentials.

A spring (constant k, mass m) obeys $y'' + \frac{k}{m}y = 0$.

Example 1

$y = \sin\sqrt{k/m}\,t$ is a solution; $y = t$ is not.

A **homogeneous linear differential equation with constant coefficients** is

$$a_n y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0, \qquad a_i \in \mathbb{C},\ a_n \neq 0. \tag{2.1}$$

We take $a_n = 1$ and seek solutions in $\mathcal{F}(\mathbb{R}, \mathbb{C})$.

Definition (Derivative of complex functions)

$x = x_1 + ix_2 \in \mathcal{F}(\mathbb{R}, \mathbb{C})$ is **differentiable** if x_1, x_2 are, with $x' = x_1' + ix_2'$.

Example 2

$x(t) = \cos 2t + i \sin 2t$ has $x'(t) = -2\sin 2t + 2i\cos 2t$; and $x^2 = \cos 4t + i \sin 4t$.

Theorem 2.27

Every solution of (2.1) has derivatives of all orders.

Proof. $x^{(n)} = -\sum_{i<n} a_i x^{(i)}$ is differentiable; inductively, if $x^{(k)}$ exists for $k \le m$ $(m \ge n)$, then $x^{(m)} = -\sum_{i<n} a_i x^{(i+m-n)}$ is differentiable. ■

Example 3

For $y'' + 4y = 0$, $y'' = -4y$, so $y''' = -4y'$, and so on.

Definition (C^∞; differential operators)

$C^\infty \subseteq \mathcal{F}(\mathbb{R}, \mathbb{C})$: functions with derivatives of all orders; $D(x) = x'$ is linear on it. For complex $p(t) = \sum a_i t^i$, $\deg p > 0$, $p(D) = \sum a_i D^i$ is a **differential operator** of order $\deg p$. The **auxiliary polynomial** of (2.1) is $p(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_0$, and (2.1) reads $p(D)(y) = 0$.

Theorem 2.28

The solutions of (2.1) are exactly $\mathrm{N}(p(D))$.

Proof. By Theorem 2.27 all solutions lie in C^∞, where the equation is $p(D)(y) = 0$. ■

Corollary

The solution set of (2.1) is a subspace of C^∞, the **solution space**.

Proof. Theorems 2.28 and 2.1. ■

Definition (Complex exponential)

For $c = a + ib$, $e^c = e^a(\cos b + i \sin b)$ (**Euler's formula**: $e^{ib} = \cos b + i \sin b$). Then $e^{c+d} = e^c e^d$ and $e^{-c} = 1/e^c$. $f(t) = e^{ct}$ is an **exponential function**.

Theorem 2.29

$\frac{d}{dt} e^{ct} = ce^{ct}$.

Proof [Supplement]. With $c = a + ib$, $e^{ct} = e^{at}\cos bt + ie^{at}\sin bt$, whose derivative is $e^{at}(a\cos bt - b\sin bt) + ie^{at}(a\sin bt + b\cos bt) = (a + ib)e^{at}(\cos bt + i\sin bt)$. ■

Theorem 2.30

The solution space of $y' + a_0 y = 0$ has dimension 1 and basis $\{e^{-a_0 t}\}$.

Proof. $e^{-a_0 t}$ solves it (Theorem 2.29). For a solution x, $(e^{a_0 t}x)' = e^{a_0 t}(a_0 x + x') = 0$, so $e^{a_0 t}x$ is constant (real and imaginary parts): $x = ke^{-a_0 t}$. ■

Corollary

For $c \in \mathbb{C}$, $\mathrm{N}(D - c\mathsf{I})$ has basis $\{e^{ct}\}$.

Proof. Theorem 2.30 with $a_0 = -c$. ■

Theorem 2.31

If c is a zero of the auxiliary polynomial $p(t)$, then e^{ct} is a solution.

Proof. $p(t) = q(t)(t-c)$ gives $p(D) = q(D)(D-c\mathsf{I})$ and $p(D)(e^{ct}) = q(D)((D-c\mathsf{I})e^{ct}) = q(D)(0) = 0$ by the Corollary. ■

Example 4

$y'' - 3y' + 2y = 0$: $p(t) = (t-1)(t-2)$, so e^t and e^{2t} are solutions.

Theorem 2.32

For any differential operator $p(D)$ of order n, $\mathrm{N}(p(D))$ is an n-dimensional subspace of C^∞.

Lemma 1

$D - c\mathsf{I} : C^\infty \to C^\infty$ is onto for every $c \in \mathbb{C}$.

Proof. For $v \in C^\infty$, antidifferentiate real and imaginary parts of ve^{-ct} to get $W' = ve^{-ct}$; with $u = We^{ct}$, $(D - c\mathsf{I})u = W'e^{ct} + cWe^{ct} - cWe^{ct} = v$. ■

Lemma 2

Let T, U be linear operators on V with U onto and $\mathrm{N}(\mathsf{T})$, $\mathrm{N}(\mathsf{U})$ finite-dimensional. Then $\mathrm{N}(\mathsf{TU})$ is finite-dimensional and $\dim \mathrm{N}(\mathsf{TU}) = \dim \mathrm{N}(\mathsf{T}) + \dim \mathrm{N}(\mathsf{U})$.

Proof. Take bases $\{u_1, \ldots, u_p\}$ of $\mathrm{N}(\mathsf{T})$, $\{w_1, \ldots, w_q\}$ of $\mathrm{N}(\mathsf{U})$, and v_i with $\mathsf{U}(v_i) = u_i$ (U onto). We show $\{v_1, \ldots, v_p, w_1, \ldots, w_q\}$ is a basis of $\mathrm{N}(\mathsf{TU})$ of $p + q$ distinct vectors.

Step 1 (inside). $\mathsf{TU}(v_i) = \mathsf{T}(u_i) = 0$ and $\mathsf{TU}(w_j) = \mathsf{T}(0) = 0$.

Step 2 (spans). If $\mathsf{TU}(v) = 0$, then $\mathsf{U}(v) = \sum a_i u_i = \mathsf{U}(\sum a_i v_i)$, so $v - \sum a_i v_i \in \mathrm{N}(\mathsf{U})$ equals $\sum b_j w_j$.

Step 3 (independent). Apply U to $\sum a_i v_i + \sum b_j w_j = 0$: $\sum a_i u_i = 0$, so $a_i = 0$, then $b_j = 0$ (distinctness similarly). ■

Proof of Theorem 2.32. Induction on n; $n = 1$: $p(D)$ is a nonzero multiple of $D - c\mathsf{I}$ (Theorem 2.30). For $n > 1$, $p(D) = q(D)(D - c\mathsf{I})$ with $\deg q = n - 1$; $D - c\mathsf{I}$ is onto (Lemma 1), $\dim \mathrm{N}(q(D)) = n - 1$ (induction), $\dim \mathrm{N}(D - c\mathsf{I}) = 1$; Lemma 2 gives n. ■

Corollary

The solution space of any nth-order equation (2.1) is an n-dimensional subspace of C^∞.

Proof. Theorems 2.28 and 2.32. ■

Theorem 2.33

For distinct $c_1, \dots, c_n \in \mathbb{C}$, $\{e^{c_1 t}, \dots, e^{c_n t}\}$ is linearly independent.

Proof. Induction on n. Apply $D - c_n \mathsf{I}$ to $\sum_i b_i e^{c_i t} = 0$: $\sum_{i<n} b_i (c_i - c_n) e^{c_i t} = 0$, so $b_i = 0$ for $i < n$ (induction), then $b_n = 0$. ■

Corollary

If the auxiliary polynomial of an nth-order equation (2.1) has n distinct zeros $c_1, \dots, c_n$, then $\{e^{c_1 t}, \dots, e^{c_n t}\}$ is a basis of the solution space.

Proof. They are n independent solutions (Theorems 2.31, 2.33) in an n-dimensional space. ■

Example 5

$y'' + 5y' + 4y = 0$: $p(t) = (t+1)(t+4)$, so the solutions are $b_1 e^{-t} + b_2 e^{-4t}$.

Example 6

$y'' + 9y = 0$: $p(t) = (t-3i)(t+3i)$, basis $\{e^{3it}, e^{-3it}\}$, or $\{\cos 3t, \sin 3t\}$ as $\cos 3t = \frac{1}{2}(e^{3it} + e^{-3it})$, $\sin 3t = \frac{1}{2i}(e^{3it} - e^{-3it})$.

Lemma

If the auxiliary polynomial is $(t-c)^n$, then $\beta = \{e^{ct}, te^{ct}, \dots, t^{n-1}e^{ct}\}$ is a basis of the solution space.

Proof.

Step 1 (solutions). $(D - c\mathsf{I})(t^k e^{ct}) = k t^{k-1} e^{ct}$, so $(D - c\mathsf{I})^n (t^k e^{ct}) = 0$ for $k < n$.

Step 2 (independent). $\sum_k b_k t^k e^{ct} = 0 \Rightarrow \sum b_k t^k \equiv 0 \Rightarrow b_k = 0$; with dim $= n$ (Theorem 2.32), β is a basis. ■

Example 7

$y^{(4)} - 4y^{(3)} + 6y^{(2)} - 4y' + y = 0$: $p(t) = (t-1)^4$, so the solutions are $(b_1 + b_2 t + b_3 t^2 + b_4 t^3) e^t$.

Theorem 2.34

If the auxiliary polynomial is $(t - c_1)^{n_1} \cdots (t - c_k)^{n_k}$ with distinct c_i, then $\{e^{c_1 t}, te^{c_1 t}, \dots, t^{n_1 - 1} e^{c_1 t}, \dots, e^{c_k t}, \dots, t^{n_k - 1} e^{c_k t}\}$ is a basis of the solution space.

Proof.

Step 1 (solutions). Factors of $p(D)$ commute, so each $t^j e^{c_i t}$ $(j < n_i)$, killed by $(D - c_i \mathsf{I})^{n_i}$ (Lemma), is killed by $p(D)$.

Step 2 (independent) [Supplement]. With $n = \sum n_i$ functions, independence suffices. Let $\sum_i q_i(t) e^{c_i t} = 0$, $\deg q_i < n_i$; induct on k. For $a \neq 0$, $(D - c\mathsf{I})(q(t) e^{(c+a)t}) = (q' + aq) e^{(c+a)t}$, and

$q \mapsto q' + aq$ preserves degree. Applying $(D - c_k \mathsf{I})^{n_k}$ kills the k-th term (Step 1) and maps the others to $\tilde{q}_i e^{c_i t}$, $\deg \tilde{q}_i = \deg q_i$. By induction $\tilde{q}_i = 0$, so $q_i = 0$ $(i < k)$; then $q_k e^{c_k t} = 0$ gives $q_k = 0$. ■

Example 8

$y''' - 4y'' + 5y' - 2y = 0$: $p(t) = (t-1)^2(t-2)$, so the solutions are $b_1 e^t + b_2 t e^t + b_3 e^{2t}$.

Chapter 3. Elementary Matrix Operations and Systems of Linear Equations

3.1 Elementary matrix operations and elementary matrices

Motivation. Each row or column operation of Gaussian elimination is multiplication by an invertible elementary matrix.

Definition (Elementary operations)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, an **elementary row [column] operation** is (1) interchanging two rows [columns]; (2) multiplying a row [column] by a nonzero scalar; (3) adding a scalar multiple of a row [column] to another row [column]. These are of **type** 1, 2, 3.

Example 1

For $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & -1 & 3 \\ 4 & 0 & 1 & 2 \end{pmatrix}$: interchanging rows $1, 2$ gives $B = \begin{pmatrix} 2 & 1 & -1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 0 & 1 & 2 \end{pmatrix}$; multiplying column 2 by 3 gives C; adding $4\times$row 3 to row 1 gives $M = \begin{pmatrix} 17 & 2 & 7 & 12 \\ 2 & 1 & -1 & 3 \\ 4 & 0 & 1 & 2 \end{pmatrix}$.

Definition (Elementary matrix)

An $n \times n$ **elementary matrix** is obtained from I_n by one elementary operation; its type is that of the operation.

Theorem 3.1

If B is obtained from $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ by an elementary row [column] operation, then $B = EA$ $[B = AE]$, where E is the $m \times m$ $[n \times n]$ elementary matrix obtained from I_m $[I_n]$ by the same operation. Conversely, if E is elementary, EA $[AE]$ is obtained from A by the corresponding row [column] operation.

Proof [Supplement]. By Theorem 2.13 (row form), row i of EA is $\sum_k E_{ik}$(row k of A), and row i of E is e_i^t unless the operation changed it. Type 1 (swap p, q): row p of E is e_q^t, so row p of EA is row q of A, and vice versa. Type 2 (row p times c): row p of EA is c·row p of A. Type 3 (add c·row q to row p): row p of E is $e_p^t + ce_q^t$, giving row $p + c$·row q. Other rows are unchanged. Columns: transpose, $(AE)^t = E^t A^t$ with E^t elementary. ■

Example 2

In Example 1, $B = EA$ with $E = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, and $C = AE'$ with $E' = \mathrm{diag}(1, 3, 1, 1)$.

Theorem 3.2

Elementary matrices are invertible, and the inverse of an elementary matrix is elementary of the same type.

Proof. The operation giving E is undone by an operation $\bar{E}$ of the same type (swap again; multiply by c^{-1}; add $-c$ times the row). By Theorem 3.1, $\bar{E}E = I_n$; then $\mathsf{L}_{\bar{E}}\mathsf{L}_E = \mathsf{I}$ makes L_E one-to-one, hence invertible (Theorem 2.5), and $E^{-1} = \bar{E}$. ■

3.2 The rank of a matrix and matrix inverses

Motivation. Rank-preserving elementary operations reduce every matrix to $\left(\begin{smallmatrix} I_r & O \\ O & O \end{smallmatrix}\right)$, giving row rank = column rank and inverses via $(A \mid I)$.

Definition (Rank of a matrix)

$\operatorname{rank}(A) = \operatorname{rank}(\mathsf{L}_A)$. Thus an $n \times n$ matrix is invertible iff its rank is n (Theorem 2.5 and Corollary 2 of Theorem 2.18).

Theorem 3.3

For linear $\mathsf{T} : V \to W$ between finite-dimensional spaces with ordered bases β, γ: $\operatorname{rank}(\mathsf{T}) = \operatorname{rank}([\mathsf{T}]_\beta^\gamma)$.

Proof [Supplement]. With $A = [\mathsf{T}]_\beta^\gamma$, $\mathsf{L}_A \phi_\beta = \phi_\gamma \mathsf{T}$ (Theorem 2.14). As ϕ_β is onto, $\mathrm{R}(\mathsf{L}_A) = \mathrm{R}(\mathsf{L}_A \phi_\beta) = \phi_\gamma(\mathrm{R}(\mathsf{T}))$, and the isomorphism ϕ_γ preserves dimension. ■

Theorem 3.4

If P, Q are invertible $m \times m$ and $n \times n$, then (a) $\operatorname{rank}(AQ) = \operatorname{rank}(A)$,
(b) $\operatorname{rank}(PA) = \operatorname{rank}(A)$,
(c) $\operatorname{rank}(PAQ) = \operatorname{rank}(A)$.

Proof.

(a) L_Q is onto, so $\mathrm{R}(\mathsf{L}_{AQ}) = \mathsf{L}_A \mathsf{L}_Q(\mathbb{F}^n) = \mathsf{L}_A(\mathbb{F}^n) = \mathrm{R}(\mathsf{L}_A)$.

(b) [Supplement] $\mathrm{R}(\mathsf{L}_{PA}) = \mathsf{L}_P(\mathrm{R}(\mathsf{L}_A))$, and the isomorphism L_P preserves dimension.

(c) Combine (a) and (b). ■

Corollary

Elementary row and column operations preserve rank.

Proof. Theorems 3.1, 3.2, 3.4. ■

Theorem 3.5

The rank of a matrix is the maximum number of linearly independent columns, i.e. the dimension of the span of its columns.

Proof. $\mathrm{R}(\mathsf{L}_A) = \operatorname{span}\{\mathsf{L}_A(e_j)\} = \operatorname{span}\{a_1, \ldots, a_n\}$ by Theorems 2.2 and 2.13(b), where $a_j = Ae_j$ is the j-th column. ■

Example 1

$A = \left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{smallmatrix}\right)$: a_1, a_2 independent, $a_3 = a_1 + a_2$, so $\operatorname{rank} A = 2$.

Example 2

For $A = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 1 & 1 & 2 \end{smallmatrix}\right)$, row operations give $\left(\begin{smallmatrix} 1 & 2 & 1 \\ 0 & -2 & 2 \\ 0 & -1 & 1 \end{smallmatrix}\right)$; clearing row 1 by column operations leaves two independent columns, so rank $A = 2$.

Theorem 3.6

If $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ has rank r, then $r \le m$, $r \le n$, and finitely many elementary row and column operations transform A into

$$D = \begin{pmatrix} I_r & O_1 \\ O_2 & O_3 \end{pmatrix} \qquad (O_i \text{ zero matrices}).$$

Proof. If $A = O$, $r = 0$ and $D = A$. Otherwise, induct on m to reach some D; then $r = \operatorname{rank} D = \operatorname{rank} A$ (Corollary) and $r \le m, n$.

Step 1 (a pivot). Move some $A_{ij} \neq 0$ to $(1,1)$ by interchanges and scale it to 1.

Step 2 (clear). Clear row 1 and column 1 by type 3 operations: $\left(\begin{smallmatrix} 1 & 0 \\ 0 & B' \end{smallmatrix}\right)$, B' of size $(m-1)\times(n-1)$.

Step 3 (recurse). If $m = 1$, done. Otherwise, by induction (trivially if $B' = O$), operations on the last $m-1$ rows and $n-1$ columns bring B' to $\left(\begin{smallmatrix} I_{r-1} & O \\ O & O \end{smallmatrix}\right)$, leaving row and column 1 intact. ■

Example 3

$A = \left(\begin{smallmatrix} 0 & 2 & 4 & 2 & 2 \\ 4 & 4 & 4 & 8 & 0 \\ 8 & 2 & 0 & 10 & 2 \\ 6 & 3 & 2 & 9 & 1 \end{smallmatrix}\right)$: swapping rows $1, 2$, scaling and clearing give $D = \left(\begin{smallmatrix} I_3 & O \\ O & O \end{smallmatrix}\right)$, so rank $A = 3$.

Corollary 1

If rank $A = r$, there are invertible B $(m \times m)$ and C $(n \times n)$ with $D = BAC$.

Proof. By Theorem 3.6, $D = E_p \cdots E_1 A G_1 \cdots G_q$ with elementary E_i, G_j (Theorem 3.1); take $B = E_p \cdots E_1$ and $C = G_1 \cdots G_q$, invertible by Theorem 3.2. ■

Corollary 2

For any $A \in \mathrm{M}_{m\times n}(\mathbb{F})$:
(a) $\operatorname{rank}(A^t) = \operatorname{rank}(A)$;
(b) the rank of A is the maximum number of linearly independent rows, i.e. the dimension of the row space;
(c) the rows and the columns of A span subspaces of the same dimension, equal to rank A.

Proof.

(a) $D^t = C^t A^t B^t$ with B^t, C^t invertible, so rank A^t = rank D^t = r = rank A (Theorem 3.4).

(b) Rows of A = columns of A^t; Theorem 3.5 and (a).

(c) Theorem 3.5 and (b). ■

Corollary 3

Every invertible matrix is a product of elementary matrices.

Proof. $\operatorname{rank} A = n$ gives $I_n = E_p \cdots E_1 A G_1 \cdots G_q$, so $A = E_1^{-1} \cdots E_p^{-1} G_q^{-1} \cdots G_1^{-1}$, elementary by Theorem 3.2. ■

Theorem 3.7

For linear $\mathsf{T} : V \to W$, $\mathsf{U} : W \to Z$ (finite-dimensional) and matrices with AB defined:
(a) $\operatorname{rank}(\mathsf{UT}) \le \operatorname{rank}(\mathsf{U})$;
(b) $\operatorname{rank}(\mathsf{UT}) \le \operatorname{rank}(\mathsf{T})$;
(c) $\operatorname{rank}(AB) \le \operatorname{rank}(A)$;
(d) $\operatorname{rank}(AB) \le \operatorname{rank}(B)$.

Proof.

(a) $\mathrm{R}(\mathsf{UT}) = \mathsf{U}(\mathsf{T}(V)) \subseteq \mathsf{U}(W) = \mathrm{R}(\mathsf{U})$.

(c) Apply (a) to $\mathsf{L}_A \mathsf{L}_B = \mathsf{L}_{AB}$.

(d) $\operatorname{rank}(AB) = \operatorname{rank}(B^t A^t) \le \operatorname{rank}(B^t) = \operatorname{rank}(B)$ by (c) and Corollary 2.

(b) With bases α, β, γ, $A = [\mathsf{U}]_\beta^\gamma$, $B = [\mathsf{T}]_\alpha^\beta$: $\operatorname{rank}(\mathsf{UT}) = \operatorname{rank}(AB) \le \operatorname{rank}(B) = \operatorname{rank}(\mathsf{T})$ by Theorems 3.3, 2.11 and (d). ■

Example 4

(a) $\left(\begin{smallmatrix} 1 & 2 & 1 & 1 \\ 2 & 1 & -1 & 1 \end{smallmatrix}\right)$: two independent rows, rank 2.
(b) $\left(\begin{smallmatrix} 1 & 3 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 3 & 0 & 0 \end{smallmatrix}\right)$: row 3 = row 1 − row 2, rank 2.
(c) $\left(\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 5 & 4 \end{smallmatrix}\right)$: subtracting gives rows $(1,2,1), (0,-3,-3), (0,3,3)$, rank 2.

The inverse of a matrix. The **augmented matrix** $(A \mid B)$ (A $m \times n$, B $m \times p$) lists the columns of A then of B; $M(A \mid B) = (MA \mid MB)$. If A is invertible, $A^{-1} = E_p \cdots E_1$ (Corollary 3), and these row operations turn $(A \mid I_n)$ into $(I_n \mid A^{-1})$. Conversely, $(A \mid I_n) \rightsquigarrow (I_n \mid B)$ gives $E_p \cdots E_1 A = I$, so $B = A^{-1}$. A zero row in the left half means $\operatorname{rank} A < n$: not invertible.

Example 5

For $A = \left(\begin{smallmatrix} 0 & 2 & 4 \\ 2 & 4 & 2 \\ 3 & 3 & 1 \end{smallmatrix}\right)$, row-reducing $(A \mid I_3)$ gives

$$A^{-1} = \begin{pmatrix} \frac{1}{8} & -\frac{5}{8} & \frac{3}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{2} \\ \frac{3}{8} & -\frac{3}{8} & \frac{1}{4} \end{pmatrix}.$$

Example 6

For $A = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 5 & 4 \end{smallmatrix}\right)$, reducing $(A \mid I_3)$ gives a zero row on the left ($\operatorname{rank} A = 2$): not invertible.

Example 7

$\mathsf{T}(f) = f + f' + f''$ on $\mathsf{P}_2(\mathbb{R})$: in the standard basis $[\mathsf{T}]_\beta = \left(\begin{smallmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ with inverse $\left(\begin{smallmatrix} 1 & -1 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{smallmatrix}\right)$. So T is invertible (Corollary 1 of Theorem 2.18) and $\mathsf{T}^{-1}(a_0 + a_1 x + a_2 x^2) = (a_0 - a_1) + (a_1 - 2a_2)x + a_2 x^2$.

3.3 Systems of linear equations: theoretical aspects

Motivation. Solutions of $Ax = 0$ form $\mathrm{N}(\mathsf{L}_A)$, those of $Ax = b$ a translate of it, and $Ax = b$ is solvable iff b lies in the column space.

A system of m equations in n unknowns is $Ax = b$, **coefficient matrix** $A \in \mathrm{M}_{m\times n}(\mathbb{F})$; a **solution** is $s \in \mathbb{F}^n$ with $As = b$; the system is **consistent** if one exists.

Example 1

(a) $x_1 + x_2 = 3$, $x_1 - x_2 = 1$ has the unique solution $(2, 1)$.
(b) $x_1 + x_2 + x_3 = 1$ has infinitely many, e.g. $(1, 0, 0)$, $(0, 0, 1)$.
(c) $x_1 + x_2 = 0$, $x_1 + x_2 = 1$ has no solution.

Definition (Homogeneous system)

$Ax = b$ is **homogeneous** if $b = 0$; it always has the solution 0.

Theorem 3.8

The solution set K of a homogeneous system $Ax = 0$ (m equations, n unknowns) is $\mathrm{N}(\mathsf{L}_A)$, a subspace of $\mathbb{F}^n$ of dimension $n - \operatorname{rank}(A)$.

Proof. $K = \{s : As = 0\} = \mathrm{N}(\mathsf{L}_A)$, and $\dim \mathrm{N}(\mathsf{L}_A) = n - \operatorname{rank} \mathsf{L}_A$ by Theorem 2.3. ∎

Corollary

If $m < n$, then $Ax = 0$ has a nonzero solution.

Proof. $\operatorname{rank} A \le m < n$, so $\dim K > 0$. ∎

Example 2

(a) $x_1 + 2x_2 + x_3 = 0$, $x_1 - x_2 - x_3 = 0$: $\operatorname{rank} A = 2$, so $\dim K = 1$, and $K = \operatorname{span}\{(1, -2, 3)\}$.
(b) $x_1 - 2x_2 + x_3 = 0$: $\dim K = 2$, with basis $\{(2, 1, 0), (-1, 0, 1)\}$.

Theorem 3.9

Let K be the solution set of $Ax = b$ and K_H that of $Ax = 0$. For any solution s of $Ax = b$, $K = \{s\} + K_H = \{s + k : k \in K_H\}$.

Proof. If $w \in K$, then $A(w - s) = b - b = 0$, so $w = s + (w - s) \in \{s\} + K_H$. Conversely $A(s + k) = b + 0 = b$. ∎

Example 3

(a) $x_1+2x_2+x_3 = 7$, $x_1-x_2-x_3 = -4$ has solution $s = (1, 1, 4)$, so $K = \{(1, 1, 4)+t(1, -2, 3) : t \in \mathbb{F}\}$.
(b) $x_1 - 2x_2 + x_3 = 4$: $K = \{(4, 0, 0) + t_1(2, 1, 0) + t_2(-1, 0, 1)\}$.

Theorem 3.10

Let $Ax = b$ have n equations and n unknowns. If A is invertible, the unique solution is $A^{-1}b$. Conversely, if the system has exactly one solution, A is invertible.

Proof. If A is invertible, $A(A^{-1}b) = b$, and $As = b \Rightarrow s = A^{-1}b$. Conversely, a unique solution gives $K_H = \{0\} = \mathrm{N}(\mathsf{L}_A)$ (Theorem 3.9), so A is invertible (Theorems 2.5, 2.18). ■

Example 4

$2x_2+4x_3 = 2$, $2x_1+4x_2+2x_3 = 3$, $3x_1+3x_2+x_3 = 1$ has the coefficient matrix of Example 5 of Section 3.2, so

$$A^{-1}\begin{pmatrix}2\\3\\1\end{pmatrix} = \begin{pmatrix}-\frac{7}{8}\\ \frac{5}{4}\\ -\frac{1}{8}\end{pmatrix}.$$

Theorem 3.11

$Ax = b$ is consistent iff $\operatorname{rank}(A) = \operatorname{rank}(A \mid b)$.

Proof. Consistent $\iff$ $b \in \mathrm{R}(\mathsf{L}_A) = \operatorname{span}\{a_1, \ldots, a_n\}$ $\iff$ $\operatorname{span}\{a_j\} = \operatorname{span}\{a_j, b\}$ $\iff$ equal dimensions (Theorem 1.11, as the first lies in the second) $\iff$ $\operatorname{rank} A = \operatorname{rank}(A \mid b)$ (Theorem 3.5). ■

Example 5

For $x_1 + x_2 = 0$, $x_1 + x_2 = 1$: $\operatorname{rank} A = 1 < 2 = \operatorname{rank}(A \mid b)$: inconsistent.

Example 6

Is $(3, 3, 2) \in \mathrm{R}(\mathsf{T})$ for $\mathsf{T}(a_1, a_2, a_3) = (a_1+a_2+a_3,\ a_1-a_2+a_3,\ a_1+a_3)$? With $A = \left(\begin{smallmatrix}1 & 1 & 1\\ 1 & -1 & 1\\ 1 & 0 & 1\end{smallmatrix}\right)$, $\operatorname{rank} A = 2$ and $\operatorname{rank}(A \mid b) = 3$. So no.

3.3.1 An application: Leontief's input–output models

Closed model. Producers (food, clothing, housing) consume fractions $A_{ij} \geq 0$ of commodity j, columns summing to 1. Incomes p must satisfy the **equilibrium condition** $(I - A)p = 0$, $p \geq 0$. ($b \geq c$ [$b > c$] is entrywise; b is **nonnegative** [**positive**] if $b \geq 0$ [$b > 0$].) Even $Ap \leq p$ forces $Ap = p$, else $\sum_i p_i > \sum_i \sum_j A_{ij}p_j = \sum_j p_j$. For

$$A = \begin{pmatrix}0.40 & 0.20 & 0.20\\ 0.10 & 0.70 & 0.20\\ 0.50 & 0.10 & 0.60\end{pmatrix}, \qquad p = \begin{pmatrix}0.25\\ 0.35\\ 0.40\end{pmatrix}$$

solves it: incomes in the ratio $5 : 7 : 8$.

Theorem 3.12

Let A be an $n \times n$ input–output matrix (nonnegative entries, columns summing to 1) of the form $A = \left(\begin{smallmatrix}B & C\\ D & E\end{smallmatrix}\right)$, with D a positive $1 \times (n-1)$ vector and C a positive $(n-1) \times 1$ vector. Then $(I - A)x = 0$ has a one-dimensional solution space generated by a nonnegative vector.

Proof [Supplement]. (The book cites the proof.) Write $x = (y, x_n)$, $y \in \mathbb{R}^{n-1}$.

Step 1 (reduction). Columns of $I - A$ sum to 0, so the last equation is minus the sum of the others: $(I - A)x = 0 \iff (I - B)y = x_n C$.

Step 2 ($I - B$ is invertible). Column sums of B are $1 - D_j < 1$. If $(I - B)y = 0$, $y \neq 0$, then $|y_i| \le \sum_j B_{ij}|y_j|$; summing, $\sum_i |y_i| \le \sum_j (1 - D_j)|y_j| < \sum_j |y_j|$, a contradiction.

Step 3 (the solution space). The solutions are $x_n((I - B)^{-1}C, 1)$, spanned by $v = (y_0, 1)$ with $(I - B)y_0 = C$.

Step 4 ($y_0 \ge 0$). Suppose $N = \{i : (y_0)_i < 0\} \neq \emptyset$. For $i \in N$, $(y_0)_i = \sum_j B_{ij}(y_0)_j + C_i > \sum_{j \in N} B_{ij}(y_0)_j$ (terms $j \notin N$ are ≥ 0, $C_i > 0$). Hence

$$\sum_{i \in N} |(y_0)_i| < \sum_{j \in N} \Big(\sum_{i \in N} B_{ij} \Big) |(y_0)_j| \le \sum_{j \in N} |(y_0)_j|,$$

a contradiction. So $v \ge 0$. ∎

Open model. With demands $d \ge 0$ and $A_{ij} \ge 0$ the amount of commodity i per unit of j (column sums ≤ 1), we need $x \ge 0$ with $(I - A)x = d$: $x = (I - A)^{-1}d$ if $(I - A)^{-1} \ge 0$ exists, e.g. when $A^k \to O$, as then $(I - A)^{-1} = I + A + A^2 + \cdots$ (Section 5.3). With

$$A = \begin{pmatrix} 0.30 & 0.20 & 0.30 \\ 0.10 & 0.40 & 0.10 \\ 0.30 & 0.20 & 0.30 \end{pmatrix}, \qquad (I - A)^{-1} = \begin{pmatrix} 2.0 & 1.0 & 1.0 \\ 0.5 & 2.0 & 0.5 \\ 1.0 & 1.0 & 2.0 \end{pmatrix},$$

$d = (30, 20, 10)$ is met by $x = (90, 60, 70)$.

3.4 Systems of linear equations: computational aspects

Motivation. Gaussian elimination brings $(A \mid b)$ to its unique reduced row echelon form, from which solutions and column dependencies are read off.

Definition (Equivalent systems)

Two systems are **equivalent** if they have the same solution set.

Theorem 3.13

If C is an invertible $m \times m$ matrix, $(CA)x = Cb$ is equivalent to $Ax = b$.

Proof. $As = b \Rightarrow CAs = Cb$, and $CAs = Cb \Rightarrow As = C^{-1}CAs = C^{-1}Cb = b$. ∎

Corollary

If $(A' \mid b')$ comes from $(A \mid b)$ by finitely many elementary row operations, $A'x = b'$ is equivalent to $Ax = b$.

Proof. $(A' \mid b') = C(A \mid b) = (CA \mid Cb)$ with C a product of elementary, hence invertible, matrices. ∎

Definition (Reduced row echelon form)

A matrix is in **reduced row echelon form** if
(a) every nonzero row precedes every zero row;
(b) the first nonzero entry of each row is the only nonzero entry in its column;
(c) the first nonzero entry of each row is 1 and lies to the right of the first nonzero entry of the preceding row.

Gaussian elimination. *Forward pass:* in the leftmost nonzero column, bring a nonzero entry to the top (type 1), scale it to 1 (type 2), clear below it (type 3); repeat on the lower-right submatrix. *Backward pass:* from the last nonzero row up, clear above each leading 1. For example

$$\begin{cases} 3x_1 + 2x_2 + 3x_3 - 2x_4 = 1 \\ x_1 + x_2 + x_3 = 3 \\ x_1 + 2x_2 + x_3 - x_4 = 2 \end{cases} \quad \rightsquigarrow \quad \left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 3 \end{array}\right),$$

so $x_1 = 1 - x_3$, $x_2 = 2$, $x_4 = 3$, with x_3 free.

Example 1

(a) The final matrix above is in reduced row echelon form.
(b) $\left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{smallmatrix}\right)$ is not: column 1 violates (b), and row 3's leading 1 violates (c).
(c) $\left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right)$ is in reduced row echelon form; $\left(\begin{smallmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right)$ is not (leading entry 2).

Theorem 3.14

Gaussian elimination transforms any matrix into reduced row echelon form.

Proof [Supplement]. After the forward pass, leading entries are 1, each strictly right of the one above, with zeros below and zero rows at the bottom: (a), (c) hold. The backward pass adds multiples of a row, which is zero left of its leading 1, so leading entries and the zeros left of them are unchanged, and (b) holds. ∎

Reading off the solution. If $(A \mid b)$ is in reduced row echelon form with r nonzero rows and no row $(0 \cdots 0 \mid 1)$ (else no solution), solve for the leading variables and set the $n - r$ free variables to parameters $t_1, \ldots, t_{n-r}$. For

$$\begin{cases} 2x_1 + 3x_2 + x_3 + 4x_4 - 9x_5 = 17 \\ x_1 + x_2 + x_3 + x_4 - 3x_5 = 6 \\ x_1 + x_2 + x_3 + 2x_4 - 5x_5 = 8 \\ 2x_1 + 2x_2 + 2x_3 + 3x_4 - 8x_5 = 14 \end{cases} \quad \rightsquigarrow \quad \left(\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & -2 & 3 \\ 0 & 1 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right),$$

the general solution is

$$x = \begin{pmatrix} 3 \\ 1 \\ 0 \\ 2 \\ 0 \end{pmatrix} + t_1 \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t_2 \begin{pmatrix} 2 \\ -1 \\ 0 \\ 2 \\ 1 \end{pmatrix}.$$

Theorem 3.15

Let $Ax = b$ have r nonzero equations in n unknowns, $\operatorname{rank} A = \operatorname{rank}(A \mid b)$, and $(A \mid b)$ in reduced row echelon form. Then (a) $\operatorname{rank} A = r$;

(b) if the procedure gives $s = s_0 + t_1u_1 + \cdots + t_{n-r}u_{n-r}$, then $\{u_1, \ldots, u_{n-r}\}$ is a basis of the homogeneous solution set K_H and s_0 is a solution.

Proof.

(a) Consistency excludes rows $(0 \cdots 0 \mid 1)$, so each of the r rows has a leading 1 in A, in a column e_i. These r columns are independent and the column space lies in $\mathbb{F}^r$, so $\operatorname{rank} A = r$.

(b) All $t_i = 0$ gives $s_0 \in K$. By Theorem 3.9, $K_H = \{-s_0\} + K = \operatorname{span}\{u_1, \ldots, u_{n-r}\}$, and $\dim K_H = n - r$ (Theorem 3.8), so the u_i form a basis. ■

What the reduced row echelon form says. If A has columns a_j and reduced form $B = CA$ (C invertible), then $b_j = Ca_j$ and, L_C being an isomorphism, the b_j satisfy exactly the linear relations of the a_j.

Theorem 3.16

Let $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ have rank $r > 0$ and reduced row echelon form B. Then
(a) B has exactly r nonzero rows;
(b) for each $i \le r$ there is a column $b_{j_i} = e_i$ ($j_1 < \cdots < j_r$);
(c) the columns $a_{j_1}, \ldots, a_{j_r}$ of A are linearly independent;
(d) if column k of B is $d_1e_1 + \cdots + d_re_r$, then $a_k = d_1a_{j_1} + \cdots + d_ra_{j_r}$.

Proof.

(a) $\operatorname{rank} B = \operatorname{rank} A = r$ (Corollary to Theorem 3.4); the nonzero rows of B are independent (each has a 1 where the others have 0), so there are r (Corollary 2 of Theorem 3.6).

(b) Let j_i be the column of the leading 1 of row i; by (b), (c) of the definition, $b_{j_i} = e_i$ and $j_1 < \cdots < j_r$.

(c) $b_{j_1}, \ldots, b_{j_r}$ are independent, and $\mathsf{L}_{C^{-1}}$ preserves independence.

(d) Every column of B lies in $\operatorname{span}\{e_1, \ldots, e_r\}$ (rows $> r$ are zero), so $b_k = \sum d_ie_i = \sum d_ib_{j_i}$; apply C^{-1}. ■

Corollary

The reduced row echelon form of a matrix is unique.

Proof. By Theorem 3.16, j_i is the first column of A not in the span of the preceding columns, and every other b_k has coordinates d_i fixed by the unique relation $a_k = \sum d_ia_{j_i}$. So A determines B. ■

Example 2

$A = \begin{pmatrix} 2&4&6&2&4\\ 1&2&3&1&1\\ 2&4&8&0&0\\ 3&6&7&5&9 \end{pmatrix}$ has reduced form $B = \begin{pmatrix} 1&2&0&4&0\\ 0&0&1&-1&0\\ 0&0&0&0&1\\ 0&0&0&0&0 \end{pmatrix}$. So $\operatorname{rank} A = 3$, columns $1, 3, 5$ of A are independent, $a_2 = 2a_1$ and $a_4 = 4a_1 - a_3$.

Example 3

$S = \{2 + x + 2x^2 + 3x^3,\ 4 + 2x + 4x^2 + 6x^3,\ 6 + 3x + 8x^2 + 7x^3,\ 2 + x + 5x^3,\ 4 + x + 9x^3\}$ has the columns of A in Example 2 as coordinate vectors, so the first, third and fifth polynomials form a basis of $\operatorname{span}(S)$.

Example 4

$V = \{x \in \mathbb{R}^5 : x_1 + 7x_2 + 5x_3 - 4x_4 + 2x_5 = 0\}$ has dimension 4, with basis $\{(-7,1,0,0,0),(-5,0,1,0,0),(4,0,0,1,0),(-2,0,0,0,1)\}$. To extend the independent $S = \{(-2,0,0,-1,-1),(1,1,-2,-1,-1),(-5,1,0,1,1)\} \subseteq V$, row-reduce the matrix with columns S then this basis: pivots $1,2,3,6$, so $S \cup \{(4,0,0,1,0)\}$ is a basis of V.

Chapter 4. Determinants

4.1 Determinants of order 2

Motivation. The number $ad - bc$ decides invertibility, gives signed area, and is linear in each row, the property that generalises.

Definition (Determinant of order 2)

For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{M}_{2\times 2}(\mathbb{F})$, $\det(A) = |A| = ad - bc$.

Example 1

For $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 3 & 2 \\ 6 & 4 \end{smallmatrix}\right)$: $\det A = -2$, $\det B = 0$, $\det(A+B) = \det\left(\begin{smallmatrix} 4 & 4 \\ 9 & 8 \end{smallmatrix}\right) = -4$: det is not linear.

Theorem 4.1

$\det : \mathrm{M}_{2\times 2}(\mathbb{F}) \to \mathbb{F}$ is linear in each row when the other row is fixed: for $u, v, w \in \mathbb{F}^2$, $k \in \mathbb{F}$,

$$\det \begin{pmatrix} u + kv \\ w \end{pmatrix} = \det \begin{pmatrix} u \\ w \end{pmatrix} + k \det \begin{pmatrix} v \\ w \end{pmatrix}, \qquad \det \begin{pmatrix} w \\ u + kv \end{pmatrix} = \det \begin{pmatrix} w \\ u \end{pmatrix} + k \det \begin{pmatrix} w \\ v \end{pmatrix}.$$

Proof. With $u = (a_1, a_2)$, $v = (b_1, b_2)$, $w = (c_1, c_2)$: $(a_1 + kb_1)c_2 - (a_2 + kb_2)c_1 = (a_1c_2 - a_2c_1) + k(b_1c_2 - b_2c_1)$. The second identity is analogous. ∎

Theorem 4.2

$A \in \mathrm{M}_{2\times 2}(\mathbb{F})$ is invertible iff $\det A \neq 0$, and then $A^{-1} = \frac{1}{\det A}\left(\begin{smallmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{smallmatrix}\right)$.

Proof. If $\det A \neq 0$, the displayed M satisfies $AM = MA = I$. Conversely, invertible A has $\operatorname{rank} A = 2$, so $A_{11} \neq 0$ or $A_{21} \neq 0$. If $A_{11} \neq 0$, adding $-A_{21}/A_{11}$·row 1 to row 2 gives $\left(\begin{smallmatrix} A_{11} & A_{12} \\ 0 & A_{22} - A_{12}A_{21}/A_{11} \end{smallmatrix}\right)$, still of rank 2 (Theorem 3.4), so $A_{22} - A_{12}A_{21}/A_{11} \neq 0$, i.e. $\det A \neq 0$. If $A_{21} \neq 0$, exchange the rows. ∎

4.1.1 The area of a parallelogram

For an ordered basis $\beta = \{u, v\}$ of $\mathbb{R}^2$, the **orientation** is $O\left(\begin{smallmatrix} u \\ v \end{smallmatrix}\right) = \det\left(\begin{smallmatrix} u \\ v \end{smallmatrix}\right) / \left|\det\left(\begin{smallmatrix} u \\ v \end{smallmatrix}\right)\right| = \pm 1$: $+1$ if β is right-handed (u turns counterclockwise by less than π to v), -1 if left-handed; $O = 1$ if $\{u, v\}$ is dependent. The **area** of the parallelogram with sides u, v is

$$\operatorname{area}\begin{pmatrix} u \\ v \end{pmatrix} = O\begin{pmatrix} u \\ v \end{pmatrix} \cdot \det\begin{pmatrix} u \\ v \end{pmatrix} = \left|\det\begin{pmatrix} u \\ v \end{pmatrix}\right|.$$

Reason. $\delta(u, v) = O \cdot \text{area}$ is linear in each row (shears preserve it, scaling a side scales it), vanishes on equal rows, and $\delta(e_1, e_2) = 1$, so $\delta = \det$ (Theorem 4.12, $n = 2$).

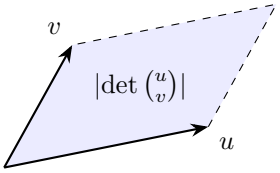


Figure. *Area* $|\det \binom{u}{v}|$*; the sign records orientation.*

Example 2

For $u = (-1, 5)$, $v = (4, -2)$: $\det\begin{pmatrix} -1 & 5 \\ 4 & -2 \end{pmatrix} = -18$: area 18, left-handed.

4.2 Determinants of order n

Motivation. Defined by first-row expansion, det is linear in each row and expandable along any row, which governs its behaviour under row operations.

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, $\tilde{A}_{ij}$ is A with row i and column j deleted.

Definition (Determinant)

For $n = 1$, $\det(A_{11}) = A_{11}$. For $n \geq 2$,

$$\det(A) = \sum_{j=1}^{n} (-1)^{1+j} A_{1j} \det(\tilde{A}_{1j}).$$

$c_{ij} = (-1)^{i+j} \det(\tilde{A}_{ij})$ is the **cofactor** of A_{ij}, and the formula is the **cofactor expansion along the first row**. For $n = 2$ it gives $A_{11}A_{22} - A_{12}A_{21}$.

Example 1

$A = \begin{pmatrix} 1 & 3 & -3 \\ -3 & -5 & 2 \\ -4 & 4 & -6 \end{pmatrix}$: $\det A = 1 \cdot (30 - 8) - 3 \cdot (18 + 8) + (-3)(-12 - 20) = 22 - 78 + 96 = 40$.

Example 2

$B = \begin{pmatrix} 0 & 1 & 3 \\ -2 & -3 & -5 \\ 4 & -4 & 4 \end{pmatrix}$: $\det B = 0 - 1 \cdot (-8 + 20) + 3 \cdot (8 + 12) = 48$.

Example 3

$C = \begin{pmatrix} 2 & 0 & 0 & 1 \\ 0 & 1 & 3 & -3 \\ -2 & -3 & -5 & 2 \\ 4 & -4 & 4 & -6 \end{pmatrix}$: only C_{11}, C_{14} contribute, with $\tilde{C}_{11} = A$, $\tilde{C}_{14} = B$, so $\det C = 2 \cdot 40 + (-1)^5 \cdot 1 \cdot 48 = 32$.

Example 4

$\det I_n = 1 \cdot \det I_{n-1} = 1$ by induction.

Theorem 4.3

The determinant is linear in each row when the other rows are fixed: if rows $a_1, \ldots, a_n$ have $a_r = u + kv$, then

$$\det(a_1, \ldots, u + kv, \ldots, a_n) = \det(a_1, \ldots, u, \ldots, a_n) + k \det(a_1, \ldots, v, \ldots, a_n).$$

Proof. Induction on n; $n = 1$ is clear. Let A, B, C have row r equal to $u + kv$, u, v, other rows equal.

Step 1 ($r = 1$). $\tilde{A}_{1j} = \tilde{B}_{1j} = \tilde{C}_{1j}$, so $\det A = \sum_j (-1)^{1+j}(u_j + kv_j) \det \tilde{A}_{1j} = \det B + k \det C$.

Step 2 ($r > 1$). $A_{1j} = B_{1j} = C_{1j}$, and $\tilde{A}_{1j}, \tilde{B}_{1j}, \tilde{C}_{1j}$ differ only in the row from row r ($u + kv$, u, v minus entry j). By induction $\det \tilde{A}_{1j} = \det \tilde{B}_{1j} + k \det \tilde{C}_{1j}$; multiply by $(-1)^{1+j} A_{1j}$ and sum. ■

Corollary

A matrix with a zero row has determinant 0.

Proof. If row r is $0 = 0 + 1 \cdot 0$, Theorem 4.3 gives $\det A = \det A + \det A$. ■

Lemma

If $n \geq 2$ and row i of $B \in \mathrm{M}_{n\times n}(\mathbb{F})$ is e_k, then $\det B = (-1)^{i+k} \det(\tilde{B}_{ik})$.

Proof. Induction on n; $n = 2$ is direct. Let $n \geq 3$; $i = 1$ is the definition, so let $i > 1$. For $j \neq k$, let C_{ij} be $\tilde{B}_{1j}$ minus row $i - 1$ (from row i of B) and the column through k. Row $i - 1$ of $\tilde{B}_{1j}$ is 0 ($j = k$), e_{k-1} ($j < k$), e_k ($j > k$), in $\mathbb{F}^{n-1}$. By induction,

$$\det \tilde{B}_{1j} = \begin{cases} 0 & j = k \\ (-1)^{(i-1)+(k-1)} \det C_{ij} & j < k \\ (-1)^{(i-1)+k} \det C_{ij} & j > k. \end{cases}$$

Hence $\det B = \sum_{j<k} (-1)^{1+j} B_{1j} (-1)^{i+k} \det C_{ij} + \sum_{j>k} (-1)^{1+j} B_{1j} (-1)^{i+k+1} \det C_{ij}$. Expanding $\det \tilde{B}_{ik}$ along its first row, B_{1j} sits in column j ($j < k$) or $j - 1$ ($j > k$) with minor C_{ij}:

$$\det \tilde{B}_{ik} = \sum_{j<k} (-1)^{1+j} B_{1j} \det C_{ij} + \sum_{j>k} (-1)^{j} B_{1j} \det C_{ij}.$$

Multiply by $(-1)^{i+k}$ to get $\det B$. ■

Theorem 4.4

The determinant can be expanded along any row: for each i, $\det A = \sum_{j=1}^{n} (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij})$.

Proof. For $i = 1$ this is the definition; let $i > 1$. Row i of A is $\sum_j A_{ij} e_j$; let B_j be A with row i replaced by e_j. By Theorem 4.3 and the Lemma, $\det A = \sum_j A_{ij} \det B_j = \sum_j A_{ij} (-1)^{i+j} \det(\tilde{B}_j)_{ij} = \sum_j (-1)^{i+j} A_{ij} \det \tilde{A}_{ij}$, as $(\tilde{B}_j)_{ij} = \tilde{A}_{ij}$. ■

Corollary

A matrix with two identical rows has determinant 0.

Proof. Induction on n; $n = 2$ is clear. For $n \geq 3$ and rows $r \neq s$ equal, expand along a row $i \notin \{r, s\}$: each $\tilde{A}_{ij}$ has two equal rows, so every term vanishes. ■

Theorem 4.5

Interchanging two rows multiplies the determinant by -1.

Proof. Let rows $r < s$ of A be u, v, interchanged in B. By Theorem 4.3 and the Corollary, $0 = \det(\dots, u+v, \dots, u+v, \dots) = \det(\dots, u, \dots, u, \dots) + \det(\dots, u, \dots, v, \dots) + \det(\dots, v, \dots, u, \dots) + \det(\dots, v, \dots, v, \dots) = 0 + \det A + \det B + 0$. ■

Theorem 4.6

Adding a multiple of one row to another leaves the determinant unchanged.

Proof. Let B be A with k·row r added to row s, and C be A with row s replaced by row r. Then $\det B = \det A + k \det C = \det A$ (Theorem 4.3; Corollary to Theorem 4.4). ■

Corollary

If $\operatorname{rank} A < n$, then $\det A = 0$.

Proof. The rows are dependent (Corollary 2 of Theorem 3.6): $a_r = \sum_{i \neq r} c_i a_i$. Subtracting each c_i·row i from row r gives a zero row with the same det (Theorem 4.6); apply the Corollary to Theorem 4.3. ■

Row operations: type 1 multiplies det by -1, type 2 by k, type 3 by 1. An upper triangular det is the diagonal product (expand along the last row repeatedly), so reduce to triangular form.

Example 5

For B of Example 2: swap rows $1, 2$, clear column 1, reach $\begin{pmatrix} -2 & -3 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 24 \end{pmatrix}$. So $\det B = -(-2)(1)(24) = 48$.

Example 6

For $\begin{pmatrix} 2 & 3 & 1 & 1 \\ -2 & -2 & 2 & -1 \\ 4 & 7 & 2 & 3 \\ -6 & -9 & 5 & 1 \end{pmatrix}$, type 3 operations give a triangular matrix with diagonal product $\det = -40$.

4.3 Properties of determinants

Motivation. det is multiplicative, detects invertibility, ignores transposition, gives Cramer's rule and measures volume.

By the row-operation rules, $\det E = -1, k, 1$ for elementary E of types $1, 2, 3$, and $\det(EA) = \det E \det A$.

Theorem 4.7

$\det(AB) = \det(A) \det(B)$ for $A, B \in \mathrm{M}_{n \times n}(\mathbb{F})$.

Proof.

Step 1 ($\operatorname{rank} A < n$). $\operatorname{rank}(AB) \leq \operatorname{rank} A < n$ (Theorem 3.7), so both sides are 0 (Corollary to Theorem 4.6).

Step 2 ($\operatorname{rank} A = n$). $A = E_m \cdots E_1$ (Corollary 3 of Theorem 3.6); by $\det(EM) = \det E \det M$, $\det(AB) = \det E_m \cdots \det E_1 \det B$ and $\det A = \det E_m \cdots \det E_1$. ■

Corollary

A is invertible iff $\det A \neq 0$, and then $\det(A^{-1}) = \frac{1}{\det A}$.

Proof. If A is not invertible, $\operatorname{rank} A < n$ and $\det A = 0$. If A is invertible, $1 = \det I = \det(AA^{-1}) = \det A \det A^{-1}$. ■

Theorem 4.8

$\det(A^t) = \det(A)$.

Proof. If A is not invertible, neither is A^t ($\operatorname{rank} A^t = \operatorname{rank} A$): both are 0. Otherwise $A = E_m \cdots E_1$, $A^t = E_1^t \cdots E_m^t$, and $\det E_i^t = \det E_i$ (types 1, 2 symmetric; type 3 transposes to type 3). By Theorem 4.7, $\det A^t = \prod \det E_i^t = \prod \det E_i = \det A$. ■

So all row statements hold for columns.

Theorem 4.9 (Cramer's rule)

If $Ax = b$ is an $n \times n$ system with $\det A \neq 0$, it has a unique solution, and

$$x_k = \frac{\det M_k}{\det A} \qquad (1 \leq k \leq n),$$

where M_k is A with column k replaced by b.

Proof. A is invertible, so the solution x is unique (Theorem 3.10). With $X_k = I_n$ with column k replaced by x, $AX_k = M_k$ (columns $Ae_j = a_j$, $j \neq k$, and $Ax = b$), and $\det X_k = x_k$ (expand along row k). So $\det M_k = \det A \cdot x_k$. ■

Example 1

$x_1 + 2x_2 + 3x_3 = 2$, $x_1 + x_3 = 3$, $x_1 + x_2 - x_3 = 1$: $\det A = 6$, $\det M_1 = 15$, $\det M_2 = -6$, $\det M_3 = 3$, so $(x_1, x_2, x_3) = (\frac{5}{2}, -1, \frac{1}{2})$.

Volume. For rows $a_1, \ldots, a_n \in \mathbb{R}^n$ of A, $|\det A|$ is the volume of the parallelepiped with sides $a_1, \ldots, a_n$ (as in Section 4.1.1, via Section 4.5).

Example 2

$a_1 = (1, -2, 1)$, $a_2 = (1, 0, -1)$, $a_3 = (1, 1, 1)$ give volume $\left|\det \begin{pmatrix} 1 & -2 & 1 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix}\right| = 6 = \sqrt{6}\sqrt{2}\sqrt{3}$ (orthogonal sides).

An ordered basis β of $\mathbb{R}^n$ (rows of A) is **right-handed** if $\det A > 0$, **left-handed** if $\det A < 0$; coordinate changes with $\det Q > 0$ preserve this.

4.4 Summary: important facts about determinants

Motivation. The facts needed later, collected in one place.

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$:

1. $\det A = \sum_j (-1)^{i+j} A_{ij} \det \tilde{A}_{ij}$ for any i, and $= \sum_i (-1)^{i+j} A_{ij} \det \tilde{A}_{ij}$ for any j (Theorems 4.4, 4.8).

2. Row (column) operations of types $1, 2, 3$ multiply det by -1, k, 1.

3. det of an upper or lower triangular matrix is the product of the diagonal entries; $\det I = 1$.

4. Two identical rows (columns), or a zero row, give det $= 0$.

5. $\det(AB) = \det A \det B$; $\det A^t = \det A$.

6. A is invertible iff $\det A \neq 0$, and then $\det A^{-1} = (\det A)^{-1}$.

7. $\det(Q^{-1}AQ) = \det Q^{-1} \det A \det Q = \det A$, so the **determinant of a linear operator** $\det \mathsf{T} = \det[\mathsf{T}]_\beta$ is independent of β.

8. For $A = \left(\begin{smallmatrix} B & C \\ O & D \end{smallmatrix}\right)$ with square B, D, $\det A = \det B \det D$.

Proof of 3 and 8 [Supplement].

3. Upper triangular: $\det A = A_{11} \det \tilde{A}_{11}$ (first column), $\tilde{A}_{11}$ upper triangular; induct. Lower: transpose.

8. Induction on the size k of B, expanding along column 1. For $k = 1$, $\det A = B_{11} \det D$. For $k > 1$, only rows $i \leq k$ contribute, and $\tilde{A}_{i1} = \left(\begin{smallmatrix} \tilde{B}_{i1} & C' \\ O & D \end{smallmatrix}\right)$, so $\det A = \sum_{i \leq k} (-1)^{i+1} B_{i1} \det \tilde{B}_{i1} \det D = \det B \det D$. ■

4.5 A characterization of the determinant *

Motivation. Row-linearity, vanishing on equal adjacent rows and $\det I = 1$ characterise det, explaining signed volume and multiplicativity.

Definition (n-linear; alternating)

$\delta : \mathrm{M}_{n \times n}(\mathbb{F}) \to \mathbb{F}$ is **n-linear** if linear in each row with the others fixed, and **alternating** if $\delta(A) = 0$ whenever two adjacent rows of A are identical.

Example 1

$\delta(A) = 0$ is n-linear.

Example 2

$\delta_j(A) = A_{1j} A_{2j} \cdots A_{nj}$ (product of column j) is n-linear (linear in A_{rj}).

Example 3

$\delta(A) = A_{11} A_{22} \cdots A_{nn}$ is n-linear.

Example 4

tr is not n-linear for $n \geq 2$: $\mathrm{tr}(2I) = 2\mathrm{tr}(I)$, not $2^n \mathrm{tr}(I)$.

det is n-linear (Theorem 4.3) and alternating (Corollary to Theorem 4.4).

Theorem 4.10

Let δ be alternating and n-linear.
(a) Interchanging two rows of A changes $\delta(A)$ to $-\delta(A)$.
(b) If A has two identical rows, $\delta(A) = 0$.

Proof.

(a) For adjacent rows u, v, Theorem 4.5's computation uses only adjacent equal rows: $0 = \delta(\dots, u+v, u+v, \dots) = \delta(A) + \delta(B)$. Swapping rows $r < s$ takes $2(s-r)-1$ adjacent swaps (row r down to s, then row s up to r), an odd number.

(b) Swaps make the identical rows adjacent, changing δ only by a sign; the result is 0. ■

Corollary 1

If B is A with a multiple of a row added to another row, then $\delta(B) = \delta(A)$.

Proof. As for Theorem 4.6, using Theorem 4.10(b). ■

Corollary 2

If $\operatorname{rank} M < n$, then $\delta(M) = 0$.

Proof. As for the Corollary to Theorem 4.6: type 3 operations (Corollary 1) produce a zero row, where $\delta = 0$ by linearity. ■

Corollary 3

For elementary E_1, E_2, E_3 of types 1, 2, 3, with E_2 multiplying a row of I by k: $\delta(E_1) = -\delta(I)$, $\delta(E_2) = k\delta(I)$, $\delta(E_3) = \delta(I)$.

Proof. Theorem 4.10(a), linearity in the scaled row, and Corollary 1, applied to I. ■

Theorem 4.11

If δ is alternating, n-linear and $\delta(I) = 1$, then $\delta(AB) = \delta(A)\delta(B)$.

Proof [Supplement]. By Theorem 4.10, linearity and Corollary 1, $\delta(EB) = \delta(E)\delta(B)$ for elementary E ($\delta(E) = -1, k, 1$ by Corollary 3). Repeat the proof of Theorem 4.7, using Corollary 2 if $\operatorname{rank} A < n$. ■

Theorem 4.12

If δ is alternating, n-linear and $\delta(I) = 1$, then $\delta(A) = \det(A)$ for all A.

Proof. If $\operatorname{rank} A < n$, both vanish (Corollary 2 and the Corollary to Theorem 4.6). Otherwise $A = E_m \cdots E_1$, and by Theorem 4.11 and Corollary 3, $\delta(A) = \prod \delta(E_i) = \prod \det(E_i) = \det A$. ■

Chapter 5. Diagonalization

5.1 Eigenvalues and eigenvectors

Motivation. A diagonalizing basis consists of eigenvectors, whose stretching factors are the zeros of $\det(A - tI)$.

Definition (Diagonalizable)

A linear operator T on finite-dimensional V is **diagonalizable** if $[\mathsf{T}]_\beta$ is diagonal for some ordered basis β. A square matrix A is diagonalizable if L_A is, i.e. if A is similar to a diagonal matrix.

Definition (Eigenvector, eigenvalue)

A nonzero $v \in V$ is an **eigenvector** of T if $\mathsf{T}(v) = \lambda v$ for some scalar λ, the corresponding **eigenvalue**. For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, eigenvectors and eigenvalues are those of L_A: $Av = \lambda v$, $v \neq 0$.

Theorem 5.1

T is diagonalizable iff V has an ordered basis β of eigenvectors of T. If $\beta = \{v_1, \ldots, v_n\}$ is such a basis and $D = [\mathsf{T}]_\beta$, then D is diagonal and D_{jj} is the eigenvalue of v_j.

Proof. $[\mathsf{T}]_\beta$ is diagonal iff each $[\mathsf{T}(v_j)]_\beta = D_{jj}e_j$, i.e. $\mathsf{T}(v_j) = D_{jj}v_j$, with $v_j \neq 0$. ■

For a basis γ of eigenvectors, $Q^{-1}AQ$ is diagonal with Q having columns γ (Corollary to Theorem 2.23).

Example 1

$A = \left(\begin{smallmatrix} 1 & 3 \\ 4 & 2 \end{smallmatrix}\right)$: $A(1,-1)^t = -2(1,-1)^t$ and $A(3,4)^t = 5(3,4)^t$. With $Q = \left(\begin{smallmatrix} 1 & 3 \\ -1 & 4 \end{smallmatrix}\right)$, $Q^{-1}AQ = \left(\begin{smallmatrix} -2 & 0 \\ 0 & 5 \end{smallmatrix}\right)$.

Example 2

Rotation of $\mathbb{R}^2$ by $\pi/2$ has no eigenvectors, so is not diagonalizable.

Example 3

On $C^\infty(\mathbb{R})$, $\mathsf{T}(f) = f'$ has $\mathsf{T}(f) = \lambda f$ iff $f = ce^{\lambda t}$ (as in Theorem 2.30): every real λ is an eigenvalue, with eigenvectors $ce^{\lambda t}$, $c \neq 0$.

Theorem 5.2

λ is an eigenvalue of $A \in \mathrm{M}_{n\times n}(\mathbb{F})$ iff $\det(A - \lambda I_n) = 0$.

Proof. λ is an eigenvalue iff $(A - \lambda I)v = 0$ for some $v \neq 0$, iff $A - \lambda I$ is not invertible, iff $\det(A - \lambda I) = 0$ (Corollary to Theorem 4.7). ■

Definition (Characteristic polynomial)

$f(t) = \det(A - tI_n)$ is the **characteristic polynomial** of A. For T on n-dimensional V it is that of $[\mathsf{T}]_\beta$, independent of β: $\det(Q^{-1}AQ - tI) = \det(Q^{-1}(A - tI)Q) = \det(A - tI)$. Write $f(t) = \det(\mathsf{T} - t\mathsf{I})$.

Example 4

$A = \left(\begin{smallmatrix} 1 & 1 \\ 4 & 1 \end{smallmatrix}\right)$: $f(t) = (1-t)^2 - 4 = (t-3)(t+1)$, so the eigenvalues are 3 and -1.

Example 5

$\mathsf{T}(f) = f + (x+1)f'$ on $\mathsf{P}_2(\mathbb{R})$: in the standard basis $A = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{smallmatrix}\right)$, so $f(t) = -(t-1)(t-2)(t-3)$ and the eigenvalues are $1, 2, 3$.

Theorem 5.3

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$: (a) $f(t)$ has degree n and leading coefficient $(-1)^n$;
(b) A has at most n distinct eigenvalues.

Proof [Supplement].

(a) Induction on n, expanding along the first row: $(A_{11} - t)\det(\widetilde{(A - tI)}_{11})$ has degree n and leading coefficient $(-1)(-1)^{n-1}$; the other minors miss two diagonal entries, of degree $\leq n - 2$.

(b) A nonzero polynomial of degree n has at most n zeros (Section 1.6.1). ■

Theorem 5.4

If λ is an eigenvalue of T, then v is an eigenvector for λ iff $v \neq 0$ and $v \in \mathrm{N}(\mathsf{T} - \lambda\mathsf{I})$.

Proof. $\mathsf{T}(v) = \lambda v$ iff $(\mathsf{T} - \lambda\mathsf{I})(v) = 0$. ■

Example 6

For A of Example 4: $\mathrm{N}(A - 3I) = \mathrm{span}\{(1,2)^t\}$ and $\mathrm{N}(A + I) = \mathrm{span}\{(1,-2)^t\}$. So $Q^{-1}AQ = \mathrm{diag}(3,-1)$ for $Q = \left(\begin{smallmatrix} 1 & 1 \\ 2 & -2 \end{smallmatrix}\right)$.

Example 7

For T of Example 5, $(A - \lambda I)x = 0$ gives eigenvectors 1, $1 + x$, $(1 + x)^2$ for $\lambda = 1, 2, 3$; in this basis $[\mathsf{T}] = \mathrm{diag}(1, 2, 3)$.

In general, v is an eigenvector of T iff $[v]_\beta$ is one of $[\mathsf{T}]_\beta$.

5.2 Diagonalizability

Motivation. Diagonalizable means: f splits and each eigenspace is as large as its multiplicity, i.e. $V = \bigoplus E_\lambda$.

Theorem 5.5

Eigenvectors $v_1, \ldots, v_k$ of T for distinct eigenvalues $\lambda_1, \ldots, \lambda_k$ are linearly independent.

Proof. Induction on k; $k = 1$ holds as $v_1 \neq 0$. If $\sum_{i=1}^{k} a_i v_i = 0$, apply $\mathsf{T} - \lambda_k \mathsf{I}$: $\sum_{i<k} a_i(\lambda_i - \lambda_k)v_i = 0$. By induction $a_i(\lambda_i - \lambda_k) = 0$, so $a_i = 0$ for $i < k$; then $a_k v_k = 0$ gives $a_k = 0$. ■

Corollary

If $\dim V = n$ and T has n distinct eigenvalues, T is diagonalizable.

Proof. One eigenvector per eigenvalue gives n independent vectors (Theorem 5.5), a basis (Corollary 2 of Theorem 1.10); apply Theorem 5.1. ■

Example 1

$A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has eigenvalues $0, 2$: diagonalizable. The converse fails: I has one eigenvalue.

Definition (Splitting)

$f \in \mathsf{P}(\mathbb{F})$ **splits over** $\mathbb{F}$ if $f(t) = c(t - a_1) \cdots (t - a_n)$ with $c, a_i \in \mathbb{F}$. E.g. $(t^2 + 1)(t - 2)$ splits over $\mathbb{C}$ but not over $\mathbb{R}$.

Theorem 5.6

The characteristic polynomial of a diagonalizable operator splits.

Proof. If $[\mathsf{T}]_\beta = D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, then $f(t) = \det(D - tI) = (\lambda_1 - t) \cdots (\lambda_n - t) = (-1)^n(t - \lambda_1) \cdots (t - \lambda_n)$. ■

Definition (Multiplicity; eigenspace)

The (algebraic) **multiplicity** of an eigenvalue λ is the largest k with $(t - \lambda)^k \mid f(t)$. The **eigenspace** is $E_\lambda = \{x : \mathsf{T}(x) = \lambda x\} = \mathrm{N}(\mathsf{T} - \lambda \mathsf{I})$.

Example 2

$A = \left(\begin{smallmatrix} 3 & 1 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 4 \end{smallmatrix}\right)$ has $f(t) = -(t - 3)^2(t - 4)$: $\lambda = 3$ has multiplicity 2, $\lambda = 4$ multiplicity 1.

Theorem 5.7

If λ is an eigenvalue of multiplicity m of T on finite-dimensional V, then $1 \leq \dim E_\lambda \leq m$.

Proof. Extend a basis $\{v_1, \ldots, v_p\}$ of E_λ ($p \geq 1$) to β. Then $[\mathsf{T}]_\beta = \left(\begin{smallmatrix} \lambda I_p & B \\ O & C \end{smallmatrix}\right)$, so $f(t) = (\lambda - t)^p \det(C - tI)$ (Section 4.4, fact 8), and $p \leq m$. ■

Example 3

$\mathsf{T}(f) = f'$ on $\mathsf{P}_2(\mathbb{R})$: $[\mathsf{T}] = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, $f(t) = -t^3$, and $\dim E_0 = 1 < 3$ (constants): not diagonalizable.

Example 4

$\mathsf{T}(a_1,a_2,a_3) = (4a_1+a_3,\ 2a_1+3a_2+2a_3,\ a_1+4a_3)$ has $f(t) = -(t-5)(t-3)^2$, $E_5 = \operatorname{span}\{(1,2,1)\}$ and $E_3 = \operatorname{span}\{(0,1,0),(-1,0,1)\}$. Their union is an eigenbasis.

Lemma

Let $\lambda_1,\dots,\lambda_k$ be distinct eigenvalues of T and $v_i \in E_{\lambda_i}$. If $v_1+\cdots+v_k = 0$, then all $v_i = 0$.

Proof. Otherwise the nonzero v_i are eigenvectors for distinct eigenvalues with a nontrivial relation (all coefficients 1), contradicting Theorem 5.5. ■

Theorem 5.8

If $\lambda_1,\dots,\lambda_k$ are distinct eigenvalues of T and each $S_i \subseteq E_{\lambda_i}$ is finite and linearly independent, then $S = S_1 \cup \cdots \cup S_k$ is linearly independent.

Proof. Let $S_i = \{v_{i1},\dots,v_{in_i}\}$ and $\sum_{i,j} a_{ij}v_{ij} = 0$. Put $w_i = \sum_j a_{ij}v_{ij} \in E_{\lambda_i}$. Then $\sum w_i = 0$, so each $w_i = 0$ by the Lemma, and independence of S_i gives $a_{ij} = 0$. ■

Theorem 5.9

Let T on finite-dimensional V have splitting characteristic polynomial and distinct eigenvalues $\lambda_1,\dots,\lambda_k$ with multiplicities m_i.
(a) T is diagonalizable iff $m_i = \dim E_{\lambda_i}$ for all i.
(b) If T is diagonalizable and β_i is an ordered basis of E_{λ_i}, then $\beta = \beta_1 \cup \cdots \cup \beta_k$ is an ordered basis of V consisting of eigenvectors.

Proof. Let $n = \dim V$, $d_i = \dim E_{\lambda_i}$. As f splits, $\sum m_i = n$.

Step 1 (only if). For an eigenbasis β, let $\beta_i' = \beta \cap E_{\lambda_i}$ have $n_i \le d_i \le m_i$ vectors; $\sum n_i = n$. So $n = \sum n_i \le \sum d_i \le \sum m_i = n$, forcing $d_i = m_i$.

Step 2 (if) and (b). If $d_i = m_i$, β in (b) is independent (Theorem 5.8) with $\sum d_i = n$ vectors: an eigenbasis. Under (b)'s hypothesis, Step 1 gives $d_i = m_i$, so this applies. ■

Test for diagonalization. T is diagonalizable iff (1) its characteristic polynomial splits and (2) for each eigenvalue, $m_i = \dim E_{\lambda_i} = n - \operatorname{rank}(\mathsf{T} - \lambda_i \mathsf{I})$.

Example 5

$A = \left(\begin{smallmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{smallmatrix}\right)$: $m_3 = 2$ but $\dim E_3 = 3 - \operatorname{rank}(A - 3I) = 1$: not diagonalizable.

Example 6

$\mathsf{T}(f) = f(1) + f'(0)x + (f'(0)+f''(0))x^2$ on $\mathsf{P}_2(\mathbb{R})$: in the standard basis $[\mathsf{T}] = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{smallmatrix}\right)$, $f(t) = -(t-1)^2(t-2)$, $E_1 = \operatorname{span}\{1, -x+x^2\}$ (dimension 2), $E_2 = \operatorname{span}\{1+x^2\}$: diagonalizable with basis $\{1, -x+x^2, 1+x^2\}$.

Example 7

$A = \left(\begin{smallmatrix} 0 & -2 \\ 1 & 3 \end{smallmatrix}\right)$ has eigenvalues $1, 2$ with eigenvectors $(-2,1)^t, (-1,1)^t$. With $Q = \left(\begin{smallmatrix} -2 & -1 \\ 1 & 1 \end{smallmatrix}\right)$, $A = QDQ^{-1}$, $D = \mathrm{diag}(1,2)$, so

$$A^n = QD^nQ^{-1} = \begin{pmatrix} 2-2^n & 2-2^{n+1} \\ -1+2^n & -1+2^{n+1} \end{pmatrix}.$$

5.2.1 Systems of differential equations

Consider $x' = Ax$ with $A = \left(\begin{smallmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ -1 & -1 & 1 \end{smallmatrix}\right)$, x a vector of real functions. Eigenvalues 2 (eigenvectors $(-1,1,0)$, $(-1,0,1)$) and 4 ($(-1,-2,1)$) give $Q^{-1}AQ = D = \mathrm{diag}(2,2,4)$. For $y = Q^{-1}x$, $y' = Dy$, so $y_i = c_i e^{\lambda_i t}$ and

$$x = Qy = e^{2t}\left(c_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}\right) + c_3 e^{4t} \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}.$$

5.2.2 Direct sums

Definition (Sum of subspaces)

$W_1 + \cdots + W_k = \sum_i W_i = \{v_1 + \cdots + v_k : v_i \in W_i\}$, a subspace.

Example 8

In $\mathbb{R}^3$, the xy-plane W_1 and the yz-plane W_2 satisfy $\mathbb{R}^3 = W_1 + W_2$, $(a,b,c) = (a,0,0) + (0,b,c)$, non-uniquely.

Definition (Direct sum)

$V = W_1 \oplus \cdots \oplus W_k$ if $V = \sum_i W_i$ and $W_j \cap \sum_{i \neq j} W_i = \{0\}$ for each j.

Example 9

$\mathbb{R}^4 = W_1 \oplus W_2 \oplus W_3$ for $W_1 = \{(a,b,0,0)\}$, $W_2 = \{(0,0,c,0)\}$, $W_3 = \{(0,0,0,d)\}$.

Theorem 5.10

For subspaces $W_1, \ldots, W_k$ of finite-dimensional V, the following are equivalent:
(a) $V = W_1 \oplus \cdots \oplus W_k$;
(b) $V = \sum W_i$, and $v_1 + \cdots + v_k = 0$ with $v_i \in W_i$ implies all $v_i = 0$;
(c) each $v \in V$ is uniquely $v = v_1 + \cdots + v_k$ with $v_i \in W_i$;
(d) if γ_i is an ordered basis of W_i for each i, then $\gamma_1 \cup \cdots \cup \gamma_k$ is an ordered basis of V;
(e) for each i there is an ordered basis γ_i of W_i with $\gamma_1 \cup \cdots \cup \gamma_k$ an ordered basis of V.

Proof.

Step (a)⇒(b). If $\sum v_i = 0$, then $-v_j = \sum_{i \neq j} v_i \in W_j \cap \sum_{i \neq j} W_i = \{0\}$.

Step (b)⇒(c). If $v = \sum v_i = \sum w_i$, then $\sum (v_i - w_i) = 0$, so $v_i = w_i$.

Step (c)⇒(d). The union spans $V = \sum W_i$. A vanishing combination groups as $\sum_i u_i = 0$, $u_i \in W_i$ a combination of γ_i; uniqueness gives $u_i = 0$, and independence of γ_i zero coefficients.

Step (d)⇒(e). Bases exist.

Step (e)⇒(a). $V = \operatorname{span}(\bigcup \gamma_i) \subseteq \sum W_i$. If $v \in W_j \cap \sum_{i \neq j} W_i$, writing v via γ_j and via $\bigcup_{i \neq j} \gamma_i$ and subtracting gives a relation in the basis $\bigcup \gamma_i$, so $v = 0$. ■

Theorem 5.11

T on finite-dimensional V is diagonalizable iff V is the direct sum of the eigenspaces of T.

Proof. If T is diagonalizable and γ_i is a basis of E_{λ_i}, $\bigcup \gamma_i$ is a basis of V (Theorem 5.9(b)), so $V = \bigoplus E_{\lambda_i}$ (Theorem 5.10(e)⇒(a)). Conversely, by (a)⇒(d) the union of bases of the E_{λ_i} is an eigenbasis. ■

Example 10

$\mathsf{T}(a, b, c, d) = (a, b, 2c, 3d)$ on $\mathbb{R}^4$: $E_1 = W_1$, $E_2 = W_2$, $E_3 = W_3$ of Example 9, and $\mathbb{R}^4 = E_1 \oplus E_2 \oplus E_3$.

5.3 Matrix limits and Markov chains *

Motivation. Eigenvalues decide whether A^m converges; for a regular transition matrix every column of the limit is the stationary vector.

Definition (Matrix limit)

$A_m \to L$ ($n \times p$, complex entries) if $(A_m)_{ij} \to L_{ij}$ for all i, j; write $\lim_m A_m = L$.

Example 1

$$A_m = \begin{pmatrix} 1 - \frac{1}{m} & (\frac{3}{4})^m \\ \frac{3m^2}{m^2+1} + i\frac{2m+1}{m-1} & (1+\frac{1}{m})^m \end{pmatrix} \to \begin{pmatrix} 1 & 0 \\ 3+2i & e \end{pmatrix}.$$

Theorem 5.12

If $A_m \to L$ ($n \times p$), then $PA_m \to PL$ and $A_m Q \to LQ$ for any $P \in \mathrm{M}_{r \times n}(\mathbb{C})$, $Q \in \mathrm{M}_{p \times s}(\mathbb{C})$.

Proof. $(PA_m)_{ij} = \sum_k P_{ik}(A_m)_{kj} \to \sum_k P_{ik} L_{kj}$, a finite sum of convergent sequences; similarly for $A_m Q$. ■

Corollary

If $A^m \to L$ and Q is invertible, then $(QAQ^{-1})^m \to QLQ^{-1}$.

Proof. $(QAQ^{-1})^m = QA^mQ^{-1}$; apply Theorem 5.12 twice. ■

Let $S = \{\lambda \in \mathbb{C} : |\lambda| < 1\} \cup \{1\}$. For $\lambda \in \mathbb{C}$, λ^m converges iff $\lambda \in S$.

Theorem 5.13

For $A \in \mathrm{M}_{n\times n}(\mathbb{C})$, $\lim A^m$ exists iff (a) every eigenvalue of A lies in S, and (b) if 1 is an eigenvalue, $\dim E_1$ equals the multiplicity of 1.

Proof [Supplement] (the book gives part of it and cites the rest; we use Theorems 5.23 and 5.24 of Section 5.4).

Step 1 (generalized eigenspaces). Over $\mathbb{C}$, $f(t) = \pm\prod_i (t-\lambda_i)^{m_i}$; $K_i = \mathrm{N}((A-\lambda_i I)^{m_i})$ is L_A-invariant. The $g_i = \prod_{j\neq i}(t-\lambda_j)^{m_j}$ have no common zero, so $\sum_i p_i g_i = 1$ for polynomials p_i (Appendix E). With $P_i = p_i(A)g_i(A)$, $\sum P_i = I$ and $(A-\lambda_i I)^{m_i}P_i = \pm p_i(A)f(A) = O$ (Cayley–Hamilton), so $\mathrm{R}\,P_i \subseteq K_i$ and $\mathbb{C}^n = \sum K_i$. If $\sum v_i = 0$, $v_i \in K_i$, then P_j kills v_i for $i \neq j$ ($(t-\lambda_i)^{m_i} \mid g_j$), so $0 = P_j v_j = \sum_i P_i v_j = v_j$. So $\mathbb{C}^n = \bigoplus K_i$ (Theorem 5.10).

Step 2 ($\dim K_i = m_i$). $A-\lambda_i I$ is nilpotent on K_i, so λ_i is the only eigenvalue of $A|_{K_i}$; as f is the product of their characteristic polynomials (Theorem 5.24), $\dim K_i = m_i$.

Step 3 (sufficiency). On K_i, $A = \lambda_i I + N$, $N^{m_i} = O$, and $A^m = \sum_{k<m_i}\binom{m}{k}\lambda_i^{m-k}N^k$. If $|\lambda_i| < 1$, each $\binom{m}{k}|\lambda_i|^{m-k} \to 0$, so $A^m \to O$ on K_i. If $\lambda_i = 1$, (b) with $E_1 \subseteq K_1$, $\dim K_1 = m_1 = \dim E_1$ gives $K_1 = E_1$, where $A = I$. In a basis adapted to $\bigoplus K_i$, A is similar to a block-diagonal matrix with convergent powers; apply the Corollary to Theorem 5.12.

Step 4 (necessity of (a)). If $Av = \lambda v$, $v \neq 0$, and $L = \lim A^m$, then $Lv = \lim A^m v = \lim \lambda^m v$ (Theorem 5.12), so λ^m converges and $\lambda \in S$.

Step 5 (necessity of (b)). If $\dim E_1 < m_1 = \dim K_1$, then $A-I$ is nilpotent but nonzero on K_1, so some $w \in K_1$ has $u = (A-I)w \neq 0$ and $(A-I)u = 0$. Then $A^m w = w + mu$ diverges. ■

Theorem 5.14

If every eigenvalue of $A \in \mathrm{M}_{n\times n}(\mathbb{C})$ lies in S and A is diagonalizable, then $\lim A^m$ exists.

Proof. $A = QDQ^{-1}$ with $D = \mathrm{diag}(\lambda_i)$, $\lambda_i \in S$, so $D^m = \mathrm{diag}(\lambda_i^m)$ converges, and $A^m = QD^mQ^{-1}$ converges by the Corollary to Theorem 5.12. ■

Markov chains. A **transition matrix** (stochastic matrix) has nonnegative real entries and column sums 1; A_{ij} is the probability of moving from state j to i. A **probability vector** has nonnegative coordinates summing to 1. From initial distribution P, A^mP is the distribution after m steps. Let $u = (1,\dots,1)^t$.

Theorem 5.15

Let M be $n\times n$ with nonnegative real entries and $v \in \mathbb{R}^n$ with nonnegative coordinates. Then
(a) M is a transition matrix iff $M^t u = u$;
(b) v is a probability vector iff $u^t v = (1)$.

Proof. $(M^t u)_j$ is the j-th column sum of M, and $u^t v$ is the coordinate sum of v. ■

Corollary

(a) Products of transition matrices are transition matrices; in particular so are their powers.
(b) A transition matrix times a probability vector is a probability vector.

Proof. Nonnegativity is preserved. (a) $(MN)^t u = N^t M^t u = N^t u = u$.
(b) $u^t(Mv) = (M^t u)^t v = u^t v = 1$. ■

Definition (Regular)

A transition matrix is **regular** if some power of it has only positive entries.

Example 2

$\left(\begin{smallmatrix} 0.90 & 0.02 \\ 0.10 & 0.98 \end{smallmatrix}\right)$ (city/suburb) is regular; one with a column e_1 (absorbing state) is not, as every power keeps it.

Definition (Gerschgorin disks)

For $A \in \mathrm{M}_{n\times n}(\mathbb{C})$, $\rho_i(A) = \sum_j |A_{ij}|$ (row sums), $\nu_j(A) = \sum_i |A_{ij}|$ (column sums), $\rho(A) = \max_i \rho_i(A)$, $\nu(A) = \max_j \nu_j(A)$. The i-th **Gerschgorin disk** C_i has center A_{ii} and radius $r_i = \rho_i(A) - |A_{ii}|$.

Example 3

For $A = \begin{pmatrix} 1 & 2 & -3 \\ 0 & 4 & 1 \\ 2 & -1 & 5 \end{pmatrix}$: $\rho_i = 6, 5, 8$ and $\nu_j = 3, 7, 9$, so $\rho(A) = 8$, $\nu(A) = 9$.

Theorem 5.16 (Gerschgorin's disk theorem)

Every eigenvalue of $A \in \mathrm{M}_{n\times n}(\mathbb{C})$ lies in some Gerschgorin disk.

Proof. Let $Av = \lambda v$, $v \neq 0$, and choose k with $|v_k|$ maximal. From $\lambda v_k = \sum_j A_{kj} v_j$, $|\lambda - A_{kk}||v_k| = |\sum_{j\neq k} A_{kj} v_j| \leq \sum_{j\neq k} |A_{kj}||v_k| = r_k |v_k|$, so $|\lambda - A_{kk}| \leq r_k$. ∎

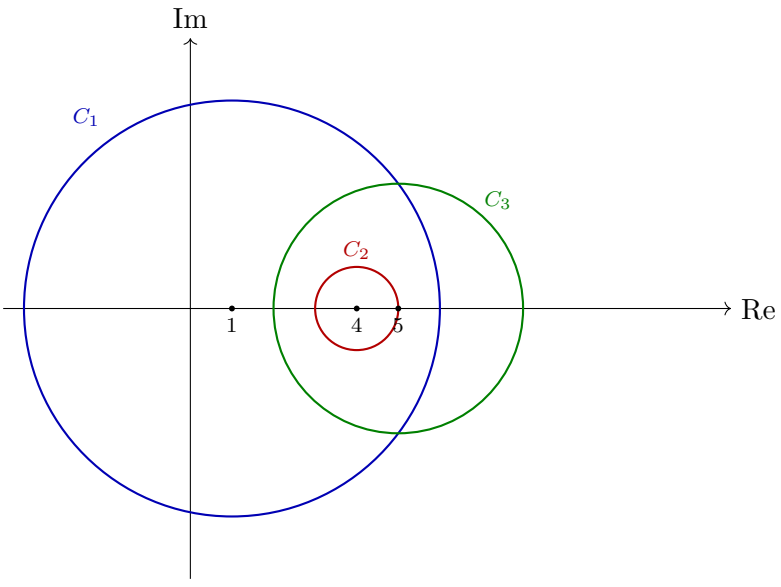


Figure. *Gerschgorin disks of Example 3 (centers* $1, 4, 5$*, radii* $5, 1, 3$*).*

Corollary 1

$|\lambda| \leq \rho(A)$ for every eigenvalue λ of A.

Proof. $|\lambda| \leq |\lambda - A_{kk}| + |A_{kk}| \leq r_k + |A_{kk}| = \rho_k(A) \leq \rho(A)$. ∎

Corollary 2

$|\lambda| \le \min\{\rho(A), \nu(A)\}$.

Proof. λ is an eigenvalue of A^t (same characteristic polynomial, Theorem 4.8), and $\rho(A^t) = \nu(A)$; apply Corollary 1 to both. ■

Corollary 3

Every eigenvalue λ of a transition matrix satisfies $|\lambda| \le 1$.

Proof. $\nu(A) = 1$; apply Corollary 2. ■

Theorem 5.17

Every transition matrix has 1 as an eigenvalue.

Proof. $A^t u = u$ (Theorem 5.15), so 1 is an eigenvalue of A^t, hence of A. ■

Theorem 5.18

Let $A \in \mathrm{M}_{n\times n}(\mathbb{C})$ have only positive entries, and let λ be an eigenvalue with $|\lambda| = \rho(A)$. Then $\lambda = \rho(A)$ and $\{u\}$ is a basis of E_λ.

Proof. Let $Av = \lambda v$, $v \neq 0$, $b = \max_j |v_j| = |v_k|$. Then

$$\rho(A)b = |\lambda||v_k| = \Big|\sum_j A_{kj}v_j\Big| \le \sum_j A_{kj}|v_j| \le \sum_j A_{kj}b = \rho_k(A)b \le \rho(A)b.$$

Equality throughout and $A_{kj} > 0$ give $|v_j| = b$ for all j and a common argument for the $A_{kj}v_j$ (triangle inequality), so $v = zu$, $z \neq 0$. Then $\lambda zu = Azu = z(\rho_1(A), \ldots, \rho_n(A))^t$, so each $\rho_i(A) = \lambda = \rho(A)$, and every eigenvector for λ is a multiple of u. ■

Corollary 1

If A has positive entries and $|\lambda| = \nu(A)$ for an eigenvalue λ, then $\lambda = \nu(A)$ and $\dim E_\lambda = 1$.

Proof. Apply Theorem 5.18 to A^t ($\rho(A^t) = \nu(A)$), and note $\operatorname{rank}(A - \lambda I) = \operatorname{rank}(A^t - \lambda I)$. ■

Corollary 2

If A is a transition matrix with positive entries, every eigenvalue $\lambda \neq 1$ has $|\lambda| < 1$, and $\dim E_1 = 1$.

Proof. $\nu(A) = 1$. By Corollary 3 of Theorem 5.16, $|\lambda| \le 1$, and if $|\lambda| = 1$ then $\lambda = 1$ by Corollary 1. Also $\dim E_1 = 1$ by Corollary 1. ■

Theorem 5.19

Let A be a regular transition matrix and λ an eigenvalue. Then (a) $|\lambda| \le 1$;
(b) if $|\lambda| = 1$, then $\lambda = 1$ and $\dim E_\lambda = 1$.

Proof.

(a) Corollary 3 of Theorem 5.16.

(b) If $A^k > 0$, so is $A^{k+1} = A \cdot A^k$ (each row of A has a positive entry, else $A^k = A \cdot A^{k-1}$ has a zero row). If $Av = \lambda v$, $|\lambda| = 1$, then λ^k, λ^{k+1} are modulus-1 eigenvalues of positive transition matrices, so both are 1 (Corollary 2 above) and $\lambda = \lambda^{k+1}/\lambda^k = 1$. Finally $E_1(A) \subseteq E_1(A^k)$, of dimension 1. ■

Corollary

A diagonalizable regular transition matrix A has a limit $\lim A^m$.

Proof. Theorems 5.19 and 5.14. ■

Theorem 5.20

Let A be an $n \times n$ regular transition matrix. Then
(a) 1 has multiplicity 1 as an eigenvalue;
(b) $L = \lim A^m$ exists;
(c) L is a transition matrix;
(d) $AL = LA = L$;
(e) all columns of L equal the unique probability vector v with $Av = v$;
(f) $\lim A^m w = v$ for every probability vector w.

Proof.

(a) [Supplement] (The book defers this to Chapter 7.) Entries of A^m lie in $[0,1]$, so $\{A^m w\}$ is bounded. If $(A-I)^2 w = 0$, then $u' = (A-I)w \in E_1$ and $A^m w = w + mu'$, forcing $u' = 0$: $\mathrm{N}((A-I)^2) = \mathrm{N}(A-I)$. Hence $\mathrm{N}(A-I) \cap \mathrm{R}(A-I) = \{0\}$ ($x = (A-I)y$, $(A-I)x = 0$ give $y \in \mathrm{N}(A-I)$, $x = 0$), and by the dimension theorem $\mathbb{C}^n = \mathrm{N}(A-I) \oplus \mathrm{R}(A-I)$, both A-invariant, with 1 not an eigenvalue on $\mathrm{R}(A-I)$. By Theorem 5.24 the multiplicity of 1 is $\dim \mathrm{N}(A-I) = 1$ (Theorem 5.19).

(b) By (a) and Theorem 5.19, the conditions of Theorem 5.13 hold.

(c) A^m is a transition matrix, so $u^t L = \lim u^t A^m = u^t$; entries stay nonnegative.

(d) $AL = \lim A^{m+1} = L$ by Theorem 5.12, and likewise $LA = L$.

(e) By (c), (d) each column of L is a probability vector in E_1, and $\dim E_1 = 1$ allows only one such v.

(f) $y = \lim A^m w = Lw$ is a probability vector with $Ay = ALw = Lw = y$, so $y = v$ by (e). ■

v is the **fixed probability vector** (**stationary vector**) of A.

Example 4

Preferences (bread, wine, "thou"): $P = (0.50, 0.30, 0.20)^t$, $A = \begin{pmatrix} 0.40 & 0.20 & 0.20 \\ 0.10 & 0.70 & 0.20 \\ 0.50 & 0.10 & 0.60 \end{pmatrix}$, positive with eigenvalues $1, 0.5, 0.2$; $AP = (0.30, 0.30, 0.40)^t$. The stationary vector $v = (0.25, 0.35, 0.40)^t$ is the long-run split for any P.

Example 5

Crop rotation (corn, soybeans, wheat; half of each crop's land to each other crop): $A = \frac{1}{2}\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, $P = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})^t$ from 300, 200, 100 acres. $A^2 > 0$, eigenvalues $1, -\frac{1}{2}, -\frac{1}{2}$, stationary vector $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})^t$: eventually 200 acres each.

An **absorbing** state (never left) makes a chain non-regular; the limit may still exist (Theorem 5.13).

5.4 Invariant subspaces and the Cayley–Hamilton theorem

Motivation. Characteristic polynomials of cyclic subspaces yield Cayley–Hamilton: every operator is annihilated by its characteristic polynomial.

Definition (Invariant subspace)

$W \subseteq V$ is **T-invariant** if $\mathsf{T}(W) \subseteq W$. Then $\mathsf{T}_W : W \to W$ denotes the restriction.

Example 1

$\{0\}$, V, $\mathrm{R}(\mathsf{T})$, $\mathrm{N}(\mathsf{T})$ and every eigenspace E_λ are T-invariant.

Example 2

For $\mathsf{T}(a,b,c) = (a+b, b+c, 0)$, the xy-plane $\{(x,y,0)\}$ and the x-axis $\{(x,0,0)\}$ are T-invariant.

Definition (Cyclic subspace)

For $x \in V$, the **T-cyclic subspace generated by** x is $W = \operatorname{span}\{x, \mathsf{T}(x), \mathsf{T}^2(x), \dots\}$, the smallest T-invariant subspace containing x.

Example 3

For $\mathsf{T}(a,b,c) = (-b+c, a+c, 3c)$: $\mathsf{T}(e_1) = e_2$, $\mathsf{T}(e_2) = -e_1$, so the cyclic subspace generated by e_1 is $\operatorname{span}\{e_1, e_2\}$.

Example 4

$\mathsf{T}(f) = f'$ on $\mathsf{P}(\mathbb{R})$: the cyclic subspace generated by x^2 is $\operatorname{span}\{x^2, 2x, 2\} = \mathsf{P}_2(\mathbb{R})$.

Theorem 5.21

If W is T-invariant (V finite-dimensional), the characteristic polynomial of T_W divides that of T.

Proof. Extend a basis γ of W (k vectors) to a basis β of V. Then $[\mathsf{T}]_\beta = \begin{pmatrix} B_1 & B_2 \\ O & B_3 \end{pmatrix}$ with $B_1 = [\mathsf{T}_W]_\gamma$, and $f(t) = \det(B_1 - tI)\det(B_3 - tI)$ (Section 4.4, fact 8). ∎

Example 5

$\mathsf{T}(a,b,c,d) = (a+b+2c-d,\ b+d,\ 2c-d,\ c+d)$ and $W = \operatorname{span}\{e_1, e_2\}$, which is invariant. $[\mathsf{T}_W] = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has characteristic polynomial $(t-1)^2$, which divides $f(t) = (t-1)^2(t^2-3t+3)$.

Theorem 5.22

Let W be the T-cyclic subspace generated by $v \neq 0$ and $k = \dim W$ (V finite-dimensional). Then
(a) $\{v, \mathsf{T}(v), \dots, \mathsf{T}^{k-1}(v)\}$ is a basis of W;

(b) if $a_0v + a_1\mathsf{T}(v) + \cdots + a_{k-1}\mathsf{T}^{k-1}(v) + \mathsf{T}^k(v) = 0$, the characteristic polynomial of T_W is $(-1)^k(a_0 + a_1t + \cdots + a_{k-1}t^{k-1} + t^k)$.

Proof.

(a) Let j be largest with $\beta = \{v, \ldots, \mathsf{T}^{j-1}(v)\}$ independent ($j \geq 1$). Then $\mathsf{T}^j(v) \in Z = \operatorname{span}(\beta)$ (Theorem 1.7), so Z is T-invariant ($\mathsf{T}^i(v) \mapsto \mathsf{T}^{i+1}(v) \in Z$ for $i < j$). Hence $Z \supseteq W$, $Z = W$, $j = k$.

(b) $[\mathsf{T}_W]_\beta$ has columns $e_2, \ldots, e_k$ and $(-a_0, \ldots, -a_{k-1})^t$; expanding $\det([\mathsf{T}_W]_\beta - tI)$ along the last column (or by induction on k) gives $(-1)^k(a_0 + a_1t + \cdots + t^k)$. ■

Example 6

For T of Example 3 and $W = \operatorname{span}\{e_1, e_2\}$: $\mathsf{T}^2(e_1) = -e_1$, so $e_1 + \mathsf{T}^2(e_1) = 0$ and the characteristic polynomial of T_W is $t^2 + 1$.

Theorem 5.23 (Cayley–Hamilton)

If $f(t)$ is the characteristic polynomial of T on finite-dimensional V, then $f(\mathsf{T}) = \mathsf{T}_0$.

Proof. Fix $v \neq 0$, its T-cyclic subspace W, $k = \dim W$, a_i as in Theorem 5.22(b). Then $g(t) = (-1)^k(a_0 + \cdots + t^k)$ is the characteristic polynomial of T_W and $g(\mathsf{T})(v) = (-1)^k(a_0v + \cdots + \mathsf{T}^k(v)) = 0$. By Theorem 5.21, $f = qg$, so $f(\mathsf{T})(v) = q(\mathsf{T})(g(\mathsf{T})(v)) = 0$. ■

Example 7

$\mathsf{T}(a, b) = (a + 2b, -2a + b)$, $A = \left(\begin{smallmatrix} 1 & 2 \\ -2 & 1 \end{smallmatrix}\right)$, $f(t) = t^2 - 2t + 5$: $A^2 - 2A + 5I = \left(\begin{smallmatrix} -3 & 4 \\ -4 & -3 \end{smallmatrix}\right) - \left(\begin{smallmatrix} 2 & 4 \\ -4 & 2 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 5 & 0 \\ 0 & 5 \end{smallmatrix}\right) = O$.

Corollary (Cayley–Hamilton for matrices)

If f is the characteristic polynomial of $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, then $f(A) = O$.

Proof. $f(\mathsf{L}_A) = \mathsf{L}_{f(A)}$ is the zero map by Theorem 5.23; apply Theorem 2.15(b). ■

5.4.1 Invariant subspaces and direct sums

Theorem 5.24

If $V = W_1 \oplus \cdots \oplus W_k$ with each W_i T-invariant and f_i the characteristic polynomial of T_{W_i}, then $f_1 \cdots f_k$ is the characteristic polynomial of T.

Proof. Induction on k; $k = 1$ is trivial. For $k = 2$, bases β_1, β_2 of W_1, W_2 give a basis $\beta = \beta_1 \cup \beta_2$ (Theorem 5.10) with $[\mathsf{T}]_\beta = \left(\begin{smallmatrix} [\mathsf{T}_{W_1}] & O \\ O & [\mathsf{T}_{W_2}] \end{smallmatrix}\right)$, whose characteristic polynomial is f_1f_2 (Section 4.4, fact 8). For $k > 2$, $W = W_1 \oplus \cdots \oplus W_{k-1}$ is T-invariant with $V = W \oplus W_k$; apply the case $k = 2$ and induction to T_W. ■

Example 8

$\mathsf{T}(a, b, c, d) = (2a - b, a + b, c - d, c + d)$, $W_1 = \operatorname{span}\{e_1, e_2\}$, $W_2 = \operatorname{span}\{e_3, e_4\}$: $f_1 = t^2 - 3t + 3$, $f_2 = t^2 - 2t + 2$, and $f = f_1f_2$.

Definition (Direct sum of matrices)

For $B_1 \in \mathrm{M}_{m\times m}(\mathbb{F})$, $B_2 \in \mathrm{M}_{n\times n}(\mathbb{F})$, $B_1 \oplus B_2 = \left(\begin{smallmatrix} B_1 & O \\ O & B_2 \end{smallmatrix}\right)$; inductively $B_1 \oplus \cdots \oplus B_k = (B_1 \oplus \cdots \oplus B_{k-1}) \oplus B_k$, block diagonal.

Example 9

$\left(\begin{smallmatrix} 1 & 2 \\ 1 & 1 \end{smallmatrix}\right) \oplus (3) \oplus \left(\begin{smallmatrix} 1 & 2 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{smallmatrix}\right)$ is the 6×6 block-diagonal matrix with these blocks.

Theorem 5.25

If $V = W_1 \oplus \cdots \oplus W_k$ with T-invariant W_i, ordered bases β_i of W_i, $\beta = \beta_1 \cup \cdots \cup \beta_k$, $A = [\mathsf{T}]_\beta$ and $B_i = [\mathsf{T}_{W_i}]_{\beta_i}$, then $A = B_1 \oplus \cdots \oplus B_k$.

Proof. Induction on k. For $k = 2$: for $v \in \beta_1$, $\mathsf{T}(v) \in W_1$ is a combination of β_1 alone, with coordinates the column of B_1; likewise for β_2. So $A = B_1 \oplus B_2$. For $k > 2$, apply this to $W = W_1 \oplus \cdots \oplus W_{k-1}$ and W_k, and induct on T_W. ■

Chapter 6. Inner Product Spaces

6.1 Inner products and norms

Motivation. An inner product, modelled on the dot product, supplies lengths, angles and orthogonality over $\mathbb{R}$ and $\mathbb{C}$.

Throughout, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$, and $\bar{c}$ is the complex conjugate.

Definition (Inner product)

An **inner product** on V assigns to $x, y \in V$ a scalar $\langle x, y\rangle \in \mathbb{F}$ such that for all $x, y, z \in V$, $c \in \mathbb{F}$:
(a) $\langle x + z, y\rangle = \langle x, y\rangle + \langle z, y\rangle$;
(b) $\langle cx, y\rangle = c\langle x, y\rangle$;
(c) $\overline{\langle x, y\rangle} = \langle y, x\rangle$;
(d) $\langle x, x\rangle > 0$ if $x \neq 0$. V with an inner product is an **inner product space**: **complex** if $\mathbb{F} = \mathbb{C}$, **real** if $\mathbb{F} = \mathbb{R}$.

By (a), (b), $\langle \sum a_i x_i, y\rangle = \sum a_i \langle x_i, y\rangle$. Over $\mathbb{R}$, (c) reads $\langle x, y\rangle = \langle y, x\rangle$.

Example 1

The **standard inner product** on $\mathbb{F}^n$: $\langle x, y\rangle = \sum_i a_i \bar{b}_i$ for $x = (a_i)$, $y = (b_i)$; for $\mathbb{F} = \mathbb{R}$ it is the dot product $x \cdot y$.

Example 2

If $\langle \cdot, \cdot \rangle$ is an inner product and $r > 0$, then $r\langle x, y\rangle$ is one; for $r \leq 0$, (d) fails.

Example 3

On $C([0,1])$ (real), $\langle f, g\rangle = \int_0^1 f(t)g(t)\,dt$ is an inner product; for (d), $f^2 \geq 0$ is continuous and $\not\equiv 0$ if $f \neq 0$.

Definition (Conjugate transpose)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $A^* \in \mathrm{M}_{n\times m}(\mathbb{F})$ has $(A^*)_{ij} = \overline{A_{ji}}$ (the **adjoint**); for real A, $A^* = A^t$.

Example 4

$\begin{pmatrix} i & 1+2i \\ 2 & 3+4i \end{pmatrix}^* = \begin{pmatrix} -i & 2 \\ 1-2i & 3-4i \end{pmatrix}$.

Example 5

On $\mathrm{M}_{n\times n}(\mathbb{F})$, $\langle A, B\rangle = \mathrm{tr}(B^*A)$ is an inner product (the **Frobenius inner product**): $\langle A, A\rangle = \sum_{i,k} |A_{ki}|^2 > 0$ for $A \neq O$.

H denotes the continuous complex functions on $[0, 2\pi]$ with

$$\langle f, g\rangle = \frac{1}{2\pi}\int_0^{2\pi} f(t)\overline{g(t)}\,dt.$$

Theorem 6.1

For $x, y, z \in V$, $c \in \mathbb{F}$:
(a) $\langle x, y+z\rangle = \langle x, y\rangle + \langle x, z\rangle$;
(b) $\langle x, cy\rangle = \bar{c}\langle x, y\rangle$;
(c) $\langle x, 0\rangle = \langle 0, x\rangle = 0$;
(d) $\langle x, x\rangle = 0$ iff $x = 0$;
(e) if $\langle x, y\rangle = \langle x, z\rangle$ for all x, then $y = z$.

Proof.

(a) $\langle x, y+z\rangle = \overline{\langle y+z, x\rangle} = \overline{\langle y, x\rangle} + \overline{\langle z, x\rangle} = \langle x, y\rangle + \langle x, z\rangle$.

(b) $\langle x, cy\rangle = \overline{c\langle y, x\rangle} = \bar{c}\langle x, y\rangle$.

(c) $\langle 0, x\rangle = \langle 0 \cdot 0, x\rangle = 0$, then conjugate.

(d) From (c) and axiom (d).

(e) $\langle x, y - z\rangle = 0$ for all x; take $x = y - z$ and use (d). ■

An inner product is linear in the first and **conjugate linear** in the second variable.

Definition (Norm)

$\|x\| = \sqrt{\langle x, x\rangle}$ is the **norm** (length) of x.

Example 6

On $\mathbb{F}^n$, $\|(a_1, \ldots, a_n)\| = \left(\sum |a_i|^2\right)^{1/2}$, the Euclidean length.

Theorem 6.2

For $x, y \in V$, $c \in \mathbb{F}$:
(a) $\|cx\| = |c|\|x\|$;
(b) $\|x\| = 0$ iff $x = 0$; always $\|x\| \geq 0$;
(c) (Cauchy–Schwarz) $|\langle x, y\rangle| \leq \|x\|\|y\|$;
(d) (triangle inequality) $\|x + y\| \leq \|x\| + \|y\|$.

Proof.

(a) $\|cx\|^2 = c\bar{c}\langle x, x\rangle$.

(b) Theorem 6.1(d).

(c) Trivial if $y = 0$. Otherwise, for $c = \langle x, y\rangle / \langle y, y\rangle$,

$$0 \leq \|x - cy\|^2 = \langle x, x\rangle - \bar{c}\langle x, y\rangle - c\langle y, x\rangle + c\bar{c}\langle y, y\rangle = \|x\|^2 - \frac{|\langle x, y\rangle|^2}{\|y\|^2}.$$

(d) $\|x + y\|^2 = \|x\|^2 + 2\,\mathrm{Re}\langle x, y\rangle + \|y\|^2 \leq \|x\|^2 + 2|\langle x, y\rangle| + \|y\|^2 \leq (\|x\| + \|y\|)^2$ by (c). ■

Example 7

On $\mathbb{F}^n$, (c) and (d) read

$$\left|\sum a_i\bar{b}_i\right| \le \left(\sum |a_i|^2\right)^{1/2}\left(\sum |b_i|^2\right)^{1/2}, \qquad \left(\sum |a_i+b_i|^2\right)^{1/2} \le \left(\sum |a_i|^2\right)^{1/2}+\left(\sum |b_i|^2\right)^{1/2}.$$

In $\mathbb{R}^2, \mathbb{R}^3$, $\langle x, y\rangle = \|x\|\|y\|\cos\theta$ (θ the angle).

Definition (Orthogonality)

x, y are **orthogonal** (perpendicular) if $\langle x, y\rangle = 0$. $S \subseteq V$ is **orthogonal** if distinct vectors of S are orthogonal, and **orthonormal** if moreover all vectors are **unit vectors** ($\|x\| = 1$). Replacing $x \neq 0$ by $x/\|x\|$ is **normalizing**.

Example 8

$\{(1,1,0),(1,-1,1),(-1,1,2)\}$ is orthogonal in $\mathbb{F}^3$; normalizing gives the orthonormal set $\{\frac{1}{\sqrt{2}}(1,1,0), \frac{1}{\sqrt{3}}(1,-1,1), \frac{1}{\sqrt{6}}(-1,1,2)\}$.

Example 9

In H, $f_n(t) = e^{int}$ ($n \in \mathbb{Z}$) form an orthonormal set: $\langle f_n, f_n\rangle = 1$ and, for $m \neq n$,

$$\langle f_m, f_n\rangle = \frac{1}{2\pi}\int_0^{2\pi} e^{i(m-n)t}\,dt = \frac{1}{2\pi i(m-n)} e^{i(m-n)t}\Big|_0^{2\pi} = 0.$$

6.2 The Gram–Schmidt orthogonalization process and orthogonal complements

Motivation. Gram–Schmidt turns any basis into an orthonormal one, in which coordinates are inner products and projections give closest points.

Definition (Orthonormal basis)

An **orthonormal basis** is an ordered basis that is orthonormal.

Example 1

The standard ordered basis of $\mathbb{F}^n$ is orthonormal.

Example 2

$\{\frac{1}{\sqrt{5}}(1,2), \frac{1}{\sqrt{5}}(2,-1)\}$ is an orthonormal basis of $\mathbb{R}^2$.

Theorem 6.3

If $S = \{v_1, \ldots, v_k\}$ is orthogonal with nonzero vectors and $y \in \operatorname{span}(S)$, then $y = \sum_{i=1}^{k} \frac{\langle y, v_i\rangle}{\|v_i\|^2} v_i$.

Proof. If $y=\sum a_i v_i$, then $\langle y,v_j\rangle=\sum a_i\langle v_i,v_j\rangle=a_j\|v_j\|^2$. ∎

Corollary 1

If S is orthonormal and $y\in\operatorname{span}(S)$, then $y=\sum\langle y,v_i\rangle v_i$.

Proof. $\|v_i\|=1$ in Theorem 6.3. ∎

Corollary 2

An orthogonal set of nonzero vectors is linearly independent.

Proof. If $\sum a_i v_i=0$, Theorem 6.3 with $y=0$ gives $a_i=\langle 0,v_i\rangle/\|v_i\|^2=0$. ∎

Example 3

By Corollary 2 the orthonormal set of Example 8 of Section 6.1 is a basis of $\mathbb{R}^3$; for $x=(2,1,3)$ the coefficients are $\frac{3}{\sqrt{2}}$, $\frac{4}{\sqrt{3}}$, $\frac{5}{\sqrt{6}}$. By Example 9 of Section 6.1, H is infinite-dimensional.

Theorem 6.4 (Gram–Schmidt)

Let $S=\{w_1,\dots,w_n\}$ be linearly independent. Put $v_1=w_1$ and

$$v_k=w_k-\sum_{j=1}^{k-1}\frac{\langle w_k,v_j\rangle}{\|v_j\|^2}v_j\qquad(2\le k\le n).\tag{6.1}$$

Then $S'=\{v_1,\dots,v_n\}$ is an orthogonal set of nonzero vectors with $\operatorname{span}(S')=\operatorname{span}(S)$.

Proof. Induction on n: with $S_k=\{w_1,\dots,w_k\}$, $S'_k=\{v_1,\dots,v_k\}$, assume S'_{k-1} is orthogonal, of nonzero vectors, with $\operatorname{span}S'_{k-1}=\operatorname{span}S_{k-1}$. If $v_k=0$, then $w_k\in\operatorname{span}S_{k-1}$, contradicting independence. For $i<k$, by (6.1) and orthogonality,

$$\langle v_k,v_i\rangle=\langle w_k,v_i\rangle-\frac{\langle w_k,v_i\rangle}{\|v_i\|^2}\|v_i\|^2=0.$$

Finally $\operatorname{span}S'_k\subseteq\operatorname{span}S_k$ by (6.1), and both have dimension k (Corollary 2), so they are equal. ∎

Example 4

In $\mathbb{R}^4$, $w_1=(1,0,1,0)$, $w_2=(1,1,1,1)$, $w_3=(0,1,2,1)$: $v_1=w_1$, $v_2=w_2-\frac{2}{2}v_1=(0,1,0,1)$, $v_3=w_3-\frac{2}{2}v_1-\frac{2}{2}v_2=(-1,0,1,0)$. Normalizing gives the orthonormal basis $\frac{1}{\sqrt{2}}(1,0,1,0)$, $\frac{1}{\sqrt{2}}(0,1,0,1)$, $\frac{1}{\sqrt{2}}(-1,0,1,0)$ of $\operatorname{span}\{w_1,w_2,w_3\}$.

Example 5

On $\mathsf{P}(\mathbb{R})$ with $\langle f,g\rangle=\int_{-1}^{1}f(t)g(t)\,dt$, Gram–Schmidt applied to $1,x,x^2$ gives 1, x, $x^2-\frac{1}{3}$, and normalizing,

$$u_1=\tfrac{1}{\sqrt{2}},\qquad u_2=\sqrt{\tfrac{3}{2}}\,x,\qquad u_3=\sqrt{\tfrac{5}{8}}\,(3x^2-1).$$

Continuing yields (multiples of) the **Legendre polynomials**.

Theorem 6.5

A nonzero finite-dimensional inner product space V has an orthonormal basis $\beta = \{v_1, \dots, v_n\}$, and $x = \sum_{i=1}^{n} \langle x, v_i \rangle v_i$ for every $x \in V$.

Proof. Apply Theorem 6.4 to any basis and normalize (Corollary 2); the expansion is Corollary 1 of Theorem 6.3. ∎

Example 6

For $f = 1 + 2x + 3x^2$ in the basis of Example 5: $\langle f, u_1 \rangle = 2\sqrt{2}$, $\langle f, u_2 \rangle = \frac{2\sqrt{6}}{3}$, $\langle f, u_3 \rangle = \frac{2\sqrt{10}}{5}$, so $f = 2\sqrt{2}u_1 + \frac{2\sqrt{6}}{3}u_2 + \frac{2\sqrt{10}}{5}u_3$.

Corollary

If $\beta = \{v_1, \dots, v_n\}$ is an orthonormal basis and $A = [\mathsf{T}]_\beta$, then $A_{ij} = \langle \mathsf{T}(v_j), v_i \rangle$.

Proof. $\mathsf{T}(v_j) = \sum_i \langle \mathsf{T}(v_j), v_i \rangle v_i$ by Theorem 6.5. ∎

Definition (Fourier coefficients)

For an orthonormal subset β (possibly infinite), the **Fourier coefficients** of x are the $\langle x, y \rangle$, $y \in \beta$.

Example 7

For $f(t) = t$ in H: $\langle f, f_n \rangle = \frac{-1}{in} = \frac{i}{n}$ $(n \neq 0)$, $\langle f, 1 \rangle = \pi$, $\|f\|^2 = \frac{4}{3}\pi^2$. Bessel's inequality ($\|x\|^2 \geq \sum |\langle x, v_i \rangle|^2$ for orthonormal $\{v_i\}$; from Theorem 6.6 below) gives

$$\tfrac{4}{3}\pi^2 \geq 2\sum_{n=1}^{k} \frac{1}{n^2} + \pi^2, \qquad \text{so} \qquad \sum_{n \geq 1} \frac{1}{n^2} \leq \frac{\pi^2}{6}$$

(in fact equality holds).

Definition (Orthogonal complement)

For nonempty $S \subseteq V$, $S^\perp = \{x \in V : \langle x, y \rangle = 0 \text{ for all } y \in S\}$, a subspace.

Example 8

$\{0\}^\perp = V$ and $V^\perp = \{0\}$.

Example 9

In $\mathbb{R}^3$, $\{e_3\}^\perp$ is the xy-plane.

Theorem 6.6

Let W be a finite-dimensional subspace of V and $y \in V$. There are unique $u \in W$, $z \in W^\perp$ with $y = u + z$. If $\{v_1, \dots, v_k\}$ is an orthonormal basis of W, then $u = \sum_{i=1}^{k} \langle y, v_i \rangle v_i$.

Proof. Put $u = \sum \langle y, v_i \rangle v_i$, $z = y - u$. Then $\langle z, v_j \rangle = \langle y, v_j \rangle - \langle y, v_j \rangle = 0$, so $z \in W^\perp$. If also $y = u' + z'$, then $u - u' = z' - z \in W \cap W^\perp = \{0\}$ (a vector orthogonal to itself is 0). ■

Corollary

u is the unique vector of W closest to y: $\|y - x\| \geq \|y - u\|$ for $x \in W$, with equality iff $x = u$.

Proof. $y - x = z + (u - x)$ with $u - x \in W$ orthogonal to z, so $\|y - x\|^2 = \|z\|^2 + \|u - x\|^2 \geq \|z\|^2 = \|y - u\|^2$ (Pythagoras: $\|a + b\|^2 = \|a\|^2 + \|b\|^2$ if $\langle a, b \rangle = 0$), with equality iff $u = x$. ■

u is the **orthogonal projection** of y on W.

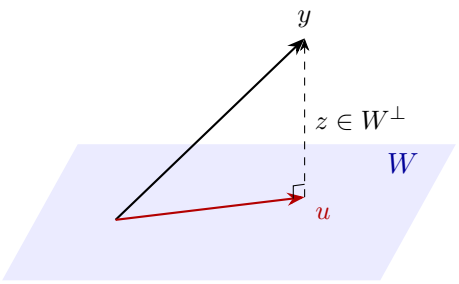


Figure. *Orthogonal projection $y = u + z$, $u \in W$, $z \perp W$.*

Example 10

In $\mathsf{P}_3(\mathbb{R})$ with $\langle f, g \rangle = \int_{-1}^{1} fg$, the orthogonal projection of x^3 on $\mathsf{P}_2(\mathbb{R})$ is $\sum \langle x^3, u_i \rangle u_i = \frac{\sqrt{6}}{5} \cdot \sqrt{\frac{3}{2}} x = \frac{3}{5} x$ ($\langle x^3, u_1 \rangle = \langle x^3, u_3 \rangle = 0$).

Theorem 6.7

Let $S = \{v_1, \dots, v_k\}$ be orthonormal in n-dimensional V.
(a) S extends to an orthonormal basis $\{v_1, \dots, v_n\}$ of V.
(b) If $W = \operatorname{span}(S)$, then $\{v_{k+1}, \dots, v_n\}$ is an orthonormal basis of $W^\perp$.
(c) For any subspace W, $\dim V = \dim W + \dim W^\perp$.

Proof.

(a) Extend S to a basis $\{v_1, \dots, v_k, w_{k+1}, \dots, w_n\}$, apply Gram–Schmidt (the orthonormal v_i are unchanged) and normalize.

(b) $S_1 = \{v_{k+1}, \dots, v_n\} \subseteq W^\perp$ is orthonormal, hence independent. If $x \in W^\perp$, then $x = \sum_{i=1}^{n} \langle x, v_i \rangle v_i = \sum_{i>k} \langle x, v_i \rangle v_i$.

(c) Apply (a), (b) to an orthonormal basis of W. ■

Example 11

For $W = \operatorname{span}\{e_1, e_2\} \subseteq \mathbb{F}^3$, $(a, b, c) \in W^\perp$ iff $a = b = 0$, so $W^\perp = \operatorname{span}\{e_3\}$.

6.3 The adjoint of a linear operator

Motivation. Writing functionals as $\langle \cdot, y\rangle$ yields the adjoint, $\langle \mathsf{T}x, y\rangle = \langle x, \mathsf{T}^* y\rangle$, with least squares and minimal solutions as first applications.

Theorem 6.8

If V is finite-dimensional and $g : V \to \mathbb{F}$ is linear, there is a unique $y \in V$ with $g(x) = \langle x, y\rangle$ for all x.

Proof. With an orthonormal basis $\{v_1, \dots, v_n\}$ put $y = \sum_i \overline{g(v_i)} v_i$. Then $h(x) = \langle x, y\rangle$ is linear with $h(v_j) = \langle v_j, y\rangle = g(v_j)$, so $h = g$ (Corollary to Theorem 2.6). Uniqueness: Theorem 6.1(e). ■

Example 1

$g(a_1, a_2) = 2a_1 + a_2$ on $\mathbb{R}^2$: $y = 2e_1 + e_2 = (2, 1)$.

Theorem 6.9

For a linear operator T on finite-dimensional V there is a unique $\mathsf{T}^* : V \to V$ with $\langle \mathsf{T}(x), y\rangle = \langle x, \mathsf{T}^*(y)\rangle$ for all x, y; T^* is linear.

Proof. For fixed y, $g(x) = \langle \mathsf{T}(x), y\rangle$ is linear, so $g(x) = \langle x, y'\rangle$ for a unique y' (Theorem 6.8); set $\mathsf{T}^*(y) = y'$. Linearity:

$$\langle x, \mathsf{T}^*(cy_1 + y_2)\rangle = \langle \mathsf{T}x, cy_1 + y_2\rangle = \bar{c}\langle \mathsf{T}x, y_1\rangle + \langle \mathsf{T}x, y_2\rangle = \langle x, c\mathsf{T}^* y_1 + \mathsf{T}^* y_2\rangle,$$

so $\mathsf{T}^*(cy_1 + y_2) = c\mathsf{T}^*(y_1) + \mathsf{T}^*(y_2)$ by Theorem 6.1(e), which also gives uniqueness. ■

T^* is the **adjoint** of T; also $\langle x, \mathsf{T}(y)\rangle = \langle \mathsf{T}^*(x), y\rangle$. (In infinite dimensions an adjoint need not exist.)

Theorem 6.10

If β is an orthonormal basis of finite-dimensional V, then $[\mathsf{T}^*]_\beta = ([\mathsf{T}]_\beta)^*$.

Proof. By the Corollary to Theorem 6.5,

$$([\mathsf{T}^*]_\beta)_{ij} = \langle \mathsf{T}^*(v_j), v_i\rangle = \overline{\langle v_i, \mathsf{T}^*(v_j)\rangle} = \overline{\langle \mathsf{T}(v_i), v_j\rangle} = \overline{([\mathsf{T}]_\beta)_{ji}}.$$

■

Corollary

$\mathsf{L}_{A^*} = (\mathsf{L}_A)^*$ for $A \in \mathrm{M}_{n\times n}(\mathbb{F})$.

Proof. With the standard (orthonormal) basis β, $[(\mathsf{L}_A)^*]_\beta = A^* = [\mathsf{L}_{A^*}]_\beta$. ■

Example 2

$\mathsf{T}(a_1, a_2) = (2ia_1 + 3a_2, a_1 - a_2)$ on $\mathbb{C}^2$: $[\mathsf{T}] = \left(\begin{smallmatrix} 2i & 3 \\ 1 & -1 \end{smallmatrix}\right)$, so $[\mathsf{T}^*] = \left(\begin{smallmatrix} -2i & 1 \\ 3 & -1 \end{smallmatrix}\right)$ and $\mathsf{T}^*(a_1, a_2) = (-2ia_1 + a_2, 3a_1 - a_2)$.

Theorem 6.11

For linear operators T, U on V having adjoints:
(a) $(\mathsf{T}+\mathsf{U})^* = \mathsf{T}^* + \mathsf{U}^*$;
(b) $(c\mathsf{T})^* = \bar{c}\mathsf{T}^*$;
(c) $(\mathsf{TU})^* = \mathsf{U}^*\mathsf{T}^*$;
(d) $\mathsf{T}^{**} = \mathsf{T}$;
(e) $\mathsf{I}^* = \mathsf{I}$.

Proof. By uniqueness in Theorem 6.9 and Theorem 6.1(e):

(a) $\langle(\mathsf{T}+\mathsf{U})x, y\rangle = \langle x, \mathsf{T}^*y + \mathsf{U}^*y\rangle$.

(b) $\langle c\mathsf{T}x, y\rangle = \langle x, \bar{c}\mathsf{T}^*y\rangle$.

(c) $\langle \mathsf{TU}x, y\rangle = \langle \mathsf{U}x, \mathsf{T}^*y\rangle = \langle x, \mathsf{U}^*\mathsf{T}^*y\rangle$.

(d) $\langle \mathsf{T}^*x, y\rangle = \overline{\langle y, \mathsf{T}^*x\rangle} = \overline{\langle \mathsf{T}y, x\rangle} = \langle x, \mathsf{T}y\rangle$.

(e) Clear. ■

Corollary

For $n \times n$ matrices: $(A+B)^* = A^* + B^*$, $(cA)^* = \bar{c}A^*$, $(AB)^* = B^*A^*$, $A^{**} = A$, $I^* = I$.

Proof. Theorem 6.11 for $\mathsf{L}_A, \mathsf{L}_B$ and the Corollary to Theorem 6.10 (or compute directly, also for rectangular matrices). ■

6.3.1 Least squares approximation

Given data $(t_1, y_1), \ldots, (t_m, y_m)$, the **least squares line** $y = ct+d$ minimises $E = \sum_i (y_i - ct_i - d)^2 = \|y - Ax\|^2$ with $A = \begin{pmatrix} t_1 & 1 \\ \vdots & \vdots \\ t_m & 1 \end{pmatrix}$, $x = \binom{c}{d}$. Write $\langle\cdot,\cdot\rangle_n$ for the standard inner product on $\mathbb{F}^n$.

Lemma 1

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $x \in \mathbb{F}^n$, $y \in \mathbb{F}^m$: $\langle Ax, y\rangle_m = \langle x, A^*y\rangle_n$.

Proof. $\langle Ax, y\rangle_m = y^*Ax = (A^*y)^*x = \langle x, A^*y\rangle_n$, viewing 1×1 matrices as scalars. ■

Lemma 2

$\operatorname{rank}(A^*A) = \operatorname{rank}(A)$ for $A \in \mathrm{M}_{m\times n}(\mathbb{F})$.

Proof. If $A^*Ax = 0$, then $0 = \langle A^*Ax, x\rangle_n = \langle Ax, Ax\rangle_m$, so $Ax = 0$. Hence $\mathrm{N}(A^*A) = \mathrm{N}(A)$, and the dimension theorem gives equal ranks. ■

Corollary

If $\operatorname{rank} A = n$, then A^*A is invertible.

Proof. A^*A is $n \times n$ of rank n by Lemma 2. ■

Theorem 6.12

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $y \in \mathbb{F}^m$, there is $x_0 \in \mathbb{F}^n$ with $(A^*A)x_0 = A^*y$ and $\|Ax_0 - y\| \le \|Ax - y\|$ for all x. If $\operatorname{rank} A = n$, then $x_0 = (A^*A)^{-1}A^*y$.

Proof. Let $W = \mathrm{R}(\mathsf{L}_A)$. By the Corollary to Theorem 6.6 some $Ax_0 \in W$ is closest to y, with $Ax_0 - y \in W^\perp$. By Lemma 1,

$$\langle Ax, Ax_0 - y\rangle_m = 0 \ \forall x \iff \langle x, A^*(Ax_0 - y)\rangle_n = 0 \ \forall x \iff A^*Ax_0 = A^*y.$$

If $\operatorname{rank} A = n$, invert A^*A (Corollary to Lemma 2). ■

For data $(1,2),(2,3),(3,5),(4,7)$: $A^*A = \left(\begin{smallmatrix}30 & 10\\ 10 & 4\end{smallmatrix}\right)$, $A^*y = (51,17)^t$, so $x_0 = \frac{1}{20}\left(\begin{smallmatrix}4 & -10\\ -10 & 30\end{smallmatrix}\right)\left(\begin{smallmatrix}51\\ 17\end{smallmatrix}\right) = \left(\begin{smallmatrix}1.7\\ 0\end{smallmatrix}\right)$. The line is $y = 1.7t$, with $E = 0.3$. Polynomial fits use columns t_i^k.

6.3.2 Minimal solutions to systems of linear equations

A solution s of $Ax = b$ is **minimal** if $\|s\| \le \|u\|$ for every solution u.

Theorem 6.13

Let $Ax = b$ be consistent, $A \in \mathrm{M}_{m\times n}(\mathbb{F})$.
(a) There is exactly one minimal solution s, and $s \in \mathrm{R}(\mathsf{L}_{A^*})$.
(b) s is the only solution in $\mathrm{R}(\mathsf{L}_{A^*})$: if $(AA^*)u = b$, then $s = A^*u$.

Proof. Let $W = \mathrm{R}(\mathsf{L}_{A^*})$ and $W' = \mathrm{N}(\mathsf{L}_A)$. By Lemma 1, $W^\perp = W'$: $x \perp A^*y$ for all y iff $\langle Ax, y\rangle = 0$ for all y, iff $Ax = 0$.

(a) For a solution u, write $u = s + z$, $s \in W$, $z \in W^\perp = W'$ (Theorem 6.6); then $As = Au = b$. Every solution is $s + z'$ with $z' \in W' = W^\perp$ (Theorem 3.9), and $\|s + z'\|^2 = \|s\|^2 + \|z'\|^2 \ge \|s\|^2$, with equality iff $z' = 0$.

(b) A solution $v \in W$ has $v - s \in W \cap W^\perp = \{0\}$. If $AA^*u = b$, then $A^*u \in W$ is a solution, so $A^*u = s$. ■

Example 3

$x + 2y + z = 4$, $x - y + 2z = -11$, $x + 5y = 19$: $AA^* = \left(\begin{smallmatrix}6 & 1 & 11\\ 1 & 6 & -4\\ 11 & -4 & 26\end{smallmatrix}\right)$, $u = (1,-2,0)$ solves $AA^*u = b$, and the minimal solution is $s = A^*u = (-1,4,-3)$.

6.4 Normal and self-adjoint operators

Motivation. An orthonormal eigenbasis exists iff T is normal (over $\mathbb{C}$) or self-adjoint (over $\mathbb{R}$), via Schur's triangularization.

Lemma

If T (on finite-dimensional V) has an eigenvector, so does T^*.

Proof. Let $\mathsf{T}v = \lambda v$, $v \ne 0$. For all x, $0 = \langle(\mathsf{T} - \lambda\mathsf{I})v, x\rangle = \langle v, (\mathsf{T}^* - \bar\lambda\mathsf{I})x\rangle$, so $v \perp \mathrm{R}(\mathsf{T}^* - \bar\lambda\mathsf{I}) \ne V$. Thus $\mathsf{T}^* - \bar\lambda\mathsf{I}$ is not onto, hence not one-to-one: $\bar\lambda$ is an eigenvalue of T^*. ■

Theorem 6.14 (Schur)

If the characteristic polynomial of T (on finite-dimensional inner product space V) splits, there is an orthonormal basis β with $[\mathsf{T}]_\beta$ upper triangular.

Proof. Induction on $n = \dim V$; $n = 1$ is clear. T has an eigenvector (splitting), so by the Lemma $\mathsf{T}^* z = \lambda z$ for a unit z. $W = \{z\}^\perp$ ($\dim n - 1$) is T-invariant: for $y \in W$, $\langle \mathsf{T}y, z\rangle = \langle y, \mathsf{T}^* z\rangle = \bar{\lambda}\langle y, z\rangle = 0$. The characteristic polynomial of T_W divides that of T (Theorem 5.21), so splits, and by induction $[\mathsf{T}_W]_\gamma$ is upper triangular for an orthonormal basis γ of W. Then $\beta = \gamma \cup \{z\}$ is orthonormal and $[\mathsf{T}]_\beta$ is upper triangular (first $n-1$ columns: $[\mathsf{T}_W]_\gamma$ padded by 0). ■

Definition (Normal)

T is **normal** if $\mathsf{T}\mathsf{T}^* = \mathsf{T}^*\mathsf{T}$; A is normal if $AA^* = A^*A$. For orthonormal β, T is normal iff $[\mathsf{T}]_\beta$ is (Theorem 6.10).

Example 1

The rotation T of $\mathbb{R}^2$ by $\theta \in (0, \pi)$, $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, has $AA^* = I = A^*A$: normal, but without eigenvectors, so normality does not suffice over $\mathbb{R}$.

Example 2

A real **skew-symmetric** A ($A^t = -A$) is normal: $AA^t = -A^2 = A^tA$.

Theorem 6.15

Let T be normal on V. Then
(a) $\|\mathsf{T}(x)\| = \|\mathsf{T}^*(x)\|$ for all x;
(b) $\mathsf{T} - c\mathsf{I}$ is normal for every $c \in \mathbb{F}$;
(c) if $\mathsf{T}(x) = \lambda x$, then $\mathsf{T}^*(x) = \bar{\lambda}x$;
(d) eigenvectors for distinct eigenvalues are orthogonal.

Proof.

(a) $\|\mathsf{T}x\|^2 = \langle \mathsf{T}^*\mathsf{T}x, x\rangle = \langle \mathsf{T}\mathsf{T}^*x, x\rangle = \|\mathsf{T}^*x\|^2$.

(b) $(\mathsf{T} - c\mathsf{I})^* = \mathsf{T}^* - \bar{c}\mathsf{I}$ commutes with $\mathsf{T} - c\mathsf{I}$.

(c) Apply (a) to the normal $\mathsf{U} = \mathsf{T} - \lambda\mathsf{I}$: $0 = \|\mathsf{U}x\| = \|\mathsf{U}^*x\| = \|(\mathsf{T}^* - \bar{\lambda}\mathsf{I})x\|$.

(d) If $\mathsf{T}x_1 = \lambda_1 x_1$, $\mathsf{T}x_2 = \lambda_2 x_2$, $\lambda_1 \neq \lambda_2$, then by (c)

$$\lambda_1\langle x_1, x_2\rangle = \langle \mathsf{T}x_1, x_2\rangle = \langle x_1, \mathsf{T}^*x_2\rangle = \langle x_1, \bar{\lambda}_2 x_2\rangle = \lambda_2\langle x_1, x_2\rangle,$$

so $\langle x_1, x_2\rangle = 0$. ■

Theorem 6.16

On a finite-dimensional complex inner product space, T is normal iff V has an orthonormal basis of eigenvectors of T.

Proof.

Step 1 (only if). Over $\mathbb{C}$ the characteristic polynomial splits, so Schur gives an orthonormal $\beta = \{v_1, \dots, v_n\}$ with $A = [\mathsf{T}]_\beta$ upper triangular. By induction on k, v_k is an eigenvector: v_1 is (A upper triangular), and if $\mathsf{T}v_j = \lambda_j v_j$ for $j < k$, then by Theorem 6.15(c)

$$A_{jk} = \langle \mathsf{T}v_k, v_j \rangle = \langle v_k, \mathsf{T}^* v_j \rangle = \langle v_k, \bar{\lambda}_j v_j \rangle = 0 \quad (j < k),$$

so $\mathsf{T}v_k = A_{kk} v_k$.

Step 2 (if). If β is an orthonormal eigenbasis, $[\mathsf{T}]_\beta = D$ is diagonal, and $[\mathsf{T}^*]_\beta = D^*$ is diagonal too; diagonal matrices commute. ■

Example 3

In $V = \operatorname{span}\{f_n : n \in \mathbb{Z}\} \subseteq \mathsf{H}$, $\mathsf{T}(f) = f_1 f$ and $\mathsf{U}(f) = f_{-1} f$ satisfy $\mathsf{T}(f_n) = f_{n+1}$, $\mathsf{U}(f_n) = f_{n-1}$, $\mathsf{U} = \mathsf{T}^*$, $\mathsf{T}\mathsf{T}^* = \mathsf{I} = \mathsf{T}^*\mathsf{T}$. So T is normal without eigenvectors: $\mathsf{T}f = \lambda f$ with $f = \sum_{i=n}^{m} a_i f_i$, $a_m \neq 0$, would put $f_{m+1} \in \operatorname{span}\{f_n, \dots, f_m\}$. Theorem 6.16 fails in infinite dimensions.

Definition (Self-adjoint)

T is **self-adjoint** (**Hermitian**) if $\mathsf{T} = \mathsf{T}^*$; A is self-adjoint if $A = A^*$. For real matrices this means symmetric.

Lemma

Let T be self-adjoint on finite-dimensional V. (a) Every eigenvalue of T is real.
(b) If V is real, the characteristic polynomial of T splits.

Proof.

(a) T is normal, so $\mathsf{T}x = \lambda x$ gives $\lambda x = \mathsf{T}x = \mathsf{T}^* x = \bar{\lambda} x$ (Theorem 6.15(c)), so $\lambda = \bar{\lambda}$.

(b) $A = [\mathsf{T}]_\beta$ (β orthonormal) is real symmetric, so L_A on $\mathbb{C}^n$ is self-adjoint. Its characteristic polynomial, that of T, splits over $\mathbb{C}$, with real zeros by (a). ■

Theorem 6.17

On a finite-dimensional real inner product space, T is self-adjoint iff V has an orthonormal basis of eigenvectors of T.

Proof. If T is self-adjoint, its characteristic polynomial splits (Lemma), Schur gives orthonormal β with $A = [\mathsf{T}]_\beta$ upper triangular, and $A = A^* = [\mathsf{T}^*]_\beta$ is also lower triangular, hence diagonal. Conversely, if $[\mathsf{T}]_\beta = D$ is real diagonal for orthonormal β, then $[\mathsf{T}^*]_\beta = D^* = D$, so $\mathsf{T}^* = \mathsf{T}$. ■

Example 4

Complex symmetric matrices need not be normal: $A = \left(\begin{smallmatrix} i & i \\ i & 1 \end{smallmatrix}\right)$ has $(AA^*)_{12} = 1 + i \neq 1 - i = (A^*A)_{12}$.

6.5 Unitary and orthogonal operators and their matrices

Motivation. Length-preserving operators (unitary over $\mathbb{C}$, orthogonal over $\mathbb{R}$) preserve inner products and orthonormal bases and have matrices with $A^*A = I$.

Definition (Unitary, orthogonal operator)

A linear T on finite-dimensional V with $\|\mathsf{T}(x)\| = \|x\|$ for all x is **unitary** if $\mathbb{F} = \mathbb{C}$ and **orthogonal** if $\mathbb{F} = \mathbb{R}$. (In infinite dimensions: an isometry; unitary/orthogonal if also onto.)

Example 1

If $h \in \mathsf{H}$ has $|h(x)| = 1$ for all x, then $\mathsf{T}(f) = hf$ satisfies $\|\mathsf{T}f\|^2 = \frac{1}{2\pi}\int |h|^2|f|^2 = \|f\|^2$.

Lemma

If U is self-adjoint on finite-dimensional V and $\langle x, \mathsf{U}(x)\rangle = 0$ for all x, then $\mathsf{U} = \mathsf{T}_0$.

Proof. Choose an orthonormal eigenbasis β (Theorems 6.16, 6.17; U is normal, and self-adjoint over $\mathbb{R}$). For $x \in \beta$, $\mathsf{U}x = \lambda x$: $0 = \langle x, \lambda x\rangle = \bar{\lambda}$, so $\mathsf{U} = \mathsf{T}_0$ on β. ■

Theorem 6.18

For linear T on finite-dimensional V the following are equivalent:
(a) $\mathsf{T}\mathsf{T}^* = \mathsf{T}^*\mathsf{T} = \mathsf{I}$;
(b) $\langle \mathsf{T}(x), \mathsf{T}(y)\rangle = \langle x, y\rangle$ for all x, y;
(c) if β is an orthonormal basis, so is $\mathsf{T}(\beta)$;
(d) there is an orthonormal basis β with $\mathsf{T}(\beta)$ orthonormal;
(e) $\|\mathsf{T}(x)\| = \|x\|$ for all x.

Proof.

Step (a)⇒(b). $\langle \mathsf{T}x, \mathsf{T}y\rangle = \langle x, \mathsf{T}^*\mathsf{T}y\rangle = \langle x, y\rangle$.

Step (b)⇒(c). $\langle \mathsf{T}v_i, \mathsf{T}v_j\rangle = \langle v_i, v_j\rangle = \delta_{ij}$, so $\mathsf{T}(\beta)$ is orthonormal with n vectors, hence a basis (Corollary 2 of Theorem 6.3).

Step (c)⇒(d). Orthonormal bases exist (Theorem 6.5).

Step (d)⇒(e). For $x = \sum a_i v_i$, $\|\mathsf{T}x\|^2 = \|\sum a_i \mathsf{T}v_i\|^2 = \sum |a_i|^2 = \|x\|^2$.

Step (e)⇒(a). $\langle x, (\mathsf{I} - \mathsf{T}^*\mathsf{T})x\rangle = \|x\|^2 - \|\mathsf{T}x\|^2 = 0$ for all x with $\mathsf{I} - \mathsf{T}^*\mathsf{T}$ self-adjoint, so $\mathsf{T}^*\mathsf{T} = \mathsf{I}$ (Lemma); in finite dimension $\mathsf{T}^{-1} = \mathsf{T}^*$, so $\mathsf{T}\mathsf{T}^* = \mathsf{I}$. ■

Hence unitary/orthogonal operators preserve inner products, angles and distances, and their eigenvalues have $|\lambda| = 1$.

Corollary 1

On a finite-dimensional real inner product space, V has an orthonormal basis of eigenvectors of T with eigenvalues of absolute value 1 iff T is both self-adjoint and orthogonal.

Proof. Self-adjoint gives an orthonormal eigenbasis (Theorem 6.17), orthogonal gives $|\lambda| = 1$. Conversely, such a basis makes T self-adjoint (Theorem 6.17), and $\mathsf{T}(\beta) = \{\pm v_i\}$ is orthonormal, so T is orthogonal (Theorem 6.18(d)). ■

Corollary 2

On a finite-dimensional complex inner product space, V has an orthonormal basis of eigenvectors of T with eigenvalues of absolute value 1 iff T is unitary.

Proof. A unitary T is normal ($\mathsf{T}\mathsf{T}^* = \mathsf{I} = \mathsf{T}^*\mathsf{T}$), so Theorem 6.16 gives the basis, with $|\lambda| = 1$. Conversely $\mathsf{T}(\beta) = \{\lambda_i v_i\}$ is orthonormal; apply Theorem 6.18(d). ■

Example 2

The rotation T of $\mathbb{R}^2$ by θ is orthogonal; $[\mathsf{T}] = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and $[\mathsf{T}]^*[\mathsf{T}] = I$.

Definition (Reflection of $\mathbb{R}^2$)

For a line L through 0, T is the **reflection about** L if $\mathsf{T}(x) = x$ for $x \in L$ and $\mathsf{T}(x) = -x$ for $x \in L^\perp$.

Example 3

A reflection is orthogonal: with unit $v_1 \in L$, $v_2 \in L^\perp$, $\mathsf{T}(\{v_1, v_2\}) = \{v_1, -v_2\}$ is orthonormal (Theorem 6.18(d)).

Definition (Orthogonal and unitary matrices)

A square A is **orthogonal** if $A^tA = AA^t = I$ and **unitary** if $A^*A = AA^* = I$. Equivalently, the rows (or columns) form an orthonormal basis of $\mathbb{F}^n$. T is unitary [orthogonal] iff $[\mathsf{T}]_\beta$ is, for orthonormal β.

Example 4

$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ is orthogonal; its rows are orthonormal.

Example 5

For a reflection $\mathsf{T} = \mathsf{L}_A$ about the line L at angle α to the x-axis, $A = Q\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)Q^{-1}$ with $Q = \left(\begin{smallmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{smallmatrix}\right)$, so $A = \left(\begin{smallmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{smallmatrix}\right)$, an orthogonal matrix.

Definition (Unitary/orthogonal equivalence)

A is **unitarily** [**orthogonally**] **equivalent** to D if $A = P^*DP$ for a unitary [orthogonal] P.

Theorem 6.19

A complex $n \times n$ matrix A is normal iff it is unitarily equivalent to a diagonal matrix.

Proof. If A is normal, Theorem 6.16 gives an orthonormal eigenbasis of L_A; the unitary Q with these columns gives a diagonal $D = Q^{-1}AQ = Q^*AQ$. Conversely, if $A = P^*DP$, then $AA^* = P^*DD^*P = P^*D^*DP = A^*A$. ■

Theorem 6.20

A real $n \times n$ matrix A is symmetric iff it is orthogonally equivalent to a real diagonal matrix.

Proof. As in Theorem 6.19, using Theorem 6.17. Conversely $(P^tDP)^t = P^tDP$. ■

Example 6

$A = \left(\begin{smallmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{smallmatrix}\right)$ has eigenvalues $2, 2, 8$. Gram–Schmidt on the eigenvectors $(-1,1,0), (-1,0,1)$ of 2, and normalizing $(1,1,1)$, gives

$$P = \begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}, \qquad P^t A P = \operatorname{diag}(2,2,8).$$

Theorem 6.21 (Schur, matrix form)

Let the characteristic polynomial of $A \in \mathrm{M}_{n\times n}(\mathbb{F})$ split over $\mathbb{F}$. (a) If $\mathbb{F} = \mathbb{C}$, A is unitarily equivalent to a complex upper triangular matrix.
(b) If $\mathbb{F} = \mathbb{R}$, A is orthogonally equivalent to a real upper triangular matrix.

Proof. Apply Theorem 6.14 to L_A; the change of coordinate matrix Q from the orthonormal β to the standard basis is unitary [orthogonal], and $Q^* A Q = [\mathsf{L}_A]_\beta$ is upper triangular. ∎

6.5.1 Rigid motions

Definition (Rigid motion; translation)

On a real inner product space V, $f : V \to V$ is a **rigid motion** if $\|f(x) - f(y)\| = \|x - y\|$ for all x, y. A **translation** by v_0 is $g(x) = x + v_0$. Orthogonal operators and translations are rigid motions, and so are their composites.

Theorem 6.22

Every rigid motion f of a finite-dimensional real inner product space is $f = g \circ \mathsf{T}$ for a unique orthogonal operator T and a unique translation g.

Proof.

Step 1 (T preserves norms and inner products). Let $\mathsf{T}(x) = f(x) - f(0)$, a rigid motion with $\mathsf{T}(0) = 0$, so $\|\mathsf{T}x\| = \|x\|$. Expanding $\|\mathsf{T}x - \mathsf{T}y\|^2 = \|x - y\|^2$ gives $\langle \mathsf{T}x, \mathsf{T}y\rangle = \langle x, y\rangle$.

Step 2 (T is linear). For $x, y \in V$, $a \in \mathbb{R}$,

$$\begin{aligned} \|\mathsf{T}(x + ay) - \mathsf{T}x - a\mathsf{T}y\|^2 &= \|\mathsf{T}(x+ay) - \mathsf{T}x\|^2 + a^2\|\mathsf{T}y\|^2 - 2a\langle \mathsf{T}(x+ay) - \mathsf{T}x, \mathsf{T}y\rangle \\ &= a^2\|y\|^2 + a^2\|y\|^2 - 2a(\langle x + ay, y\rangle - \langle x, y\rangle) = 0, \end{aligned}$$

by Step 1. So T is linear and orthogonal, and $f = g \circ \mathsf{T}$ with g the translation by $f(0)$.

Step 3 (uniqueness). If $\mathsf{T}x + u_0 = \mathsf{U}x + v_0$ for all x, then $x = 0$ gives $u_0 = v_0$, and then $\mathsf{T} = \mathsf{U}$. ∎

Theorem 6.23

For an orthogonal operator T on $\mathbb{R}^2$ with $A = [\mathsf{T}]_\beta$ (β standard), exactly one holds:
(a) T is a rotation and $\det A = 1$;
(b) T is a reflection about a line through 0 and $\det A = -1$.

Proof. $\mathsf{T}(e_1) = (\cos\theta, \sin\theta)$ for a unique $\theta \in [0, 2\pi)$, and the unit vector $\mathsf{T}(e_2) \perp \mathsf{T}(e_1)$ is $(-\sin\theta, \cos\theta)$ or $(\sin\theta, -\cos\theta)$. In the first case A is the rotation matrix, $\det A = 1$; in the

second $A = \left(\begin{smallmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{smallmatrix}\right)$, $\det A = -1$, the reflection about the line at angle $\theta/2$ (Example 5). The determinants differ, so exactly one case occurs. ■

Corollary

Every rigid motion of $\mathbb{R}^2$ is a rotation followed by a translation, or a reflection about a line through 0 followed by a translation.

Proof. Theorems 6.22 and 6.23. ■

Example 7

$A = \frac{1}{\sqrt{5}}\left(\begin{smallmatrix} 1 & 2 \\ 2 & -1 \end{smallmatrix}\right)$ is orthogonal with $\det A = -\frac{1}{5} - \frac{4}{5} = -1$, so L_A is a reflection; its axis is $E_1 = \operatorname{span}\{(1+\sqrt{5}, 2)\}$, the line $y = \frac{\sqrt{5}-1}{2}x$.

6.5.2 Conic sections

A quadratic equation $ax^2 + 2bxy + cy^2 + dx + ey + f = 0$ has quadratic part X^tAX with $X = \binom{x}{y}$, $A = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$. By Theorem 6.20, $P^tAP = \operatorname{diag}(\lambda_1, \lambda_2)$ for an orthogonal P, with $\det P = 1$ (a rotation) after replacing P by $P \cdot \operatorname{diag}(1, -1)$ if needed. With $X = PX'$, $X^tAX = \lambda_1(x')^2 + \lambda_2(y')^2$: rotating the axes removes the xy-term. E.g. $2x^2 - 4xy + 5y^2 - 36 = 0$: $A = \left(\begin{smallmatrix} 2 & -2 \\ -2 & 5 \end{smallmatrix}\right)$, eigenvalues $1, 6$, $P = \frac{1}{\sqrt{5}}\left(\begin{smallmatrix} 2 & -1 \\ 1 & 2 \end{smallmatrix}\right)$, giving the ellipse $(x')^2 + 6(y')^2 = 36$.

6.6 Orthogonal projections and the spectral theorem

Motivation. A normal (complex) or self-adjoint (real) operator is $\mathsf{T} = \sum \lambda_i \mathsf{T}_i$, with T_i the orthogonal projections onto its mutually orthogonal eigenspaces.

If $V = W_1 \oplus W_2$, the **projection on** W_1 **along** W_2 is $\mathsf{T}(x_1 + x_2) = x_1$; then $\mathrm{R}(\mathsf{T}) = W_1$, $\mathrm{N}(\mathsf{T}) = W_2$, and T is a projection iff $\mathsf{T}^2 = \mathsf{T}$.

Definition (Orthogonal projection)

A projection T on an inner product space is **orthogonal** if $\mathrm{R}(\mathsf{T})^\perp = \mathrm{N}(\mathsf{T})$ and $\mathrm{N}(\mathsf{T})^\perp = \mathrm{R}(\mathsf{T})$ (in finite dimensions one of these suffices). For finite-dimensional W, the orthogonal projection on W is $x \mapsto u$ of Theorem 6.6.

Theorem 6.24

T is an orthogonal projection iff T has an adjoint and $\mathsf{T}^2 = \mathsf{T} = \mathsf{T}^*$.

Proof.

Step 1 (only if). $\mathsf{T}^2 = \mathsf{T}$ as T is a projection. Write $x = x_1 + x_2$, $y = y_1 + y_2$ with $x_1, y_1 \in \mathrm{R}\,\mathsf{T}$, $x_2, y_2 \in \mathrm{N}\,\mathsf{T}$ (mutually orthogonal). Then

$$\langle x, \mathsf{T}y \rangle = \langle x_1 + x_2, y_1 \rangle = \langle x_1, y_1 \rangle = \langle x_1, y_1 + y_2 \rangle = \langle \mathsf{T}x, y \rangle,$$

so T^* exists and equals T.

Step 2 (if). $\mathsf{T}^2 = \mathsf{T}$ makes T the projection on $\mathrm{R}\,\mathsf{T}$ along $\mathrm{N}\,\mathsf{T}$ ($x = \mathsf{T}x + (x - \mathsf{T}x)$). For $x \in \mathrm{R}\,\mathsf{T}$, $y \in \mathrm{N}\,\mathsf{T}$: $\langle x, y\rangle = \langle \mathsf{T}x, y\rangle = \langle x, \mathsf{T}y\rangle = 0$, so $\mathrm{R}\,\mathsf{T} \subseteq \mathrm{N}(\mathsf{T})^\perp$. For $y \in \mathrm{N}(\mathsf{T})^\perp$:

$$\|y - \mathsf{T}y\|^2 = \langle y - \mathsf{T}y, y\rangle - \langle y - \mathsf{T}y, \mathsf{T}y\rangle = 0 - \langle \mathsf{T}(y - \mathsf{T}y), y\rangle = 0,$$

so $y \in \mathrm{R}\,\mathsf{T}$. Hence $\mathrm{N}(\mathsf{T})^\perp = \mathrm{R}\,\mathsf{T}$; the other equality is analogous. ■

Theorem 6.25 (Spectral theorem)

Let T on finite-dimensional V be normal ($\mathbb{F} = \mathbb{C}$) or self-adjoint ($\mathbb{F} = \mathbb{R}$), with distinct eigenvalues $\lambda_1, \ldots, \lambda_k$, eigenspaces W_i, and T_i the orthogonal projection on W_i. Then
(a) $V = W_1 \oplus \cdots \oplus W_k$;
(b) $W_i^\perp = W_i' = \bigoplus_{j \neq i} W_j$;
(c) $\mathsf{T}_i \mathsf{T}_j = \delta_{ij} \mathsf{T}_i$;
(d) $\mathsf{I} = \mathsf{T}_1 + \cdots + \mathsf{T}_k$;
(e) $\mathsf{T} = \lambda_1 \mathsf{T}_1 + \cdots + \lambda_k \mathsf{T}_k$.

Proof.

(a) T is diagonalizable (Theorems 6.16, 6.17); apply Theorem 5.11.

(b) Eigenspaces are mutually orthogonal (Theorem 6.15(d)), so $W_i' \subseteq W_i^\perp$; both have dimension $\dim V - \dim W_i$ (by (a) and Theorem 6.7(c)).

(c) For $i \neq j$, $\mathrm{R}\,\mathsf{T}_j = W_j \subseteq W_i^\perp = \mathrm{N}\,\mathsf{T}_i$, so $\mathsf{T}_i \mathsf{T}_j = \mathsf{T}_0$; and $\mathsf{T}_i^2 = \mathsf{T}_i$.

(d) For $x = \sum x_j$, $x_j \in W_j$, $\mathsf{T}_i x = x_i$ by (b), so $\sum \mathsf{T}_i x = x$.

(e) $\mathsf{T}x = \sum \mathsf{T}x_i = \sum \lambda_i x_i = \sum \lambda_i \mathsf{T}_i x$. ■

$\{\lambda_i\}$ is the **spectrum**, (d) the **resolution of the identity**, and (e) the **spectral decomposition** of T. Note $g(\mathsf{T}) = \sum g(\lambda_i)\mathsf{T}_i$ for any polynomial g, by (c).

Corollary 1

If $\mathbb{F} = \mathbb{C}$, T is normal iff $\mathsf{T}^* = g(\mathsf{T})$ for some polynomial g.

Proof. If T is normal, $\mathsf{T}^* = \sum \bar{\lambda}_i \mathsf{T}_i$ (each T_i is self-adjoint by Theorem 6.24). Lagrange interpolation gives g with $g(\lambda_i) = \bar{\lambda}_i$, so $g(\mathsf{T}) = \sum \bar{\lambda}_i \mathsf{T}_i = \mathsf{T}^*$. Conversely $g(\mathsf{T})$ commutes with T. ■

Corollary 2

If $\mathbb{F} = \mathbb{C}$, T is unitary iff T is normal and $|\lambda| = 1$ for every eigenvalue.

Proof. Unitary operators are normal with $|\lambda| = 1$. Conversely $\mathsf{T}\mathsf{T}^* = \sum \lambda_i \bar{\lambda}_i \mathsf{T}_i = \sum \mathsf{T}_i = \mathsf{I}$ by (c), (d). ■

Corollary 3

If $\mathbb{F} = \mathbb{C}$ and T is normal, T is self-adjoint iff every eigenvalue is real.

Proof. $\mathsf{T}^* = \sum \bar{\lambda}_i \mathsf{T}_i$ equals $\mathsf{T} = \sum \lambda_i \mathsf{T}_i$ iff $\bar{\lambda}_i = \lambda_i$ for all i (apply both sides to W_j). ■

Corollary 4

In the spectral decomposition $\mathsf{T} = \sum \lambda_i \mathsf{T}_i$, each T_j is a polynomial in T.

Proof. Choose g_j (Lagrange) with $g_j(\lambda_i) = \delta_{ij}$; then $g_j(\mathsf{T}) = \sum g_j(\lambda_i)\mathsf{T}_i = \mathsf{T}_j$. ∎

6.7 The singular value decomposition and the pseudoinverse *

Motivation. Every linear map sends some orthonormal basis to an orthogonal set, $\mathsf{T}v_i = \sigma_i u_i$, which yields the polar decomposition and the pseudoinverse.

Theorem 6.26 (Singular value theorem for linear transformations)

Let $\mathsf{T} : V \to W$ (finite-dimensional inner product spaces) have rank r. There are orthonormal bases $\{v_1, \dots, v_n\}$ of V, $\{u_1, \dots, u_m\}$ of W and $\sigma_1 \geq \cdots \geq \sigma_r > 0$ with $\mathsf{T}(v_i) = \sigma_i u_i$ for $i \leq r$ and $\mathsf{T}(v_i) = 0$ for $i > r$. Conversely, given such bases and scalars, the v_i are eigenvectors of $\mathsf{T}^*\mathsf{T}$ with eigenvalues σ_i^2 ($i \leq r$) and 0 ($i > r$); hence the σ_i are uniquely determined by T.

Proof.

Step 1 (existence). $\mathsf{T}^* : W \to V$ exists (as in Theorem 6.9); $\mathsf{T}^*\mathsf{T}$ is self-adjoint of rank r (Lemma 2 of Section 6.3). Take an orthonormal eigenbasis $\{v_i\}$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_r > 0 = \lambda_{r+1} = \dots$, all ≥ 0 as $\lambda_i = \langle \mathsf{T}^*\mathsf{T}v_i, v_i\rangle = \|\mathsf{T}v_i\|^2$. Put $\sigma_i = \sqrt{\lambda_i}$, $u_i = \frac{1}{\sigma_i}\mathsf{T}(v_i)$ ($i \leq r$). Then

$$\langle u_i, u_j\rangle = \frac{1}{\sigma_i\sigma_j}\langle \mathsf{T}^*\mathsf{T}v_i, v_j\rangle = \frac{\lambda_i}{\sigma_i\sigma_j}\delta_{ij} = \delta_{ij};$$

extend to an orthonormal basis of W. For $i > r$, $\|\mathsf{T}v_i\|^2 = \lambda_i = 0$.

Step 2 (converse). For $i \leq r$ (and with 0 for $i > r$),

$$\mathsf{T}^*(u_i) = \sum_j \langle \mathsf{T}^*u_i, v_j\rangle v_j = \sum_j \langle u_i, \mathsf{T}v_j\rangle v_j = \sigma_i v_i,$$

so $\mathsf{T}^*\mathsf{T}v_i = \sigma_i^2 v_i$ ($i \leq r$) and $\mathsf{T}^*\mathsf{T}v_i = 0$ ($i > r$). ∎

The σ_i are the **singular values** of T (extended by zeros up to $\min(m, n)$).

Example 1

$\mathsf{T}(f) = f'$ from $\mathsf{P}_2(\mathbb{R})$ to $\mathsf{P}_1(\mathbb{R})$, $\langle f, g\rangle = \int_{-1}^{1} fg$. In the orthonormal bases $\alpha = \{\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}x, \sqrt{\frac{5}{8}}(3x^2 - 1)\}$ and $\alpha' = \{\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}x\}$, $A = [\mathsf{T}] = \begin{pmatrix} 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{15} \end{pmatrix}$ and $A^*A = \operatorname{diag}(0, 3, 15)$. So $\sigma_1 = \sqrt{15}$, $\sigma_2 = \sqrt{3}$, with $v_1 = \sqrt{\frac{5}{8}}(3x^2 - 1)$, $v_2 = \sqrt{\frac{3}{2}}x$, $v_3 = \frac{1}{\sqrt{2}}$, $u_1 = \sqrt{\frac{3}{2}}x$, $u_2 = \frac{1}{\sqrt{2}}$.

Example 2

For invertible T on $\mathbb{R}^2$ with $\mathsf{T}v_i = \sigma_i u_i$, the unit circle $\{x_1v_1 + x_2v_2 : x_1^2 + x_2^2 = 1\}$ maps to $\{y_1u_1 + y_2u_2 : \frac{y_1^2}{\sigma_1^2} + \frac{y_2^2}{\sigma_2^2} = 1\}$, an ellipse with semi-axes σ_1, σ_2 along u_1, u_2.

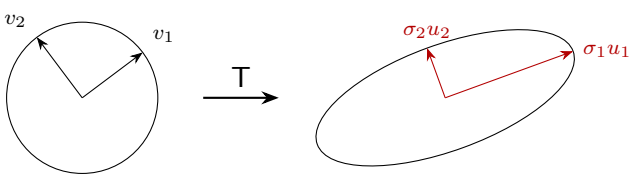


Figure. *Example 2: the unit circle maps onto an ellipse with semi-axes $\sigma_1 u_1$, $\sigma_2 u_2$.*

The singular values of a matrix A are those of L_A.

Theorem 6.27 (Singular value decomposition for matrices)

Let $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ have rank r and positive singular values $\sigma_1 \geq \cdots \geq \sigma_r$, and let $\Sigma \in \mathrm{M}_{m\times n}$ have $\Sigma_{ii} = \sigma_i$ $(i \leq r)$ and all other entries 0. Then $A = U\Sigma V^*$ for some unitary U $(m \times m)$ and V $(n \times n)$.

Proof. Apply Theorem 6.26 to L_A; let V, U have columns v_j, u_i. Then AV has columns $Av_j = \sigma_j u_j$ (or 0), which are the columns of $U\Sigma$. So $AV = U\Sigma$ and $A = U\Sigma V^*$. ∎

Such a factorization is a **singular value decomposition** of A.

Example 3

$A = \left(\begin{smallmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{smallmatrix}\right)$: A^*A has eigenvalues $6, 0, 0$ with orthonormal eigenvectors $v_1 = \frac{1}{\sqrt{3}}(1, 1, -1)$, $v_2 = \frac{1}{\sqrt{2}}(1, -1, 0)$, $v_3 = \frac{1}{\sqrt{6}}(1, 1, 2)$. So $\sigma_1 = \sqrt{6}$, $u_1 = \frac{1}{\sqrt{6}}Av_1 = \frac{1}{\sqrt{2}}(1, 1)$, $u_2 = \frac{1}{\sqrt{2}}(1, -1)$, and $A = U\Sigma V^*$ with $\Sigma = \left(\begin{smallmatrix} \sqrt{6} & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$.

Theorem 6.28 (Polar decomposition)

Every square A is $A = WP$ with W unitary and P positive semidefinite (P self-adjoint with nonnegative eigenvalues). If A is invertible, the representation is unique.

Proof. With $A = U\Sigma V^*$, put $W = UV^*$ (unitary), $P = V\Sigma V^*$ (self-adjoint, eigenvalues $\sigma_i \geq 0$); then $WP = A$. If A is invertible and $A = WP = ZQ$, then $P^2 = P^*W^*WP = A^*A = Q^2$. Positive semidefinite square roots are unique: P commutes with P^2 and has eigenvalues ≥ 0, so on each eigenspace of A^*A (eigenvalue μ) both P and Q act as $\sqrt{\mu}$. So $P = Q$ is invertible and $W = Z = AP^{-1}$. ∎

Example 4

$A = \left(\begin{smallmatrix} 11 & -5 \\ -2 & 10 \end{smallmatrix}\right)$: A^*A has eigenvalues $200, 50$ with $v_1 = \frac{1}{\sqrt{2}}(1, -1)$, $v_2 = \frac{1}{\sqrt{2}}(1, 1)$; $u_1 = \frac{1}{5}(4, -3)$, $u_2 = \frac{1}{5}(3, 4)$. Then

$$W = UV^* = \frac{1}{5\sqrt{2}}\begin{pmatrix} 7 & -1 \\ 1 & 7 \end{pmatrix}, \qquad P = V\Sigma V^* = \frac{5}{\sqrt{2}}\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}.$$

6.7.1 The pseudoinverse

Definition (Pseudoinverse)

For linear $\mathsf{T} : V \to W$, let $\mathsf{L} : \mathrm{N}(\mathsf{T})^\perp \to \mathrm{R}(\mathsf{T})$ be the restriction of T (an isomorphism). The **pseudoinverse** (Moore–Penrose generalized inverse) $\mathsf{T}^\dagger : W \to V$ is $\mathsf{T}^\dagger(y) = \mathsf{L}^{-1}(y)$ for $y \in \mathrm{R}(\mathsf{T})$ and $\mathsf{T}^\dagger(y) = 0$ for $y \in \mathrm{R}(\mathsf{T})^\perp$. For a matrix, $A^\dagger$ is the matrix with $\mathsf{L}_{A^\dagger} = (\mathsf{L}_A)^\dagger$. If T is invertible, $\mathsf{T}^\dagger = \mathsf{T}^{-1}$.

In the setting of Theorem 6.26, $\mathsf{T}^\dagger(u_i) = \frac{1}{\sigma_i}v_i$ for $i \leq r$ and $\mathsf{T}^\dagger(u_i) = 0$ for $i > r$ ($\{v_1, \ldots, v_r\}$ spans $\mathrm{N}(\mathsf{T})^\perp$ and $\{u_1, \ldots, u_r\}$ spans $\mathrm{R}\,\mathsf{T}$).

Example 5

For $\mathsf{T} = d/dx$ of Example 1: $\mathsf{T}^\dagger(u_1) = \frac{1}{\sqrt{15}}v_1$, $\mathsf{T}^\dagger(u_2) = \frac{1}{\sqrt{3}}v_2$; e.g. $\mathsf{T}^\dagger(1) = \mathsf{T}^\dagger(\sqrt{2}u_2) = \sqrt{\frac{2}{3}}\sqrt{\frac{3}{2}}x = x$.

Theorem 6.29

If $A = U\Sigma V^*$ is a singular value decomposition and $\Sigma^\dagger \in \mathrm{M}_{n\times m}$ has $\Sigma^\dagger_{ii} = \frac{1}{\sigma_i}$ $(i \le r)$ and other entries 0, then $A^\dagger = V\Sigma^\dagger U^*$, and this is a singular value decomposition of $A^\dagger$.

Proof. $V\Sigma^\dagger U^* u_i = V\Sigma^\dagger e_i = \frac{1}{\sigma_i}v_i$ for $i \le r$ and 0 for $i > r$, matching $(\mathsf{L}_A)^\dagger$ on $\{u_i\}$. The factors are unitary and $\Sigma^\dagger$ has the required form (values $\frac{1}{\sigma_r} \ge \cdots \ge \frac{1}{\sigma_1}$ after reordering). ■

Example 6

For A of Example 3, $A^\dagger = V\Sigma^\dagger U^* = \frac{1}{\sqrt{6}}v_1 u_1^* = \frac{1}{6}\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \end{pmatrix}$.

Lemma

For linear $\mathsf{T} : V \to W$ (finite-dimensional): (a) $\mathsf{T}^\dagger\mathsf{T}$ is the orthogonal projection of V on $\mathrm{N}(\mathsf{T})^\perp$;
(b) $\mathsf{T}\mathsf{T}^\dagger$ is the orthogonal projection of W on $\mathrm{R}(\mathsf{T})$.

Proof.

(a) For $x \in \mathrm{N}(\mathsf{T})^\perp$, $\mathsf{T}^\dagger\mathsf{T}x = \mathsf{L}^{-1}\mathsf{L}x = x$; for $x \in \mathrm{N}(\mathsf{T})$, $\mathsf{T}^\dagger\mathsf{T}x = 0$.

(b) For $y \in \mathrm{R}\,\mathsf{T}$, $\mathsf{T}\mathsf{T}^\dagger y = \mathsf{L}\mathsf{L}^{-1}y = y$; for $y \in \mathrm{R}(\mathsf{T})^\perp$, $\mathsf{T}^\dagger y = 0$. ■

Theorem 6.30

For $Ax = b$ $(A \in \mathrm{M}_{m\times n}(\mathbb{F}))$ let $z = A^\dagger b$.
(a) If $Ax = b$ is consistent, z is its unique solution of minimal norm.
(b) If $Ax = b$ is inconsistent, z is the unique best approximation of minimal norm: $\|Az - b\| \le \|Ay - b\|$ for all y, with equality iff $Az = Ay$, and then $\|z\| \le \|y\|$, with equality iff $z = y$.

Proof.

(a) $Az = AA^\dagger b$ is the orthogonal projection of b on $\mathrm{R}\,\mathsf{L}_A$ (Lemma), which is b itself. $z \in \mathrm{R}(\mathsf{L}_{A^\dagger}) = \mathrm{N}(\mathsf{L}_A)^\perp = \mathrm{R}(\mathsf{L}_{A^*})$, so z is the minimal solution by Theorem 6.13.

(b) Az, the projection of b on $\mathrm{R}\,\mathsf{L}_A$, is the closest point (Corollary to Theorem 6.6): $\|Az - b\| \le \|Ay - b\|$, with equality iff $Ay = Az$. Among the solutions y of $Ax = Az$, z has minimal norm by (a). ■

Example 7

With A of Example 3: $z = A^\dagger(1,1)^t = \frac{1}{3}(1,1,-1)$ is the minimal solution of $Ax = (1,1)^t$, and $z = A^\dagger(1,2)^t = \frac{1}{2}(1,1,-1)$ the minimal-norm best approximation for the inconsistent $Ax = (1,2)^t$.

6.8 Bilinear and quadratic forms *

Motivation. Matrices of bilinear forms change by congruence Q^tAQ; symmetric forms diagonalize, and by Sylvester's law the signs on the diagonal are invariants.

Definition (Bilinear form)

$H : V \times V \to \mathbb{F}$ is a **bilinear form** if it is linear in each variable when the other is fixed: $H(ax_1 + x_2, y) = aH(x_1, y) + H(x_2, y)$ and $H(x, ay_1 + y_2) = aH(x, y_1) + H(x, y_2)$. $\mathcal{B}(V)$ is the set of bilinear forms on V.

Example 1

$H(a, b) = 2a_1b_1 + 3a_1b_2 + 4a_2b_1 - a_2b_2$ on $\mathbb{R}^2$ equals x^tAy with $A = \left(\begin{smallmatrix} 2 & 3 \\ 4 & -1 \end{smallmatrix}\right)$, hence is bilinear.

Example 2

$H(x, y) = x^tAy$ is bilinear on $\mathbb{F}^n$ for any $A \in \mathrm{M}_{n\times n}(\mathbb{F})$. A real inner product is bilinear; a complex one is not (conjugate linear in y).

Properties: $H(0, x) = H(x, 0) = 0$; for fixed x, $L_x(y) = H(x, y)$ and $R_x(y) = H(y, x)$ are linear.

Theorem 6.31

Sums and scalar multiples of bilinear forms are bilinear forms, and $\mathcal{B}(V)$ is a vector space under $(H_1 + H_2)(x, y) = H_1(x, y) + H_2(x, y)$, $(aH)(x, y) = aH(x, y)$.

Proof. Pointwise operations preserve bilinearity, so $\mathcal{B}(V)$ is a subspace of $\mathcal{F}(V \times V, \mathbb{F})$ (it contains the zero form). ■

Definition (Matrix of a bilinear form)

For an ordered basis $\beta = \{v_1, \ldots, v_n\}$, $\psi_\beta(H) = A$ with $A_{ij} = H(v_i, v_j)$ is the **matrix representation** of H.

Example 3

For H of Example 1 and $\beta = \{(1, 1), (1, -1)\}$: $\psi_\beta(H) = \left(\begin{smallmatrix} 8 & 4 \\ 2 & -6 \end{smallmatrix}\right)$ (e.g. $B_{11} = 2 + 3 + 4 - 1$); in the standard basis $\psi(H) = \left(\begin{smallmatrix} 2 & 3 \\ 4 & -1 \end{smallmatrix}\right)$.

Theorem 6.32

$\psi_\beta : \mathcal{B}(V) \to \mathrm{M}_{n\times n}(\mathbb{F})$ is an isomorphism ($\dim V = n$).

Proof. ψ_β is linear. If $\psi_\beta(H) = O$, then $H(v_i, \cdot) = 0$ on β, so $H(v_i, y) = 0$ for all y, i.e. $H(\cdot, y) = 0$ on β; hence $H = 0$. Onto: given A, $H(x, y) = [\phi_\beta(x)]^tA[\phi_\beta(y)]$ is bilinear with $H(v_i, v_j) = e_i^tAe_j = A_{ij}$. ■

Corollary 1

$\dim \mathcal{B}(V) = n^2$.

Proof. Theorems 6.32 and 2.19. ■

Corollary 2

$\psi_\beta(H) = A$ iff $H(x,y) = [\phi_\beta(x)]^t A[\phi_\beta(y)]$ for all x, y.

Proof. The form on the right has matrix A (proof of Theorem 6.32); apply injectivity of ψ_β. ■

Corollary 3

For every $H \in \mathcal{B}(\mathbb{F}^n)$ there is a unique A with $H(x,y) = x^t A y$, namely $A = \psi_\beta(H)$ for the standard β.

Proof. Corollary 2 with $\phi_\beta = \mathsf{I}$. ■

Example 4

$H(a,b) = \det\begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix} = a_1 b_2 - a_2 b_1$ has $A_{ij} = H(e_i, e_j)$, i.e. $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.

Definition (Congruence)

B is **congruent** to A if $B = Q^t A Q$ for some invertible Q; congruence is an equivalence relation.

Theorem 6.33

If Q changes γ-coordinates into β-coordinates, then $\psi_\gamma(H) = Q^t \psi_\beta(H) Q$ for every $H \in \mathcal{B}(V)$.

Proof. $w_j = \sum_k Q_{kj} v_k$, so $H(w_i, w_j) = \sum_{k,l} Q_{ki} H(v_k, v_l) Q_{lj} = (Q^t \psi_\beta(H) Q)_{ij}$. ■

Corollary

If $B = Q^t \psi_\beta(H) Q$ with Q invertible, then $B = \psi_\gamma(H)$ for the ordered basis γ with $w_j = \sum_i Q_{ij} v_i$ (so Q changes γ- into β-coordinates).

Proof. γ is a basis since Q is invertible; apply Theorem 6.33. ■

Definition (Symmetric, diagonalizable forms)

H is **symmetric** if $H(x,y) = H(y,x)$; it is **diagonalizable** if $\psi_\beta(H)$ is diagonal for some β.

Theorem 6.34

H is symmetric iff $\psi_\beta(H)$ is symmetric.

Proof. If H is symmetric, $A_{ij} = H(v_i, v_j) = H(v_j, v_i) = A_{ji}$. Conversely, if A is symmetric, $\bar{H}(x,y) = H(y,x)$ is bilinear with $\psi_\beta(\bar{H}) = A^t = A$, so $\bar{H} = H$. ■

Corollary

A diagonalizable bilinear form is symmetric.

Proof. Diagonal matrices are symmetric; apply Theorem 6.34. ■

Example 5

Over $\mathbb{F} = \mathbb{Z}_2$, $H(a,b) = a_1b_2 + a_2b_1$ on $\mathbb{F}^2$ is symmetric with $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ of rank 2, but not diagonalizable: a congruent diagonal matrix would be I (1 is the only nonzero scalar), yet $H(x,x) = 2a_1a_2 = 0$ for all x, while matrix I in a basis $\{v_1, v_2\}$ means $H(v_1, v_1) = 1$.

Lemma

If $H \neq 0$ is a symmetric bilinear form on V over a field of characteristic $\neq 2$, then $H(x,x) \neq 0$ for some x.

Proof. Take u, v with $H(u,v) \neq 0$. If $H(u,u) = H(v,v) = 0$, then $H(u+v, u+v) = 2H(u,v) \neq 0$. ■

Theorem 6.35

Over a field of characteristic $\neq 2$, every symmetric bilinear form on finite-dimensional V is diagonalizable.

Proof. Induction on $n = \dim V$; $n = 1$ and $H = 0$ are trivial. Otherwise take $H(x,x) \neq 0$ (Lemma). $L_x(y) = H(x,y)$ is a nonzero functional, so $\dim \mathrm{N}(L_x) = n - 1$ and $x \notin \mathrm{N}(L_x)$. By induction the symmetric restriction of H to $\mathrm{N}(L_x)$ has a basis $\{v_1, \ldots, v_{n-1}\}$ with $H(v_i, v_j) = 0$ $(i \neq j)$. With $v_n = x$, $H(v_i, v_n) = H(x, v_i) = 0$ $(i < n)$, so $\{v_1, \ldots, v_n\}$ diagonalizes H. ■

Corollary

Over a field of characteristic $\neq 2$, every symmetric matrix is congruent to a diagonal matrix.

Proof. Apply Theorem 6.35 to $H(x,y) = x^tAy$ on $\mathbb{F}^n$ and use Theorem 6.33. ■

Computing. Write $Q = E_1 \cdots E_k$ (elementary). Then $D = Q^tAQ = E_k^t \cdots E_1^t A E_1 \cdots E_k$: each elementary column operation is followed by the same row operation.

Example 6

$A = \left(\begin{smallmatrix} 1 & -1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & 1 \end{smallmatrix}\right)$: the column operations $C_2 + C_1$, $C_3 - 3C_1$, $C_3 - 4C_2$, each followed by the same row operation, give

$$Q = \begin{pmatrix} 1 & 1 & -7 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{pmatrix}, \qquad Q^tAQ = \operatorname{diag}(1, 1, -24).$$

Definition (Quadratic form)

$K : V \to \mathbb{F}$ is a **quadratic form** if $K(x) = H(x,x)$ for a symmetric $H \in \mathcal{B}(V)$. In characteristic $\neq 2$, H is recovered: $H(x,y) = \frac{1}{2}(K(x+y) - K(x) - K(y))$.

Example 7

A homogeneous polynomial of degree 2, $f(t_1,\dots,t_n)=\sum_{i\le j}a_{ij}t_it_j$, is a quadratic form on $\mathbb{R}^n$: $f(x)=x^tAx$ with $A_{ii}=a_{ii}$ and $A_{ij}=A_{ji}=\frac{1}{2}a_{ij}$ for $i<j$.

Theorem 6.36

For a symmetric bilinear form H on a finite-dimensional real inner product space there is an orthonormal basis β with $\psi_\beta(H)$ diagonal.

Proof. For orthonormal γ, $A=\psi_\gamma(H)$ is symmetric, so $Q^tAQ=D$ for an orthogonal Q (Theorem 6.20). The basis β given by Q (Corollary to Theorem 6.33), $w_j=\sum_i Q_{ij}v_i$, is orthonormal (orthonormal columns of Q), and $\psi_\beta(H)=D$. ■

Corollary

For a quadratic form K on a finite-dimensional real inner product space there are an orthonormal basis $\{v_1,\dots,v_n\}$ and scalars λ_i with $K(\sum s_iv_i)=\sum\lambda_is_i^2$. (The λ_i are the eigenvalues of the matrix of K in any orthonormal basis.)

Proof. Theorem 6.36: $K(x)=H(x,x)=\sum_{i,j}s_is_jH(v_i,v_j)=\sum\lambda_is_i^2$. ■

Example 8

$f(t_1,t_2)=5t_1^2+2t_2^2+4t_1t_2$ has $A=\left(\begin{smallmatrix}5&2\\2&2\end{smallmatrix}\right)$ with eigenvalues $6,1$ and unit eigenvectors $x_1=\frac{1}{\sqrt{5}}(2,1)$, $x_2=\frac{1}{\sqrt{5}}(-1,2)$. In coordinates $(t_1,t_2)=s_1x_1+s_2x_2$, $f=6s_1^2+s_2^2$.

Example 9

The surface $2t_1^2+6t_1t_2+5t_2^2-2t_2t_3+2t_3^2+3t_1-2t_2-t_3+14=0$: the quadratic part has matrix $\left(\begin{smallmatrix}2&3&0\\3&5&-1\\0&-1&2\end{smallmatrix}\right)$ with eigenvalues $2,7,0$ and unit eigenvectors $v_1=\frac{1}{\sqrt{10}}(1,0,3)$, $v_2=\frac{1}{\sqrt{35}}(-3,-5,1)$, $v_3=\frac{1}{\sqrt{14}}(-3,2,1)$. In coordinates s_i for $\{v_1,v_2,v_3\}$ the linear part is $-\sqrt{14}\,s_3$, giving the elliptic paraboloid $s_3=\frac{2s_1^2+7s_2^2}{\sqrt{14}}+\sqrt{14}$.

6.8.1 The second derivative test

For $f:\mathbb{R}^n\to\mathbb{R}$ with continuous third partial derivatives, p is a **critical point** if all $\frac{\partial f}{\partial t_i}(p)=0$; the **Hessian** $A(p)$ has entries $\frac{\partial^2 f}{\partial t_i\partial t_j}(p)$ and is symmetric.

Theorem 6.37 (Second derivative test)

Let p be a critical point of f with Hessian $A(p)$.
(a) If all eigenvalues of $A(p)$ are positive, f has a local minimum at p.
(b) If all are negative, f has a local maximum at p.
(c) If $A(p)$ has a positive and a negative eigenvalue, p is a saddle point (no local extremum).
(d) If rank $A(p)<n$ and $A(p)$ does not have eigenvalues of both signs, the test is inconclusive.

Proof. Assume $p = 0$, $f(0) = 0$. By Taylor, $f(x) = \frac{1}{2}x^t A(0)x + S(x)$ with $S(x)/\|x\|^2 \to 0$ as $x \to 0$. With an orthonormal eigenbasis $\{v_i\}$ of $A(0)$ (Theorem 6.20), eigenvalues λ_i, and $x = \sum s_i v_i$: $\frac{1}{2}x^t A(0)x = \frac{1}{2}\sum \lambda_i s_i^2$.

(a) Let $\lambda = \min \lambda_i > 0$ and $|S(x)| < \frac{\lambda}{4}\|x\|^2$ for $0 < \|x\| < \varepsilon$. Then $f(x) \geq \frac{\lambda}{2}\|x\|^2 - \frac{\lambda}{4}\|x\|^2 > 0 = f(0)$.

(b) Apply (a) to $-f$.

(c) If $\lambda_1 > 0 > \lambda_2$, then for small $s \neq 0$, $f(sv_1) = \frac{1}{2}\lambda_1 s^2 + S(sv_1) > 0$ and $f(sv_2) = \frac{1}{2}\lambda_2 s^2 + S(sv_2) < 0$, so 0 is no local extremum.

(d) $t_1^2 + t_2^4$, $t_1^2 - t_2^4$, $t_1^2 + t_2^3$ all have Hessian $\operatorname{diag}(2, 0)$ at 0, with a minimum, no extremum, no extremum there. ∎

6.8.2 Sylvester's law of inertia

Theorem 6.38 (Sylvester's law of inertia)

For a symmetric bilinear form H on a finite-dimensional real vector space, the numbers of positive and of negative diagonal entries are the same in every diagonal matrix representation of H.

Proof. The number of nonzero diagonal entries is the rank, invariant under congruence (Theorem 3.4); so treat the positive entries. Let $\beta = \{v_i\}$, $\gamma = \{w_i\}$ diagonalize H, with $H(v_i, v_i) > 0$ exactly for $i \leq p$, $H(w_i, w_i) > 0$ exactly for $i \leq q$, and suppose $p < q$. With $r = \operatorname{rank}$, let

$$L(x) = (H(x, v_1), \ldots, H(x, v_p), H(x, w_{q+1}), \ldots, H(x, w_r)), \qquad L : V \to \mathbb{R}^{p+r-q}.$$

As $p + r - q < r$, $\operatorname{nullity} L > n - r$, so $L(x_0) = 0$ for some $x_0 \notin \operatorname{span}\{v_{r+1}, \ldots, v_n\}$. Write $x_0 = \sum a_j v_j = \sum b_j w_j$; $H(x_0, v_i) = a_i H(v_i, v_i) = 0$ gives $a_i = 0$ $(i \leq p)$, and likewise $b_i = 0$ $(q < i \leq r)$. Hence

$$H(x_0, x_0) = \sum_{j=p+1}^{r} a_j^2 H(v_j, v_j) \leq 0 \quad \text{and} \quad H(x_0, x_0) = \sum_{j=1}^{q} b_j^2 H(w_j, w_j) \geq 0,$$

so both vanish. The first gives $a_j = 0$ for $p < j \leq r$, so $x_0 \in \operatorname{span}\{v_{r+1}, \ldots, v_n\}$, a contradiction. By symmetry $p = q$. ∎

Definition (Index, signature)

The number of positive diagonal entries in a diagonal representation is the **index**; (positive) − (negative) is the **signature**. Rank, index and signature are the **invariants**; any two determine the third. The same terms apply to real symmetric matrices via congruence.

Example 10

The form of Example 9 has a diagonal representation $\operatorname{diag}(2, 7, 0)$: rank 2, index 2, signature 2.

Example 11

$K(x, y) = x^2 - y^2$ has matrix $\operatorname{diag}(1, -1)$, and in the basis $\{(1, 1), (1, -1)\}$ the matrix $\left(\begin{smallmatrix} 0 & 2 \\ 2 & 0 \end{smallmatrix}\right)$ (not diagonal); rank 2, index 1, signature 0.

Corollary 1 (Sylvester's law for matrices)

For a real symmetric A, the numbers of positive and negative diagonal entries of a diagonal matrix congruent to A do not depend on the choice of the diagonal matrix.

Proof. Apply Theorem 6.38 to $H(x,y) = x^tAy$ and use the Corollary to Theorem 6.33. ■

Corollary 2

Two real symmetric $n \times n$ matrices are congruent iff they have the same invariants.

Proof. Congruent matrices are congruent to the same diagonal matrices, so have the same invariants (Corollary 1). Conversely, a real symmetric matrix of index p, rank r is congruent to $D = \operatorname{diag}(d_1, \ldots, d_n)$ with $d_i > 0$ $(i \le p)$, $d_i < 0$ $(p < i \le r)$, $d_i = 0$ $(i > r)$ (reorder by a permutation), and $Q = \operatorname{diag}(|d_i|^{-1/2}$ or $1)$ gives $Q^tDQ = J_{pr} = \operatorname{diag}(I_p, -I_{r-p}, O)$. So matrices with the same invariants are both congruent to J_{pr}. ■

Corollary 3

A real symmetric $n \times n$ matrix has index p and rank r iff it is congruent to J_{pr}.

Proof. The proof of Corollary 2, and the fact that J_{pr} has index p and rank r. ■

Example 12

A of Example 6 is congruent to $\operatorname{diag}(1,1,-24)$: rank 3, index 2. $B = \left(\begin{smallmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{smallmatrix}\right)$ has eigenvalues $2, -1, -1$: rank 3, index 1. $C = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & -1 \end{smallmatrix}\right)$ has eigenvalues $1, \frac{1 \pm \sqrt{13}}{2}$: rank 3, index 2. So A and C are congruent, B to neither (eigenvalues give a congruent diagonal matrix by Theorem 6.20).

6.9 Einstein's special theory of relativity *

Motivation. A constant speed of light forces coordinate changes between inertial frames to preserve the light cone $x^2 + y^2 + z^2 - t^2 = 0$, which determines the Lorentz transformation.

Frames S, S' have parallel axes, S' moves along the x-axis of S with velocity $0 < v < 1$ (light speed 1), and clocks C, C' read 0 when the origins coincide. An event has **space–time coordinates** $(x,y,z,t)^t$ relative to S, C and $(x',y',z',t')^t$ relative to S', C'; $\mathsf{T}_v : \mathbb{R}^4 \to \mathbb{R}^4$ maps the former to the latter.

Axioms. (R1) Light has speed 1 in either frame. (R2) T_v is an isomorphism. (R3) $y' = y$ and $z' = z$. (R4) x', t' do not depend on y, z. (R5) The origin of S moves along the x'-axis with velocity $-v$ as measured in S'. T_v is the **Lorentz transformation** in direction x.

Theorem 6.39

On $\mathbb{R}^4$ (standard inner product):
(a) $\mathsf{T}_v(e_i) = e_i$ for $i = 2, 3$;
(b) $\operatorname{span}\{e_2, e_3\}$ is T_v-invariant;
(c) $\operatorname{span}\{e_1, e_4\}$ is T_v-invariant;
(d) both are T_v^*-invariant;
(e) $\mathsf{T}_v^*(e_i) = e_i$ for $i = 2, 3$.

Proof.

(a) $\mathsf{T}_v(0)=0$, so by (R4) the first and fourth coordinates of $\mathsf{T}_v(0,a,b,0)^t$ are 0 for all a,b, and by (R3) $\mathsf{T}_v(e_2)=e_2$, $\mathsf{T}_v(e_3)=e_3$.

(b) Follows from (a).

(c) [Supplement] By (R3) the second and third coordinates of $\mathsf{T}_v(x,0,0,t)^t$ are 0.

(d) [Supplement] $W=\operatorname{span}\{e_2,e_3\}$ and $W^\perp=\operatorname{span}\{e_1,e_4\}$ are T_v-invariant. For $x\in W^\perp$, $w\in W$: $\langle \mathsf{T}_v^* x,w\rangle=\langle x,\mathsf{T}_v w\rangle=0$, so $\mathsf{T}_v^*(W^\perp)\subseteq W^\perp$; symmetrically for W.

(e) For $j\neq 2$, $\langle \mathsf{T}_v^* e_2,e_j\rangle=\langle e_2,\mathsf{T}_v e_j\rangle=0$ by (a), (c); so $\mathsf{T}_v^* e_2=ke_2$ with $k=\langle \mathsf{T}_v^* e_2,e_2\rangle=\langle e_2,\mathsf{T}_v e_2\rangle=1$. Similarly for e_3. ■

A light flash at the common origin at time 0 is seen at (x,y,z) at time $t\geq 0$ iff $x^2+y^2+z^2-t^2=0$, and by (R1) the same holds in primed coordinates. Let $A=\operatorname{diag}(1,1,1,-1)$, so $\langle \mathsf{L}_A(w),w\rangle=x^2+y^2+z^2-t^2$.

Theorem 6.40

If $\langle \mathsf{L}_A(w),w\rangle=0$, then $\langle \mathsf{T}_v^*\mathsf{L}_A\mathsf{T}_v(w),w\rangle=0$.

Proof. If $t\geq 0$, w is an event of the flash, so $0=\langle \mathsf{L}_A\mathsf{T}_v w,\mathsf{T}_v w\rangle=\langle \mathsf{T}_v^*\mathsf{L}_A\mathsf{T}_v w,w\rangle$. If $t<0$, use $-w$. ■

Let $w_1=(1,0,0,1)^t$, $w_2=(1,0,0,-1)^t$, an orthogonal basis of $\operatorname{span}\{e_1,e_4\}$, which is invariant under $\mathsf{T}_v^*\mathsf{L}_A\mathsf{T}_v$ by Theorem 6.39.

Theorem 6.41

There are nonzero scalars a,b with (a) $\mathsf{T}_v^*\mathsf{L}_A\mathsf{T}_v(w_1)=aw_2$;
(b) $\mathsf{T}_v^*\mathsf{L}_A\mathsf{T}_v(w_2)=bw_1$.

Proof.

(a) $\langle \mathsf{L}_A w_1,w_1\rangle=0$, so $\mathsf{T}_v^*\mathsf{L}_A\mathsf{T}_v(w_1)\in\operatorname{span}\{w_1,w_2\}$ is $\perp w_1$ (Theorem 6.40), i.e. equals aw_2; $a\neq 0$ as $\mathsf{T}_v^*\mathsf{L}_A\mathsf{T}_v$ is invertible.

(b) Same, with $\langle \mathsf{L}_A w_2,w_2\rangle=0$. ■

Corollary

With $B_v=[\mathsf{T}_v]_\beta$ (β standard): (a) $B_v^* A B_v=A$;
(b) $\mathsf{T}_v^*\mathsf{L}_A\mathsf{T}_v=\mathsf{L}_A$.

Proof [Supplement] (left as an exercise in the book). $M=\mathsf{T}_v^*\mathsf{L}_A\mathsf{T}_v$ is self-adjoint, so $2a=\langle Mw_1,w_2\rangle=\langle w_1,Mw_2\rangle=2b$, i.e. $a=b$. Since $Aw_1=w_2$ and $Aw_2=w_1$, $M=a\mathsf{L}_A$ on $\operatorname{span}\{e_1,e_4\}$, and $M=\mathsf{L}_A$ on $\operatorname{span}\{e_2,e_3\}$ by Theorem 6.39. So $\langle A\mathsf{T}_v w,\mathsf{T}_v w\rangle=a\langle Aw,w\rangle$ for $w\in\operatorname{span}\{e_1,e_4\}$. The axioms do not force $a=1$ (rescaling T_v on $\operatorname{span}\{e_1,e_4\}$ by $\sqrt{a}$ preserves (R1)–(R5)), so we add *reciprocity*: $\mathsf{T}_{-v}=\mathsf{T}_v^{-1}$ and $\mathsf{T}_{-v}=R\mathsf{T}_v R$ for $R=\operatorname{diag}(-1,1,1,1)$ (reversing the x-axes). Then $a(-v)=a(v)$ (R preserves $\langle A\cdot,\cdot\rangle$) and $a(v)a(-v)=1$ ($\mathsf{T}_{-v}\mathsf{T}_v=\mathsf{I}$), so $a^2=1$. Also $a>0$: by (R5), $\mathsf{T}_v(e_4)=(-vt'',0,0,t'')^t$ with $t''>0$, and $-a=a\langle Ae_4,e_4\rangle=(v^2-1)t''^2<0$. Hence $a=1$, which is (b); (a) is its matrix form. ■

One second (on C) after the origins coincide, the origin of S' has S-coordinates $(v,0,0,1)^t$ and S'-coordinates $(0,0,0,t')^t$, $t'>0$. By the Corollary,

$$v^2-1=\langle \mathsf{L}_A(v,0,0,1)^t,(v,0,0,1)^t\rangle=\langle \mathsf{L}_A(0,0,0,t')^t,(0,0,0,t')^t\rangle=-t'^2,$$

so $t' = \sqrt{1-v^2}$ and $\mathsf{T}_v(v,0,0,1)^t = (0,0,0,\sqrt{1-v^2})^t$. Likewise (R5) gives $\mathsf{T}_v(e_4) = (-vt'',0,0,t'')^t$ with $t'' = 1/\sqrt{1-v^2}$.

Theorem 6.42

In the standard basis,

$$[\mathsf{T}_v]_\beta = B_v = \begin{pmatrix} \frac{1}{\sqrt{1-v^2}} & 0 & 0 & \frac{-v}{\sqrt{1-v^2}} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{-v}{\sqrt{1-v^2}} & 0 & 0 & \frac{1}{\sqrt{1-v^2}} \end{pmatrix}.$$

Proof. Columns $2,3$ are e_2, e_3 (Theorem 6.39(a)); column 4 is $\mathsf{T}_v(e_4)$ above. Column 1 is $\mathsf{T}_v(e_1) = \frac{1}{v}(\mathsf{T}_v(v,0,0,1)^t - \mathsf{T}_v(e_4)) = \frac{1}{v}(v/\sqrt{1-v^2}, 0, 0, \sqrt{1-v^2} - 1/\sqrt{1-v^2})^t = (\frac{1}{\sqrt{1-v^2}}, 0, 0, \frac{-v}{\sqrt{1-v^2}})^t$. ∎

Time contraction. A ship leaving the solar system with velocity v has S-coordinates $(vt,0,0,t)^t$ and S'-coordinates $(0,0,0,t')^t$. From $B_v(vt,0,0,t)^t = (0,0,0,t')^t$,

$$t' = \frac{-v^2t + t}{\sqrt{1-v^2}} = t\sqrt{1-v^2} \qquad \left(\text{general units: } t' = t\sqrt{1-v^2/c^2}\right).$$

6.10 Conditioning and the Rayleigh quotient *

Motivation. Relative errors in b are amplified in x by at most $\operatorname{cond}(A) = \|A\|\|A^{-1}\|$, computed via the Rayleigh quotient.

Example 1

$x_1 + x_2 = 5$, $x_1 - x_2 = 1 + h$ has solution $(3 + \frac{h}{2}, 2 - \frac{h}{2})$. The relative changes are $\frac{\|\delta b\|}{\|b\|} = \frac{|h|}{\sqrt{26}} = \frac{\|\delta x\|}{\|x\|}$: the system is **well-conditioned**.

Example 2

$x_1 + x_2 = 3$, $x_1 + 1.00001x_2 = 3.00001 + h$ has solution $(2 - 10^5 h,\ 1 + 10^5 h)$, so $\frac{\|\delta x\|}{\|x\|} = 10^5\sqrt{\frac{2}{5}}\,|h| \geq 10^4|h|$ while $\frac{\|\delta b\|}{\|b\|} \approx \frac{|h|}{3\sqrt{2}}$: **ill-conditioned** (nearly parallel lines).

Definition (Matrix norm; Rayleigh quotient)

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, $\|A\| = \max_{x\neq 0} \frac{\|Ax\|}{\|x\|}$. For self-adjoint B, the **Rayleigh quotient** is $R(x) = \frac{\langle Bx, x\rangle}{\|x\|^2}$ $(x \neq 0)$.

Theorem 6.43

For self-adjoint B, $\max_{x\neq 0} R(x)$ is the largest and $\min_{x\neq 0} R(x)$ the smallest eigenvalue of B.

Proof. Take an orthonormal eigenbasis $\{v_i\}$, $Bv_i = \lambda_i v_i$, $\lambda_1 \geq \cdots \geq \lambda_n$ (real). For $x = \sum a_i v_i$, $R(x) = \frac{\sum \lambda_i |a_i|^2}{\sum |a_i|^2} \leq \lambda_1$, with $R(v_1) = \lambda_1$. The minimum is analogous. ∎

Corollary 1

For any square A, $\|A\| = \sqrt{\lambda}$, where λ is the largest eigenvalue of A^*A.

Proof. $\frac{\|Ax\|^2}{\|x\|^2} = \frac{\langle A^*Ax, x\rangle}{\|x\|^2}$ is the Rayleigh quotient of the self-adjoint A^*A; apply Theorem 6.43. ■

Lemma

For square A, λ is an eigenvalue of A^*A iff it is an eigenvalue of AA^*.

Proof. If $\lambda \neq 0$ and $A^*Ax = \lambda x$, $x \neq 0$, then $Ax \neq 0$ and $AA^*(Ax) = \lambda Ax$; symmetrically. For $\lambda = 0$: A^*A is singular iff A is (Lemma 2 of Section 6.3), iff A^* is, iff AA^* is. ■

Corollary 2

If A is invertible, $\|A^{-1}\| = \frac{1}{\sqrt{\lambda}}$, where λ is the smallest eigenvalue of A^*A.

Proof. By Corollary 1, $\|A^{-1}\|^2$ is the largest eigenvalue of $(A^{-1})^*A^{-1} = (AA^*)^{-1}$, the reciprocal of the smallest eigenvalue of AA^*, i.e. of A^*A (Lemma). ■

Example 3

For $A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$, $B = A^*A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ has eigenvalues $3, 3, 0$, so $\|A\| = \sqrt{3}$; indeed $R(x) = \frac{2(a^2+b^2+c^2-ab+ac+bc)}{a^2+b^2+c^2} \leq 3$.

The **condition number** of invertible A is $\text{cond}(A) = \|A\| \cdot \|A^{-1}\|$. From $\|Ax\| \leq \|A\|\|x\|$: $\|b\| \leq \|A\|\|x\|$, $\|\delta x\| \leq \|A^{-1}\|\|\delta b\|$, $\|\delta b\| \leq \|A\|\|\delta x\|$, $\|x\| \leq \|A^{-1}\|\|b\|$.

Theorem 6.44

For $Ax = b$ with A invertible and $b \neq 0$:
(a) for any norm, $\frac{1}{\text{cond}(A)} \frac{\|\delta b\|}{\|b\|} \leq \frac{\|\delta x\|}{\|x\|} \leq \text{cond}(A) \frac{\|\delta b\|}{\|b\|}$;
(b) for the Euclidean norm, $\text{cond}(A) = \sqrt{\lambda_1/\lambda_n}$, the largest and smallest eigenvalues of A^*A.

Proof.

(a) Multiply the inequalities above: $\frac{\|\delta x\|}{\|x\|} \leq \frac{\|A^{-1}\|\|\delta b\|}{\|b\|/\|A\|}$ and $\frac{\|\delta x\|}{\|x\|} \geq \frac{\|\delta b\|/\|A\|}{\|A^{-1}\|\|b\|}$.

(b) Corollaries 1 and 2. ■

$\text{cond}(A) \geq 1$, with equality iff A is a scalar multiple of a unitary (orthogonal) matrix; a large value signals possible loss of accuracy.

6.11 The geometry of orthogonal operators *

Motivation. An orthogonal operator splits into rotations of mutually orthogonal planes and at most one reflection.

Definition (Rotation; reflection)

An operator T on finite-dimensional real V is a **rotation** if $\mathsf{T} = \mathsf{I}$ or there are a 2-dimensional subspace W, an orthonormal basis $\{x_1, x_2\}$ of W and θ with $\mathsf{T}(x_1) = (\cos\theta)x_1 + (\sin\theta)x_2$, $\mathsf{T}(x_2) = (-\sin\theta)x_1 + (\cos\theta)x_2$, and $\mathsf{T}(y) = y$ for $y \in W^\perp$ (a rotation of W about $W^\perp$). T is a **reflection** if for some 1-dimensional W, $\mathsf{T}(x) = -x$ on W and $\mathsf{T}(y) = y$ on $W^\perp$ (a reflection about $W^\perp$). Both are orthogonal.

Example 1

On a 1-dimensional real space an orthogonal operator is $\mathsf{T} = \pm\mathsf{I}$ ($|\lambda| = 1$): the identity (a rotation) or a reflection.

Example 2

(a) $\mathsf{T}(a, b) = (-a, b)$ is the reflection of $\mathbb{R}^2$ about the y-axis ($W = \operatorname{span}\{e_1\}$).
(b) $\mathsf{T}(a, b, c) = (a, b, -c)$ is the reflection of $\mathbb{R}^3$ about the xy-plane.

Theorem 6.45

An orthogonal operator T on a 2-dimensional real inner product space is a rotation or a reflection; it is a rotation iff $\det \mathsf{T} = 1$ and a reflection iff $\det \mathsf{T} = -1$.

Proof. As in Theorem 6.23, in an orthonormal basis the matrix is $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ (a rotation, $\det = 1$) or $\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ (a symmetric orthogonal matrix with eigenvalues $1, -1$ and orthogonal eigenvectors: a reflection, $\det = -1$). ■

Corollary

On a 2-dimensional real inner product space, a composite of a reflection and a rotation (in either order) is a reflection.

Proof. It is orthogonal with determinant $(-1) \cdot 1 = -1$; apply Theorem 6.45. ■

Lemma

Every linear operator T on a nonzero finite-dimensional real vector space has an invariant subspace W with $1 \le \dim W \le 2$.

Proof [Supplement]. Factor the characteristic polynomial f over $\mathbb{R}$ into linear and irreducible quadratic factors $g_1 \cdots g_k$. By Cayley–Hamilton $f(\mathsf{T}) = g_1(\mathsf{T}) \cdots g_k(\mathsf{T}) = \mathsf{T}_0$, so some $g_j(\mathsf{T})$ is not one-to-one; take $x \neq 0$ with $g_j(\mathsf{T})x = 0$. If $g_j = t - c$, $W = \operatorname{span}\{x\}$; if $g_j = t^2 + bt + c$, then $\mathsf{T}^2 x = -b\mathsf{T}x - cx$, so $W = \operatorname{span}\{x, \mathsf{T}x\}$ is invariant. ■

Theorem 6.46

For an orthogonal operator T on a nonzero finite-dimensional real inner product space V, there are pairwise orthogonal T-invariant subspaces $W_1, \ldots, W_m$ with (a) $1 \le \dim W_i \le 2$ and (b) $V = W_1 \oplus \cdots \oplus W_m$.

Proof. Induction on $\dim V$. The Lemma gives an invariant W_1 with $\dim W_1 \in \{1,2\}$; T is injective, so $\mathsf{T}(W_1) = W_1$. Then $W_1^\perp$ is invariant: for $y \in W_1^\perp$, $x = \mathsf{T}x' \in W_1$ ($x' \in W_1$), $\langle \mathsf{T}y, x\rangle = \langle \mathsf{T}y, \mathsf{T}x'\rangle = \langle y, x'\rangle = 0$. Induction applies to the orthogonal $\mathsf{T}_{W_1^\perp}$, and $V = W_1 \oplus W_1^\perp$. ■

Theorem 6.47

In Theorem 6.46:
(a) the number of W_i on which T_{W_i} is a reflection is even or odd according as $\det \mathsf{T} = 1$ or -1;
(b) the decomposition can be chosen so that this number is 0 or 1 according as $\det \mathsf{T} = 1$ or -1, and so that every 1-dimensional W_i on which T is a rotation (i.e. the identity) is orthogonal to all the others.

Proof.

(a) $\det \mathsf{T} = \prod \det \mathsf{T}_{W_i}$ (Theorem 5.25: block-diagonal matrix), and $\det \mathsf{T}_{W_i} = -1$ exactly for reflections (Example 1, Theorem 6.45).

(b) Split each 2-dimensional reflection piece W_i into the orthogonal lines E_1, E_{-1} of T_{W_i}. All reflection pieces are now lines with $\mathsf{T} = -\mathsf{I}$; pair them into planes on which $\mathsf{T} = -\mathsf{I}$, a rotation by π. At most one such line remains, as decided by (a). ■

Corollary

Every orthogonal operator T on a finite-dimensional real inner product space is a product $\mathsf{T} = \mathsf{T}_1 \cdots \mathsf{T}_m$ of commuting orthogonal operators T_i, each a rotation or a reflection, at most one of them a reflection.

Proof. With $V = W_1 \oplus \cdots \oplus W_m$ as in Theorem 6.47(b), let $\mathsf{T}_i = \mathsf{T}$ on W_i and $\mathsf{T}_i = \mathsf{I}$ on $W_i^\perp = \bigoplus_{j \neq i} W_j$. Each T_i is a rotation or a reflection, they commute (different summands), and $\mathsf{T}_1 \cdots \mathsf{T}_m = \mathsf{T}$ on every W_i. ■

Example 3

For orthogonal T on a 3-dimensional space, Theorem 6.47(b) gives $m = 2$ (a line W_1 and a plane W_2 with T_{W_2} a rotation) or $m = 3$ (three lines). If $m = 2$, T is a rotation about the axis W_1, followed by a reflection in $W_1^\perp$ if $\mathsf{T}_{W_1} = -\mathsf{I}$; if $m = 3$, at most one line carries $\mathsf{T} = -\mathsf{I}$, so T is the identity or a reflection. So an orthogonal operator on $\mathbb{R}^3$ is a rotation about an axis, composed with a reflection iff $\det \mathsf{T} = -1$.

Appendices. Sets, Functions, Fields, Complex Numbers, Polynomials

A Sets

Motivation. The language of sets, unions, intersections and equivalence relations used throughout the notes.

A **set** is a collection of objects (**elements**), $x \in A$. $A = B$ if they have the same elements; $A \subseteq B$ (**subset**) if every element of A is in B; $A \subsetneq B$ is a **proper** subset; $\emptyset$ is the **empty set**. Sets are given by listing, $\{1, 2, 3\}$, or by a property, $\{x \in \mathbb{R} : 1 < x < 2\}$.

Example 1

$\{x \in \mathbb{R} : 1 < x < 2\}$ is the reals between 1 and 2; $\{1, 2\} = \{2, 1\}$ (order and repetition do not matter).

The **union** $A \cup B = \{x : x \in A \text{ or } x \in B\}$, the **intersection** $A \cap B = \{x : x \in A \text{ and } x \in B\}$; A, B are **disjoint** if $A \cap B = \emptyset$. For a family $\{A_\alpha : \alpha \in \Lambda\}$, $\bigcup_\alpha A_\alpha$ and $\bigcap_\alpha A_\alpha$ are defined likewise.

Example 2

$A = \{1, 3, 5\}$, $B = \{1, 5, 7, 8\}$: $A \cup B = \{1, 3, 5, 7, 8\}$, $A \cap B = \{1, 5\}$.

Example 3

For $\alpha > 1$ let $A_\alpha = \{x \in \mathbb{R} : -\frac{1}{\alpha} \leq x \leq 1 + \alpha\}$. Then $\bigcup_{\alpha>1} A_\alpha = \{x : -1 < x\}$ and $\bigcap_{\alpha>1} A_\alpha = \{x : 0 \leq x \leq 2\}$.

A **relation** on A is a set S of ordered pairs in A, $x \sim y$ iff $(x, y) \in S$; it is an **equivalence relation** if it is **reflexive** ($x \sim x$), **symmetric** ($x \sim y \Rightarrow y \sim x$) and **transitive** ($x \sim y$, $y \sim z \Rightarrow x \sim z$). Examples: equality; congruence mod m on $\mathbb{Z}$; similarity and congruence of matrices; isomorphism of vector spaces.

B Functions

Motivation. One-to-one, onto and invertible functions, composition and restriction.

A **function** $f : A \to B$ assigns to each $x \in A$ a unique $f(x) \in B$; A is the **domain**, B the **codomain**, $f(A)$ the **range**, $f(S)$ ($S \subseteq A$) the **image** and $f^{-1}(T) = \{x : f(x) \in T\}$ ($T \subseteq B$) the **preimage**. $f = g$ if $f(x) = g(x)$ for all x.

Example 1

$f : [-10, 10] \to \mathbb{R}$, $f(x) = x^2 + 1$: the range is $[1, 101]$, $f([2, 3]) = [5, 10]$, and $f^{-1}([5, 10]) = [-3, -2] \cup [2, 3]$.

f is **one-to-one** if $f(x) = f(y)$ implies $x = y$, and **onto** if $f(A) = B$. The **restriction** f_S of f to $S \subseteq A$ is f on S.

Example 2

$f : [-1, 1] \to [0, 1]$, $f(x) = x^2$ is onto but not one-to-one ($f(-1) = f(1)$); $f_{[0,1]}$ is one-to-one and onto.

The **composite** of $f : A \to B$ and $g : B \to C$ is $g \circ f : A \to C$, $(g \circ f)(x) = g(f(x))$; composition is associative, not commutative. f is **invertible** if $g \circ f = \mathrm{I}_A$, $f \circ g = \mathrm{I}_B$ for some (unique) $g = f^{-1} : B \to A$; this holds iff f is one-to-one and onto. $(f^{-1})^{-1} = f$, and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ for invertible f, g.

Example 3

$f(x) = 3x + 1$ on $\mathbb{R}$ is one-to-one and onto, with $f^{-1}(x) = \frac{x-1}{3}$.

C Fields

Motivation. Scalars may come from any field, where one adds, subtracts, multiplies and divides by nonzero elements.

Definition (Field)

A **field** $\mathbb{F}$ is a set with operations $+$ and $\cdot$ such that for all $a, b, c \in \mathbb{F}$: (F1) $a + b = b + a$, $ab = ba$; (F2) $(a + b) + c = a + (b + c)$, $(ab)c = a(bc)$; (F3) there are distinct $0, 1$ with $0 + a = a$, $1 \cdot a = a$; (F4) each a has c with $a + c = 0$, and each $b \neq 0$ has d with $bd = 1$; (F5) $a(b + c) = ab + ac$.

Example 1

$\mathbb{R}$ is a field.

Example 2

$\mathbb{Q}$ is a field.

Example 3

$\{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is a field: $(a + b\sqrt{2})^{-1} = \frac{a - b\sqrt{2}}{a^2 - 2b^2}$ ($a^2 \neq 2b^2$ for rationals not both zero).

Example 4

$\mathbb{Z}_2 = \{0, 1\}$ with $0 + 0 = 1 + 1 = 0$, $0 + 1 = 1$, $0 \cdot 0 = 0 \cdot 1 = 0$, $1 \cdot 1 = 1$ is a field.

Example 5

$\mathbb{N}$ and $\mathbb{Z}$ are not fields: (F4) fails ($2^{-1} \notin \mathbb{Z}$).

Theorem C.1 (Cancellation laws)

(a) If $a+b=c+b$, then $a=c$.
(b) If $ab=cb$ and $b\neq 0$, then $a=c$.

Proof.

(a) Add d with $b+d=0$: $a=a+(b+d)=(a+b)+d=(c+b)+d=c$.

(b) Multiply by b^{-1} similarly. ■

Corollary

The elements 0, 1 of (F3) and the c, d of (F4) are unique.

Proof. If $0'$ also works, $0'+b=b=0+b$, so $0'=0$ by (a); if $a+c=0=a+c'$, then $c=c'$ by (a). Similarly for 1 and d using (b). ■

Write $-a$ and b^{-1}; $a-b=a+(-b)$, $a/b=ab^{-1}$.

Theorem C.2

(a) $a\cdot 0=0$.
(b) $(-a)b=a(-b)=-(ab)$.
(c) $(-a)(-b)=ab$.

Proof.

(a) $0+a\cdot 0=a\cdot 0=a(0+0)=a\cdot 0+a\cdot 0$; cancel.

(b) $ab+(-a)b=(a+(-a))b=0\cdot b=0$, so $(-a)b=-(ab)$; similarly $a(-b)$.

(c) By (b) twice, $(-a)(-b)=-(a(-b))=-(-(ab))=ab$. ■

Corollary

0 has no multiplicative inverse.

Proof. $0\cdot d=0\neq 1$ by (a) and (F3). ■

The **characteristic** of $\mathbb{F}$ is the least $p\geq 1$ with $1+\cdots+1$ (p summands) $=0$, or 0 if none exists; it is 2 for $\mathbb{Z}_2$, 0 for $\mathbb{Q},\mathbb{R},\mathbb{C}$, and prime if nonzero. In characteristic 2, $x+x=0$, so 2 is not invertible; some results (e.g. Theorem 6.35) exclude this case.

D Complex numbers

Motivation. In $\mathbb{C}$ every nonconstant polynomial has a zero, which makes eigenvalue theory complete.

Definition (Complex numbers)

A **complex number** is $z=a+bi$ with $a,b\in\mathbb{R}$ (real and imaginary parts); $(a+bi)+(c+di)=(a+c)+(b+d)i$ and $(a+bi)(c+di)=(ac-bd)+(bc+ad)i$. Then $i^2=-1$, and $\mathbb{R}\subseteq\mathbb{C}$ via $a=a+0i$.

Example 1

$z = 3 - 5i$, $w = 9 + 7i$: $z + w = 12 + 2i$, $zw = (27 + 35) + (-45 + 21)i = 62 - 24i$.

Example 2

$(-5 + 2i)(1 - 3i) = -5 + 15i + 2i - 6i^2 = 1 + 17i$.

Theorem D.1

$\mathbb{C}$ is a field.

Proof [Supplement]. (F1), (F2), (F5) are direct computations; $0 = 0 + 0i$, $1 = 1 + 0i$; $-(a + bi) = -a - bi$; and for $a + bi \neq 0$, $(a + bi)\frac{a-bi}{a^2+b^2} = 1$. ■

Definition (Conjugate)

$\overline{a + bi} = a - bi$.

Example 3

$\overline{-3 + 2i} = -3 - 2i$, $\overline{4 - 7i} = 4 + 7i$, $\bar{6} = 6$.

Theorem D.2

(a) $\bar{\bar{z}} = z$;
(b) $\overline{z + w} = \bar{z} + \bar{w}$;
(c) $\overline{zw} = \bar{z}\bar{w}$;
(d) $\overline{z/w} = \bar{z}/\bar{w}$ $(w \neq 0)$;
(e) $z \in \mathbb{R}$ iff $\bar{z} = z$.

Proof. Direct computation with $z = a + bi$, $w = c + di$; e.g. (c): $\overline{zw} = (ac - bd) - (bc + ad)i = (a - bi)(c - di)$. (d) follows from (c) applied to $(z/w)w = z$. ■

Definition (Absolute value)

$|a + bi| = \sqrt{a^2 + b^2}$; then $z\bar{z} = |z|^2$, and $\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$.

Example 4

$\frac{1+4i}{3-2i} = \frac{(1+4i)(3+2i)}{9+4} = \frac{-5+14i}{13}$.

Theorem D.3

(a) $|zw| = |z||w|$;
(b) $|z/w| = |z|/|w|$ $(w \neq 0)$;
(c) $|z + w| \leq |z| + |w|$;
(d) $|z| - |w| \leq |z + w|$.

Proof.

(a) $|zw|^2 = zw\overline{zw} = z\bar{z}w\bar{w}$.

(b) From (a).

(c) $|z+w|^2 = |z|^2 + 2\operatorname{Re}(z\bar{w}) + |w|^2 \le |z|^2 + 2|z||w| + |w|^2$, since $\operatorname{Re} u \le |u|$.

(d) $|z| = |(z+w)+(-w)| \le |z+w| + |w|$. ■

Polar form: $z = |z|(\cos\theta + i\sin\theta) = |z|e^{i\theta}$; products multiply absolute values and add angles, and $z^n = |z|^n e^{in\theta}$, so n-th roots exist.

Theorem D.4 (Fundamental theorem of algebra)

Every $p(z) = a_n z^n + \cdots + a_0 \in \mathsf{P}(\mathbb{C})$ of degree $n \ge 1$ has a zero.

Proof.

Step 1 (a minimum exists). Let $\mu = \inf_z |p(z)|$. For $|z| = s > R$ (R large), $|p(z)| \ge |a_n|s^n - \sum_{k<n}|a_k|s^k > \mu + 1$. On the compact disk $|z| \le R$, $|p|$ attains its minimum at some z_0, and $|p(z_0)| = \mu$.

Step 2 (d'Alembert). Suppose $p(z_0) \ne 0$. Then $q(z) = p(z+z_0)/p(z_0) = 1 + b_k z^k + \cdots + b_n z^n$ ($b_k \ne 0$) has $|q| \ge 1$ everywhere. Choose θ with $e^{ik\theta}b_k = -|b_k|$ (k-th roots exist). For $r > 0$ with $r^k|b_k| < 1$,

$$|q(re^{i\theta})| \le 1 - r^k|b_k| + r^{k+1}\sum_{j>k}|b_j|r^{j-k-1} = 1 - r^k\Big(|b_k| - r\sum_{j>k}|b_j|r^{j-k-1}\Big) < 1$$

for r small enough, contradicting $|q| \ge 1$. So $p(z_0) = 0$. ■

Corollary

Every $p \in \mathsf{P}(\mathbb{C})$ of degree $n \ge 1$ factors as $p(z) = a_n(z-c_1)\cdots(z-c_n)$.

Proof. Induction on n: by Theorem D.4 and Corollary 1 of Theorem E.1, $p(z) = (z-c_1)q(z)$ with $\deg q = n-1$ and leading coefficient a_n. ■

A field in which every nonconstant polynomial splits is **algebraically closed**; $\mathbb{C}$ is, $\mathbb{R}$ is not.

E Polynomials

Motivation. Division with remainder, $q_1 f_1 + q_2 f_2 = 1$ for relatively prime f_1, f_2, and unique factorization are the polynomial facts used for operators.

$f \mid g$ (f **divides** g) if $g = fq$ for some polynomial q.

Theorem E.1 (Division algorithm)

If $\deg f = n$ and $\deg g = m \ge 0$, there are unique q, r with $f = qg + r$ and $\deg r < m$.

Proof.

Step 1 (existence). Induction on n. If $n < m$, take $q = 0$, $r = f$. Otherwise, with leading coefficients a_n of f and b_m of g, $h = f - a_n b_m^{-1} x^{n-m} g$ has degree $< n$, so $h = q_1 g + r$ by induction, and $f = (q_1 + a_n b_m^{-1} x^{n-m})g + r$.

Step 2 (uniqueness). If $qg + r = q'g + r'$, then $(q-q')g = r' - r$ has degree $< m$, which forces $q = q'$ (otherwise the left side has degree $\ge m$) and then $r = r'$. ■

Corollary 1

For $\deg f > 0$ and $a \in \mathbb{F}$: $f(a) = 0$ iff $(x - a) \mid f$.

Proof. $f = q \cdot (x - a) + r$ with r constant; evaluating at a, $r = f(a)$. ■

Corollary 2

A polynomial of degree $n \geq 1$ has at most n distinct zeros.

Proof. Induction: if $f(a) = 0$, $f = (x - a)q$ with $\deg q = n - 1$, and every other zero of f is a zero of q (a field has no zero divisors). ■

Definition (Relatively prime)

Nonzero f, g are **relatively prime** if no polynomial of positive degree divides both. E.g. $x^2(x-1)$ and $(x-1)(x-2)$ are not; $x^2(x-1)$ and $(x-2)(x-3)$ are.

Theorem E.2

If f_1, f_2 are relatively prime, there are q_1, q_2 with $q_1 f_1 + q_2 f_2 = 1$.

Proof. $I = \{q_1 f_1 + q_2 f_2\}$ is closed under addition and multiplication by polynomials; let $d \in I$ be nonzero of least degree. Dividing, $f_1 = qd + r$ with $\deg r < \deg d$ and $r = f_1 - qd \in I$, so $r = 0$ and $d \mid f_1$; likewise $d \mid f_2$. Hence $d = c \neq 0$ is constant, and $c^{-1}d = 1 \in I$. ■

Example 1

$f_1 = x^3 - x^2 + 1$ and $f_2 = (x-1)^2$ are relatively prime; indeed $(-x+2)f_1 + (x^2 - x - 1)f_2 = 1$.

Definition (Polynomials of operators)

For $f(x) = a_0 + a_1 x + \cdots + a_n x^n$, $f(\mathsf{T}) = a_0 \mathsf{I} + a_1 \mathsf{T} + \cdots + a_n \mathsf{T}^n$ and $f(A) = a_0 I + a_1 A + \cdots + a_n A^n$.

Example 2

$\mathsf{T}(a, b) = (2a + b, a - b)$ and $f(x) = x^2 + 2x - 3$: $\mathsf{T}^2(a, b) = (5a + b, a + 2b)$, so $f(\mathsf{T})(a, b) = (5a + b, a + 2b) + (4a + 2b, 2a - 2b) - (3a, 3b) = (6a + 3b, 3a - 3b)$.

Theorem E.3

(a) $f(\mathsf{T})$ is a linear operator.
(b) If β is a finite ordered basis and $A = [\mathsf{T}]_\beta$, then $[f(\mathsf{T})]_\beta = f(A)$.

Proof.

(a) Linear combinations of composites of linear maps are linear (Theorems 2.7, 2.9).

(b) $[\mathsf{T}^k]_\beta = A^k$ (Corollary to Theorem 2.11), and Theorem 2.8. ■

Theorem E.4

$f_1(\mathsf{T})f_2(\mathsf{T}) = f_2(\mathsf{T})f_1(\mathsf{T})$ and $f_1(A)f_2(A) = f_2(A)f_1(A)$.

Proof. Both equal $(f_1 f_2)(\mathsf{T})$, since powers of T commute and composition distributes over sums (Theorem 2.10). ■

Theorem E.5

If f_1, f_2 are relatively prime, there are q_1, q_2 with (a) $q_1(\mathsf{T})f_1(\mathsf{T}) + q_2(\mathsf{T})f_2(\mathsf{T}) = \mathsf{I}$; (b) $q_1(A)f_1(A) + q_2(A)f_2(A) = I$.

Proof. Apply $g \mapsto g(\mathsf{T})$, which respects sums and products, to the identity of Theorem E.2. ■

Definition (Monic, irreducible)

f is **monic** if its leading coefficient is 1; it is **irreducible** if $\deg f > 0$ and f is not a product of two polynomials of positive degree. (Irreducibility depends on $\mathbb{F}$: x^2+1 is irreducible over $\mathbb{R}$, not over $\mathbb{C}$.) Degree-one polynomials are irreducible.

Theorem E.6

If ϕ is irreducible and $\phi \nmid f$, then ϕ and f are relatively prime.

Proof. A common divisor g of positive degree divides ϕ, so $\phi = cg$ with c constant (irreducibility); then $\phi \mid f$, a contradiction. ■

Theorem E.7

Two distinct irreducible monic polynomials are relatively prime.

Proof. If $\phi_1 \mid \phi_2$, then $\phi_2 = c\phi_1$ (irreducible), and $c = 1$ (monic), so $\phi_1 = \phi_2$. Hence $\phi_1 \nmid \phi_2$; apply Theorem E.6. ■

Theorem E.8

If ϕ is irreducible and $\phi \mid fg$, then $\phi \mid f$ or $\phi \mid g$.

Proof. If $\phi \nmid f$, then $q_1\phi + q_2 f = 1$ (Theorems E.6, E.2), so $g = q_1\phi g + q_2 fg$ is divisible by ϕ. ■

Corollary

If irreducible monic ϕ divides $\phi_1 \cdots \phi_n$ (irreducible monic), then $\phi = \phi_i$ for some i.

Proof. Induction on n: $\phi \mid \phi_1 \cdots \phi_{n-1}$ (then induction) or $\phi \mid \phi_n$ (then $\phi = \phi_n$ by Theorem E.7). ■

Theorem E.9 (Unique factorization)

Every f of positive degree is uniquely $f = c\,\phi_1^{n_1} \cdots \phi_k^{n_k}$ with c a constant, ϕ_i distinct irreducible monic polynomials and $n_i \geq 1$.

Proof.

Step 1 (existence). Induction on $\deg f$. If f is irreducible, $f = c \cdot (c^{-1}f)$. Otherwise $f = gh$ with $0 < \deg g, \deg h < \deg f$; factor g and h by induction and collect equal factors.

Step 2 (uniqueness). c is the leading coefficient. If $\prod \phi_i^{n_i} = \prod \psi_j^{m_j}$, then ϕ_1 divides the right side, so $\phi_1 = \psi_j$ for some j (Corollary to Theorem E.8). Cancel one factor (no zero divisors) and induct on the total degree. ■

Theorem E.10

If $\mathbb{F}$ is infinite and $f(a) = g(a)$ for all $a \in \mathbb{F}$, then $f = g$.

Proof. $h = f - g$ has infinitely many zeros, so $\deg h \geq 1$ is impossible (Corollary 2 of Theorem E.1); h is a constant with $h(a) = 0$, so $h = 0$. ■

Over a finite field this fails: x^2 and x define the same function on $\mathbb{Z}_2$.